UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement 3.3V MOSFET SPICE Model Document

Version 1.0 Phase 2
2013/04/12

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1 Introduction

1.1 Revision history

Version	Date	Owner	Comment
0.2_P1	2010/05/11	Kelly Pao	1. Copy from G-05-LOGIC/MIXED_MODE11-MMC/L110E-SPICE and revise the Idlin corner. 2. Revise cgc fitting 3. Lmin=0.144um to 0.16um
0.3_p1	2010/10/22	Yealing Lin	1. Add Twell NMOS model (copy from Twin well NMOS).
0.4_p1	2010/12/24	Laney Fan	1. Re-center the model to align EDR target 2. Add the enhancement of thermal noise model
0.4_p2	2012/01/17	Laney Fan	1. Add the note in the general information
1.0_p1	2012/09/14	Laney Fan	1.Version up to 1.0 based on 6 pcs Si data.
1.0_p2	2013/04/12	Laney Fan	1. Modify the verilog file for avoiding the noise as device is off.

1.2 General information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11um LOGIC/MIXED_MODE Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110E-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

3.3V n-ch MOSFET
3.3V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.24

For detailed descriptions of the compact model equations and parameters, please refer to this material.

BSIM3V3.24 MOSFET Model User Manual

In BSIM3V3.24 , there are two types of models that can be used. One is a binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is a global model, which means one scalable parameter set is provided and it can cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for their many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the hardware data, which can be found in the following sections.

Note: Three choices of thick oxide IO devices are available, 2.5V, 3.3V and HG 3.3V, which cannot coexist on the same chip. If any design requirement is necessary, please discuss in advance with UMC through CE department

1.3 Model usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.9
Cadence Spectre version 6.2.1.299
Mentor Graphics Eldo version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below. Please note that 'fast' here means lower threshold voltage, higher leakage and driving current, and 'slow' means higher threshold voltage, lower leakage and driving current.

TT: Typical n-ch MOSFET and Typical p-ch MOSFET
FF: Fast n-ch MOSFET and Fast p-ch MOSFET
SS: Slow n-ch MOSFET and Slow p-ch MOSFET
FNSP: Fast n-ch MOSFET and Slow p-ch MOSFET
SNFP: Slow n-ch MOSFET and Fast p-ch MOSFET

Follow the description for net-list of LOD effect, and these steps to invoke the model file during simulation.

Example:

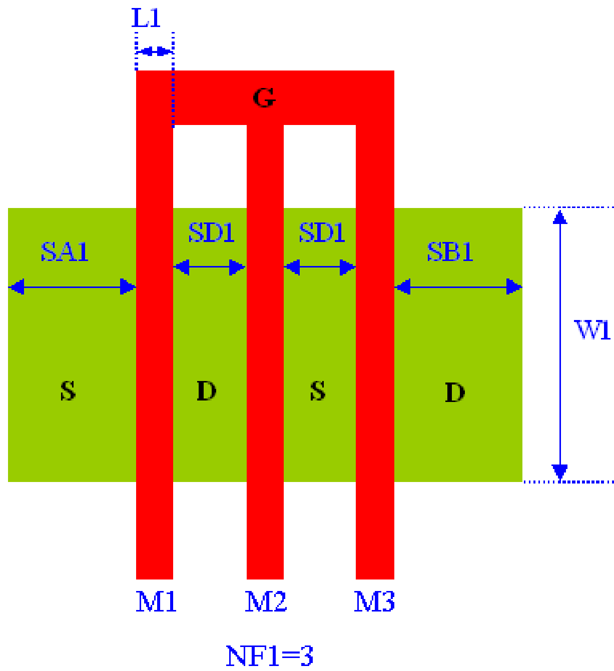

Figure 1-1 Layout example for LOD effect

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.

- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

```
.Option  scale=0.9
.lib "./ModelFileName.lib" CaseName
```

, where CaseName could be TT, FF, SS, FNSP, or SNFP.

- Define device condition by 'Mx' statement;

Example : (Refer to Figure 1-1)

```
M1  Vdd  Vgg  Vss  Vbb  DeviceName  W=W1  L=L1
+ SA=SA1
+ SB=SD1*2+L1*2+SB1
+ PS='2*(SA1+W1)'
+ PD='2*(SD1*2+L1*2+SB1+W1)'
+ AS='SA1*W1'
+ AD='(SD1*2+L1*2+SB1)*W1'

M2  Vdd  Vgg  Vss  Vbb  DeviceName  W=W1  L=L1
+ SA=SA1+SD1*1+L1*1  SB=SD1*1+L1*1+SB1
+ PS='2*(SA1+SD1*1+L1*1+W1)'
+ PD='2*(SB1+SD1*1+L1*1+W1)'
+ AS='(SA1+SD1*1+L1*1)*W1'
```

```
+ AD='(SB1+SD1*1+L1*1)*W1'

M3 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1+SD1*2+L1*2 SB=SB1
+ PS='2*(SD1*2+L1*2+SA1+W1)'
+ PD='2*(SB1+W1)'
+ AS='(SD1*2+L1*2+SA1+W1)*W1'
+ AD='SB1*W1'
```

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

```
SetOption1 options scale=0.9
include "./ModelFileName.lib.scs" section=CaseName
```

, where CaseName could be TT, FF, SS, FNSP, or SNFP.

- Define device condition by 'Mxx' statement;

Example : (Refer to Figure 1-1)

```
M1 (Vdd Vgg Vss Vbb) DeviceName w=W1 l=L1
+ sa=SA1
+ sb=SD1*2+L1*2+SB1
+ ps='2*(SA1+W1)'
+ pd='2*(SD1*2+L1*2+SB1+W1)'
+ as='SA1*W1'
+ ad='(SD1*2+L1*2+SB1)*W1'


M2 (Vdd Vgg Vss Vbb) DeviceName w=W1 l=L1
+ sa=SA1+SD1*1+L1*1 sb=SD1*1+L1*1+SB1
+ ps='2*(SA1+SD1*1+L1*1+W1)'
+ pd='2*(SB1+SD1*1+L1*1+W1)'
+ as='(SA1+SD1*1+L1*1)*W1'
+ ad='(SB1+SD1*1+L1*1)*W1'

M3 (Vdd Vgg Vss Vbb) DeviceName w=W1 l=L1
+ sa=SA1+SD1*2+L1*2 sb=SB1
+ ps='2*(SD1*2+L1*2+SA1+W1)'
+ pd='2*(SB1+W1)'
+ as='(SD1*2+L1*2+SA1+W1)*W1'
+ ad='SB1*W1'
```

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

```
.Option scale=0.9
```

.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName could be TT, FF, SS, FNSP, or SNFP.
- Define device condition by 'Mx' statement;

Example : (Refer to Figure 1-1)

```
M1 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1
+ SB=SD1*2+L1*2+SB1
+ PS='2*(SA1+W1)'
+ PD='2*(SD1*2+L1*2+SB1+W1)'
+ AS='SA1*W1'
+ AD='(SD1*2+L1*2+SB1)*W1'

M2 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1+SD1*1+L1*1 SB=SD1*1+L1*1+SB1
+ PS='2*(SA1+SD1*1+L1*1+W1)'
+ PD='2*(SB1+SD1*1+L1*1+W1)'
+ AS='(SA1+SD1*1+L1*1)*W1'
+ AD='(SB1+SD1*1+L1*1)*W1'

M3 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1
+ SA=SA1+SD1*2+L1*2 SB=SB1
+ PS='2*(SD1*2+L1*2+SA1+W1)'
+ PD='2*(SB1+W1)'
+ AS='(SD1*2+L1*2+SA1+W1)*W1'
+ AD='SB1*W1'
```

ModelFileName for 3.3V MOSFET: l110ae_33_v102
DeviceName for 3.3V n-ch MOSFET: n_33_l110e
DeviceName for 3.3V p-ch MOSFET: p_33_l110e
DeviceName for 3.3V n-ch MOSFET: n_33bpw_l110e (for Twell model)

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range(After 90% shrink):

For both 3.3V n-ch and p-ch MOSFET:

$0.306\ \mu m \leqq L_{DES} \leqq 50\ \mu m$

$0.16\ \mu m \leqq W_{DES} \leqq 100\ \mu m$

Adopt multi-finger structures if $W_{DES} > 9\ \mu m$.

$0.324\ \mu m \leqq SA,SB \leqq 1.98\ \mu m$

Note: SAREF=SBREF=1.98µm is the reference dimension of the STI-stress (LOD) effect.

Voltage range:

For 3.3V n-ch MOSFET:

$0V \leqq V_{GS} \leqq 3.3V$ (*1.1)

$0V \leqq V_{DS} \leqq 3.3V$ (*1.1)

$-3.3V \leqq V_{BS} \leqq 0V$

For 3.3V p-ch MOSFET:

$0V \geqq V_{GS} \geqq -3.3V$ (*1.1)

$0V \geqq V_{DS} \geqq -3.3V$ (*1.1)

$3.3V \geqq V_{BS} \geqq 0V$

Temperature range:

For 3.3V n-ch and p-ch MOSFET:

-50 ℃ ~ +125 ℃

UMC L110E is UMC L130E 90% direct shrunk option.The following table give users (including designers) a dimension correlation between Drawing vs. SPICE/Resistor dimensions.

Feature\ GDSII	Drawing in L130E GDSII database (unit : um)	L110 GDSII =0.9*(L130E GDSII) Dimension (unit : um)	L110 SPICE Dimension (unit : um)
Core (W/L)	10/0.12	9/0.108	(L sizing back 12 nm) 9/0.12
Salicide poly resistor	(Width/Length)	0.9*(Width/Length)	
3.3V I/O (W/L)	10/0.34	9/0.306	(L sizing back 12 nm) 9/0.318
contact size	0.16*0.16	0.144*0.144	
Via size(1X;2X)	0.2*0.2; 0.4*0.4	0.18*0.18; 0.36*0.36	
Metal size	Width/Space	0.9*(Width/Space)	
Sloting inserting	metal line width>5.556um	mwtal line width>5um	
N-well,salicide and non-salicide resistor	(Width/Length)	0.9*(Width/Length)	

Note:

1.Because process bias make drawn and physical dimension are different, please refer to interconnect capacitance model (G-04 document in individual DSM) for precise interconnection-capacitance extraction.

2.Please refer to SPICE model (G-05 document in individual DSM) for precise device characteristics.

2 3.3 V n-ch MOSFET

2.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	QGSML	QGSML	QGSML	QGSML
Lot ID	D1A0BFF3	D1A0BFF3	D1A0BFF3	D1A0BFF3
Wafer Number	12	12	12	12

2.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for NMOS and I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for PMOS, respectively. The dimension is after 90% shrink and 12nm sizing channel length back.
All the normalized current are divided with W_{DES}.

All parameters are given for T=25 ℃, V_{BS}=0 V unless specified otherwise.

Table 2-2 Electrical Parameters for NMOS

All parameters are given for T=25 ^{o}C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.608	0.609	0.608	0.608
I_{ON}	V_{DS} = 3.3 V	μA/ μm	39.8	40.31	39.47	39.48
Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.32 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.637	0.629	0.627	0.630
V_{TSAT}	V_{DS} = 3.3 V	V	0.470	0.463	0.467	0.470
Body Effect V_T shift	V_{DS} = 3.3 V V_{BS}= 0 ~ -3.3 V	V	--	0.097	0.152	0.152
DIBL V_T shift	V_{DS}= 0.1~3.3 V	V	0.167	0.167	0.160	0.160
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	635	634.67	628.33	635
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	3E-12	2.0E-12	4.7E-12	4.3E-12
Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.617	0.617	0.614	0.614
V_{TSAT}	V_{DS}= 3.3 V	V	--	0.611	0.606	0.606
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	35.77	35.41	35.20	35.21
Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μm / 0.32 μm)						
V_{TLIN}	V_{DS} = 0.1 V	V	0.585	0.576	0.582	0.585
V_{TSAT}	V_{DS}= 3.3 V	V	0.427	0.420	0.430	0.433
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	857.14	855.34	844.57	853.71
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	1.29E-11	1.9E-12	1.7E-11	1.6E-11
Capacitance						
CJ	V_J = 0 V	F/m^2	7.84E-04	7.71E-04	7.80E-04	7.80E-04
CJSW	V_J = 0 V	F/m	1.08E-10	1.19E-10	1.08E-10	1.08E-10
CJSWG	V_J = 0 V	F/m	2.35E-10	2.56E-10	2.30E-10	2.30E-10

2.3 MOSFET DC model

2.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below(The dimensions are after 90% shrink and 12nm sizing channel length back.).

Table 2-3 Geometry of NMOS devices measured for DC parameter extraction

W\L	10	1	0.8	0.68	0.5	0.44	0.34	0.32
10	X	X	X	X	X	X	X	X
1	X							X
0.8						X		X
0.6	X	X		X		X		X
0.43	X							X
0.35	X					X		X
0.25	X					X		X
0.16	X							X
0.14	X	X	X	X	X	X	X	X

2.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 2-4 Test conditions for IV curves of NMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	0.1	3.3	3.2
V_{GS} (V)	-0.5	3.6	0.05
V_{BS} (V)	0	-3.3	-0.825

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	3.6	0.05
V_{GS} (V)	0.7	3.3	0.65
V_{BS} (V)	0	-3.3	-3.3

2.3.3 Model verification plots

The simulation results (solid lines) obtained from the extracted model were plotted together with the measurement results (dotted lines) for comparison. The L and W in the plots refer to (L_{DES}*0.9+12nm and W_{DES}*0.9) respectively.

I_D *vs.* V_{DS} *Plots*

The following plots show measured and simulated output I-V curves for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

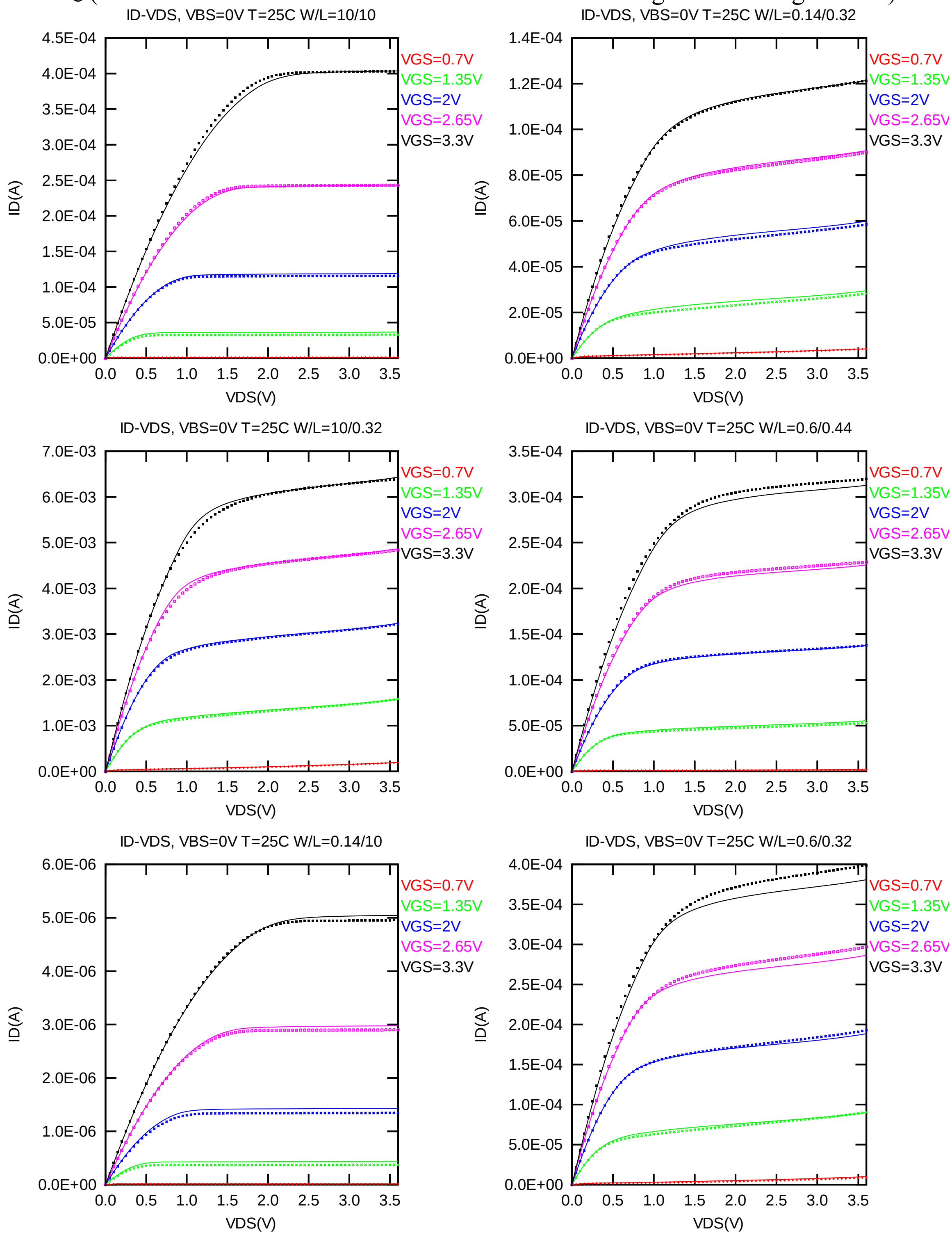

Figure 2-1 I_D vs. V_{DS} at V_{BS} = 0 V for NMOS

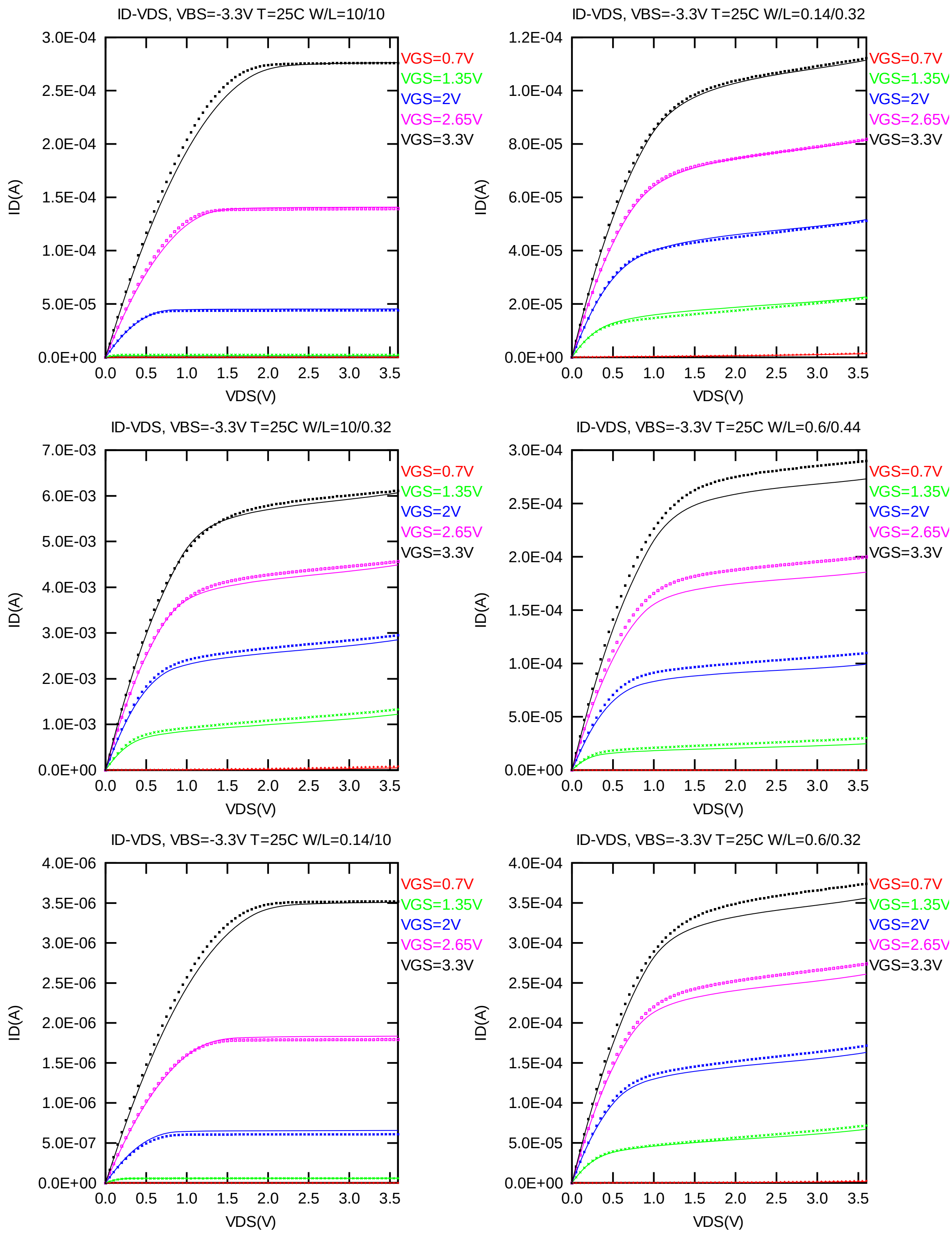

Figure 2-2 I_D vs. V_{DS} at V_{BS}=-3.3 V for NMOS

G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

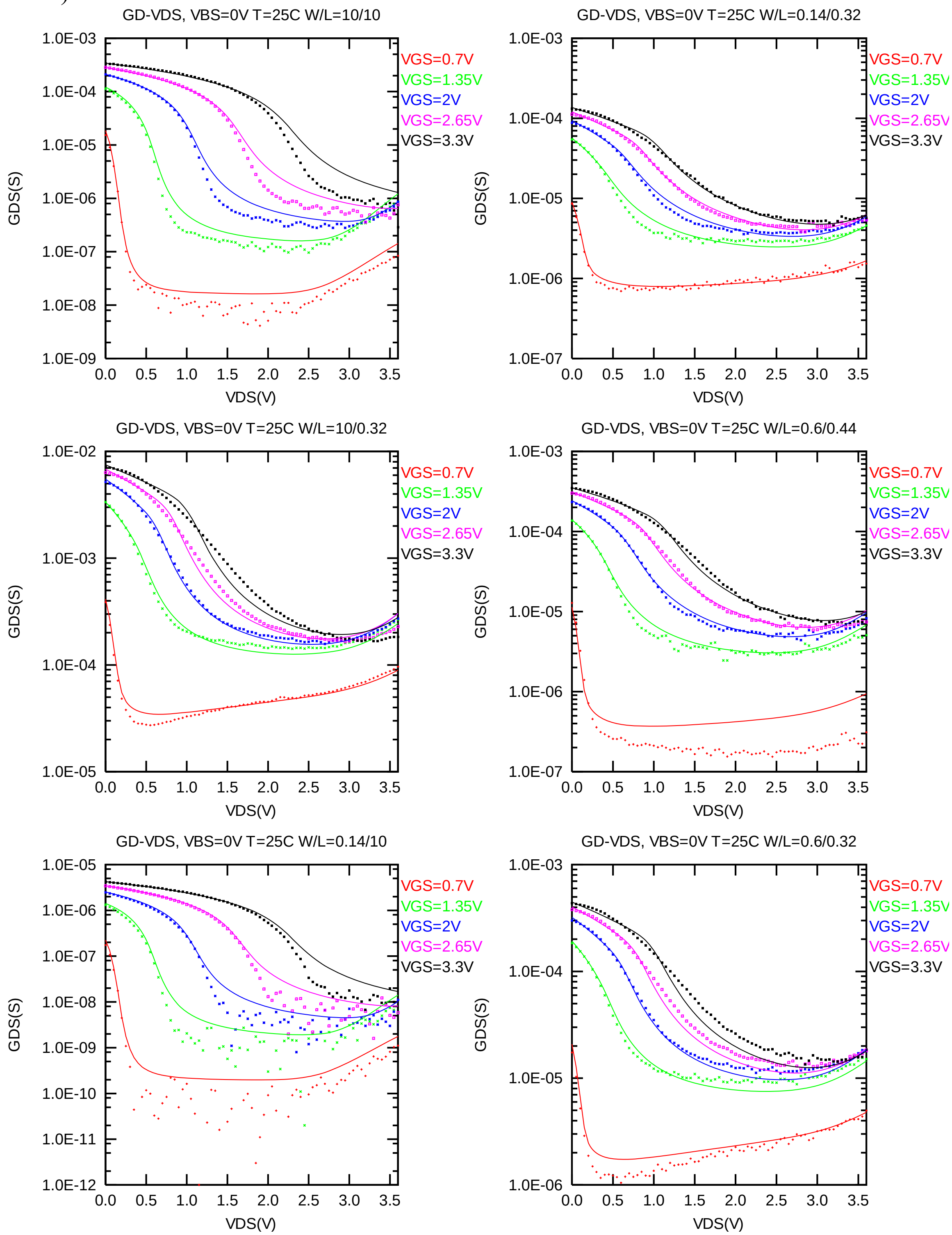

Figure 2-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for NMOS

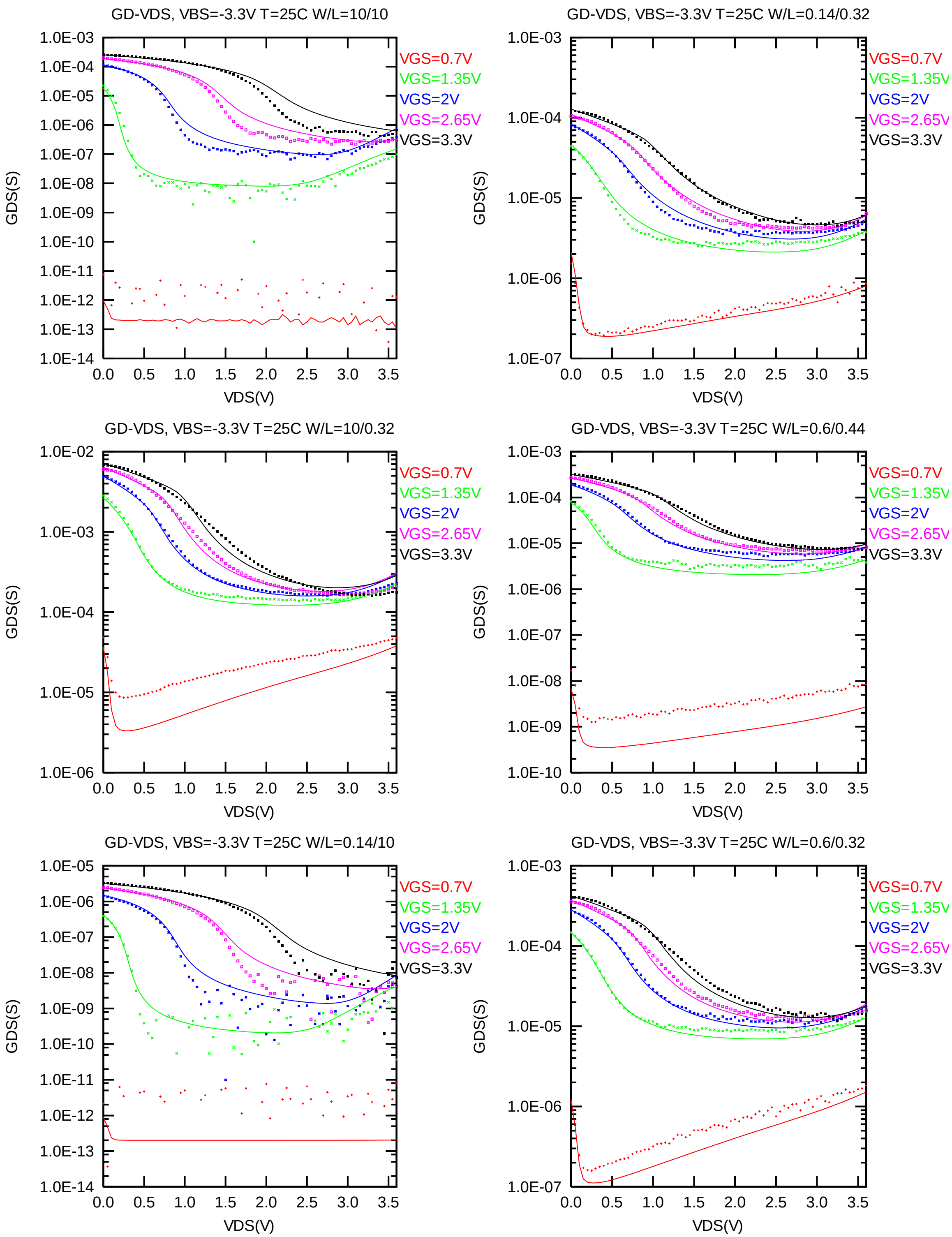

Figure 2-4 G_{DS} vs. V_{DS} at V_{BS} = -3.3 V for NMOS

I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

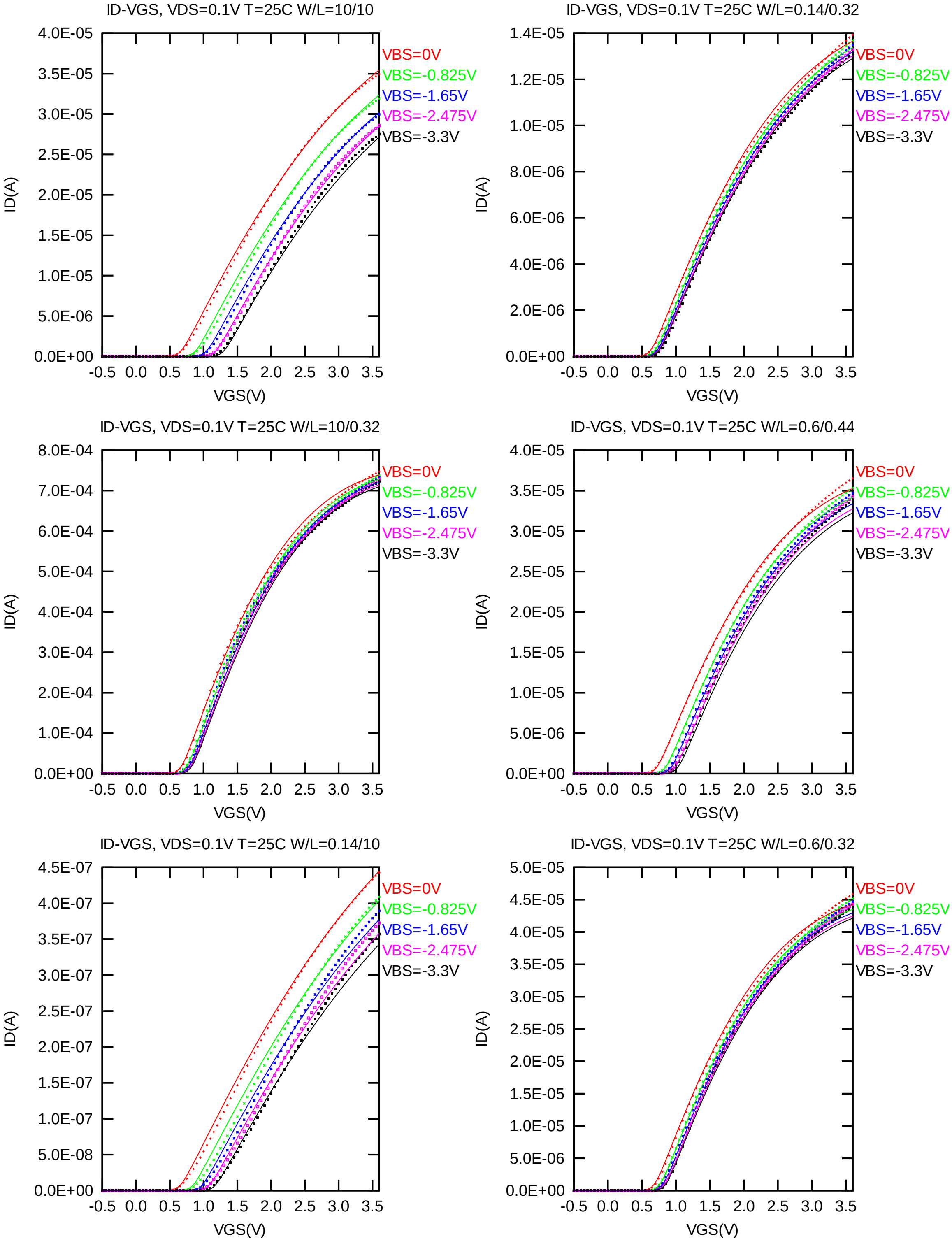

Figure 2-5 I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

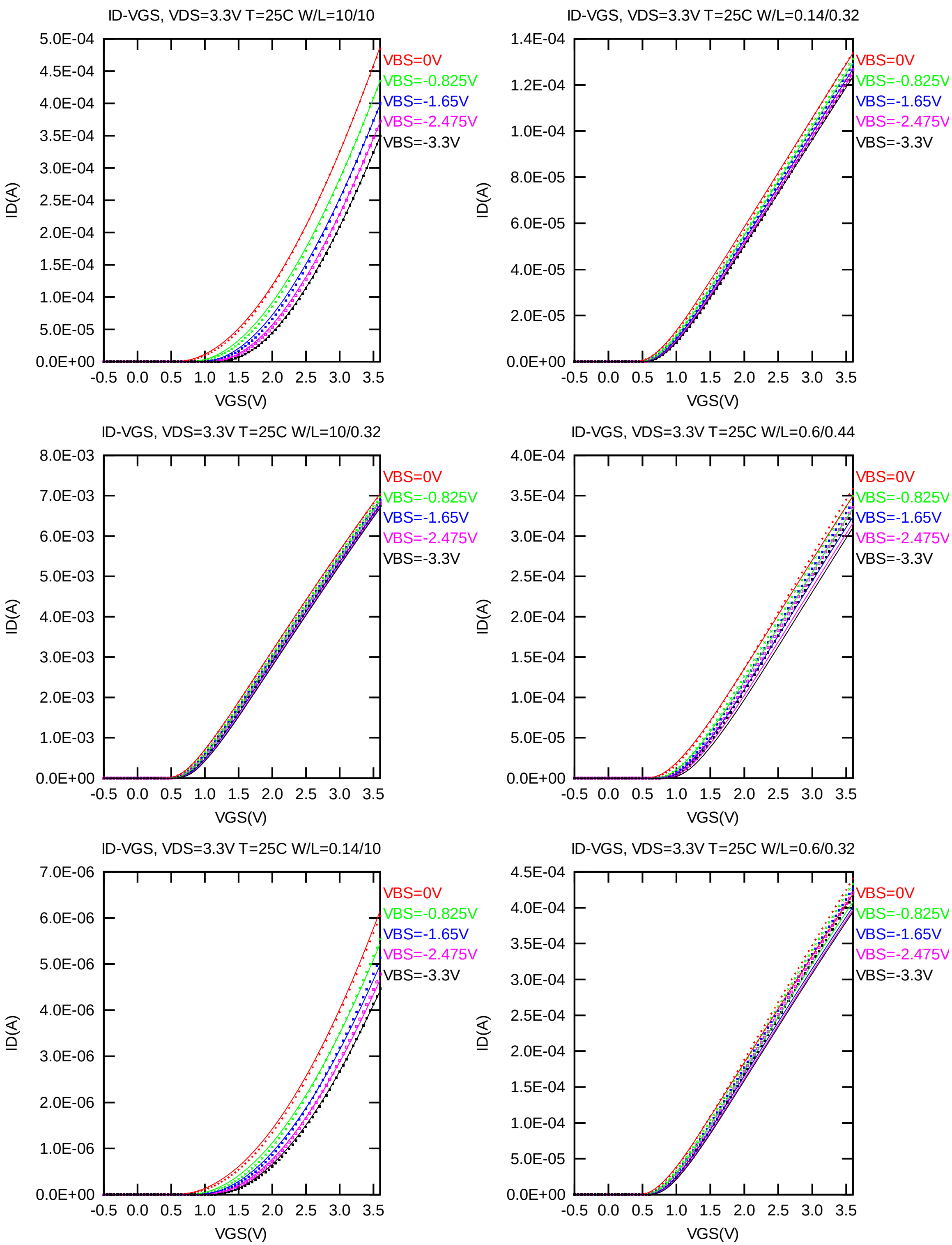

Figure 2-6 I_D vs. V_{GS} at V_{DS} = 3.3 V for NMOS

I_D *vs.* V_{GS} *Plots (logarithmic scale)*

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

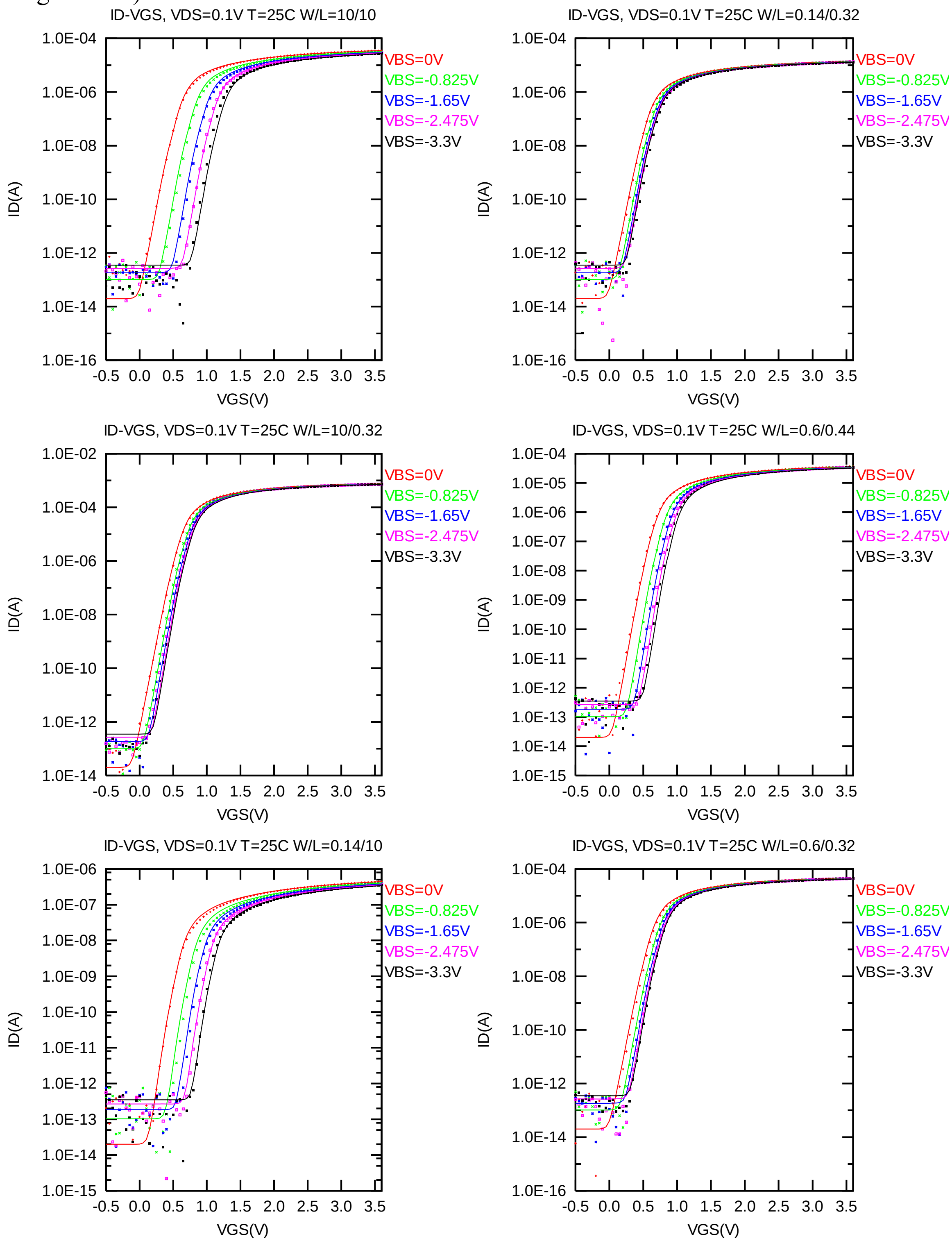

Figure 2-7 Log scaled I_D vs. V_{GS} at V_{DS} = 0.1 V for NMOS

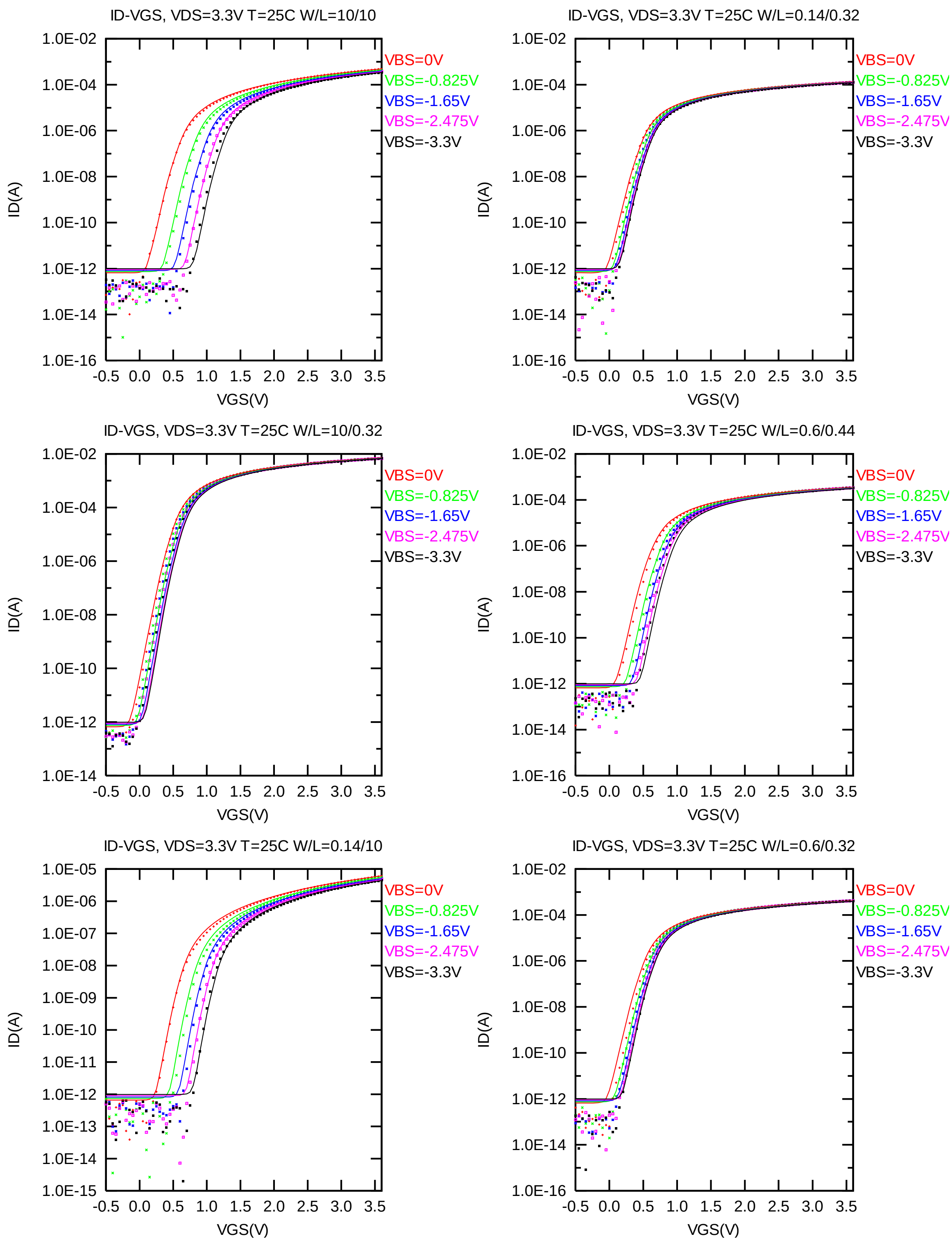

Figure 2-8 Log scaled I_D vs. V_{GS} at V_{DS} = 3.3 V for NMOS

G_M vs. V_{GS} Plots

The following plots show measured and simulated transfer conductance curves in linear scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

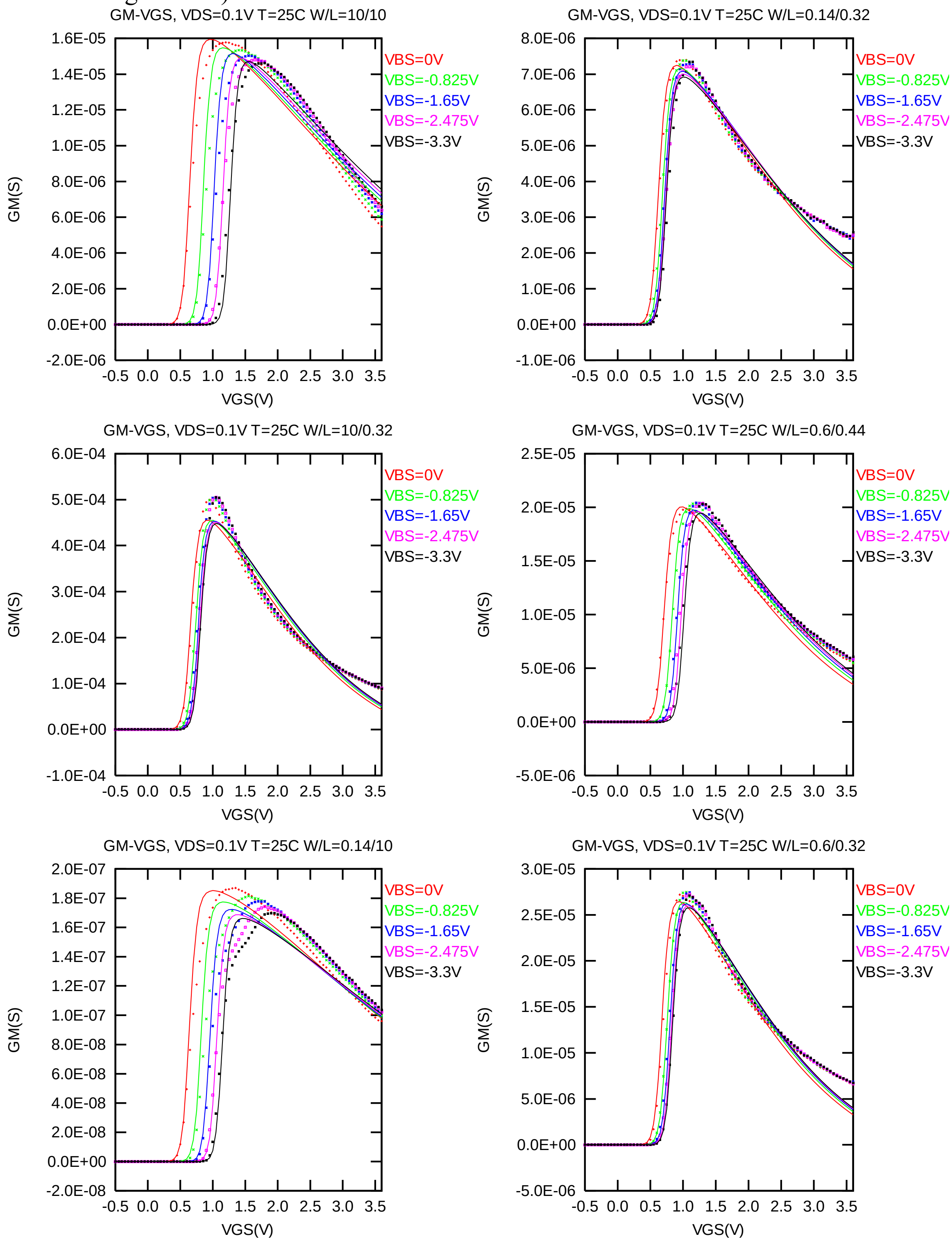

Figure 2-9 G_M vs. V_{GS} at V_{DS} = 0.1 V for NMOS

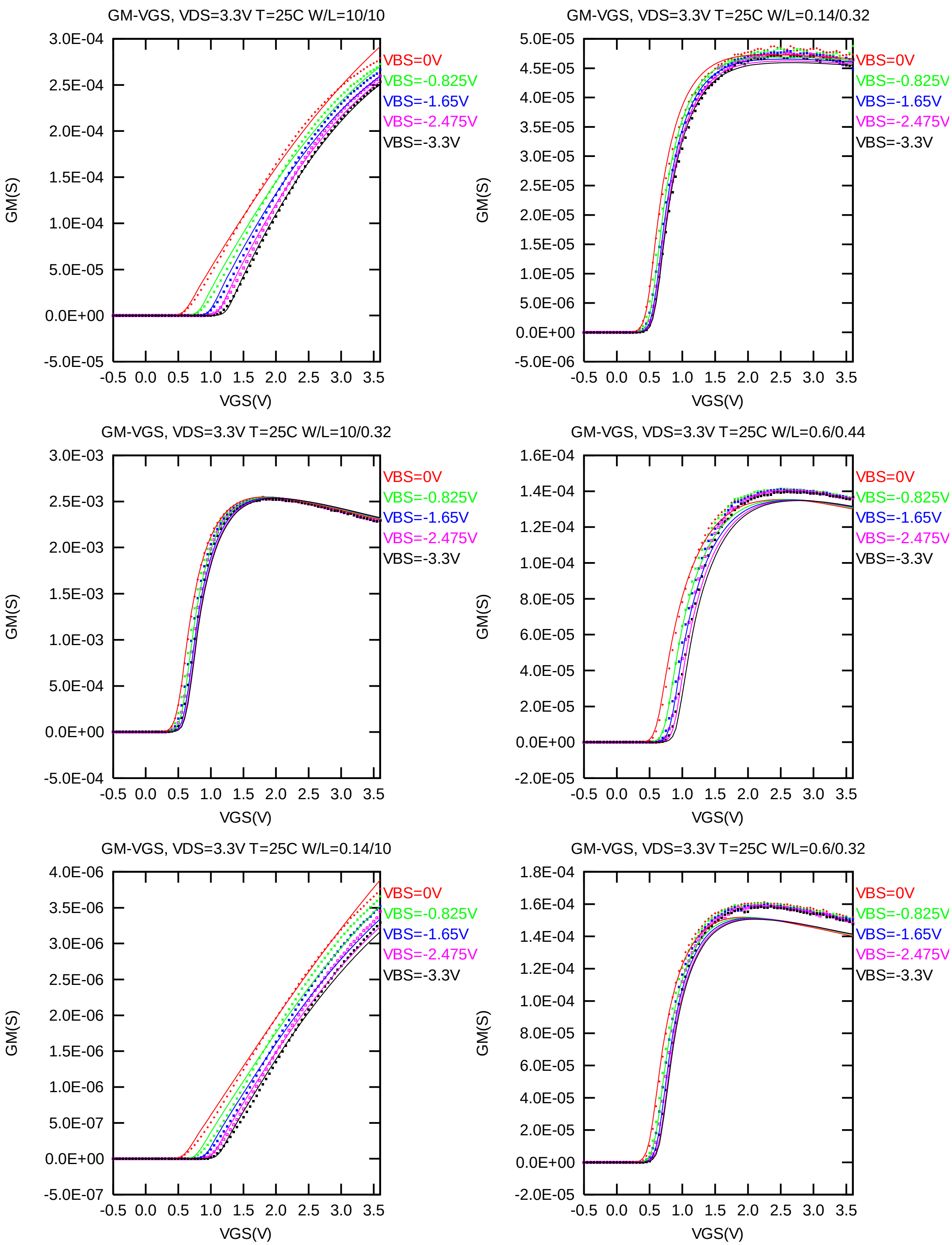

Figure 2-10 G_M vs. V_{GS} at V_{DS} = 3.3 V for NMOS

V_{TLIN} & V_{TSAT} vs. L_{DES} & W_{DES}

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

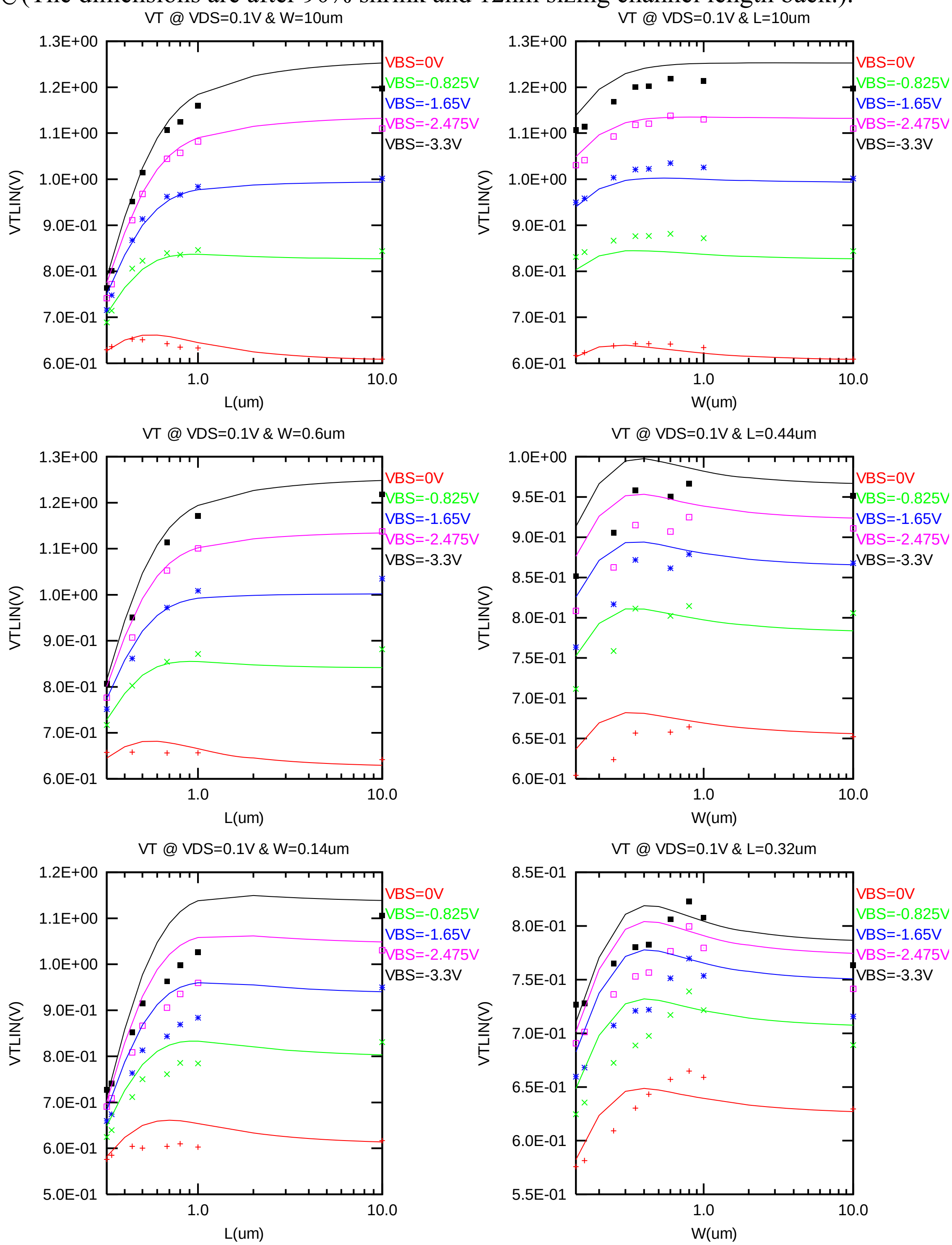

Figure 2-11 V_{TLIN} vs. channel length & channel width for low drain bias (V_{DS} = 0.1 V) for NMOS

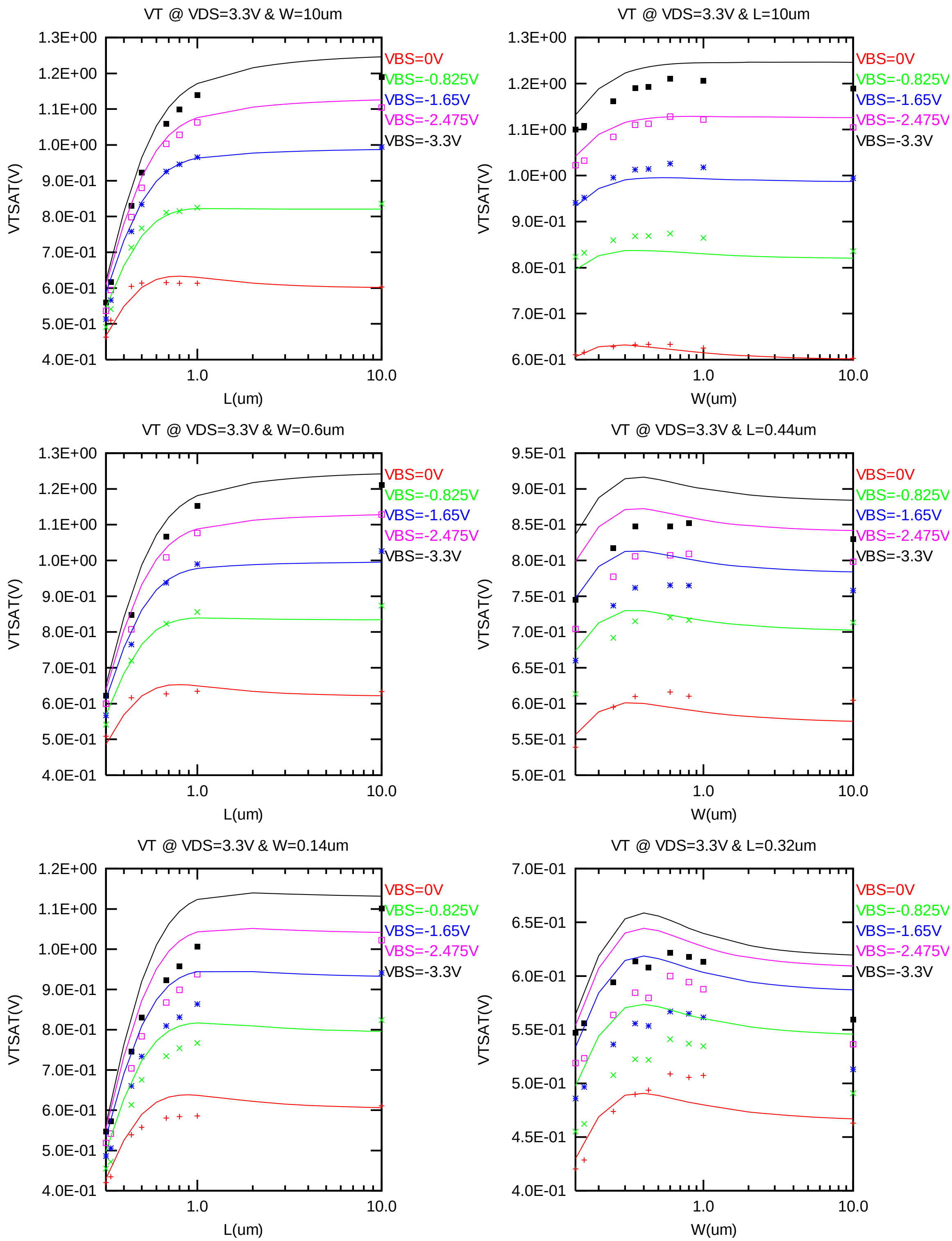

Figure 2-12 V_{TSAT} vs. channel length & channel width for high drain bias (V_{DS} = 3.3 V) for NMOS

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = 3.3$ V for $V_{BS} = 0$ V and $V_{BS} = -3.3$ V at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

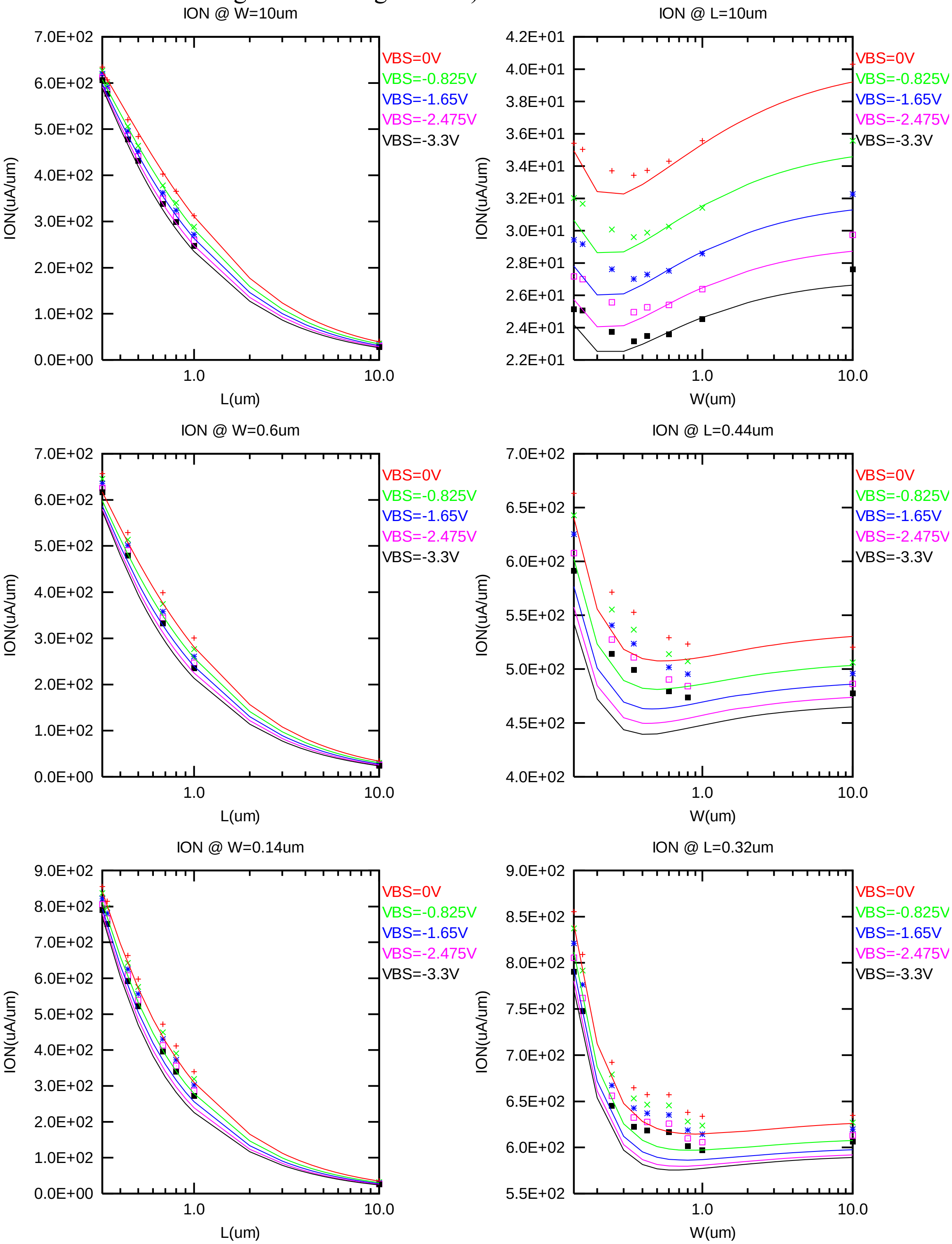

Figure 2-13 Saturation current I_{ON} vs. channel length and channel width for NMOS

I_{OFF} vs. L_{DES} & I_{OFF} vs. W_{DES}

The following plots show the channel length and width dependence of I_{OFF} extracted at $V_{DS} = V_{CC}$ and $V_{GS} = 0V$ for $V_{BS} = 0$ V to -3.3V at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.). A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.

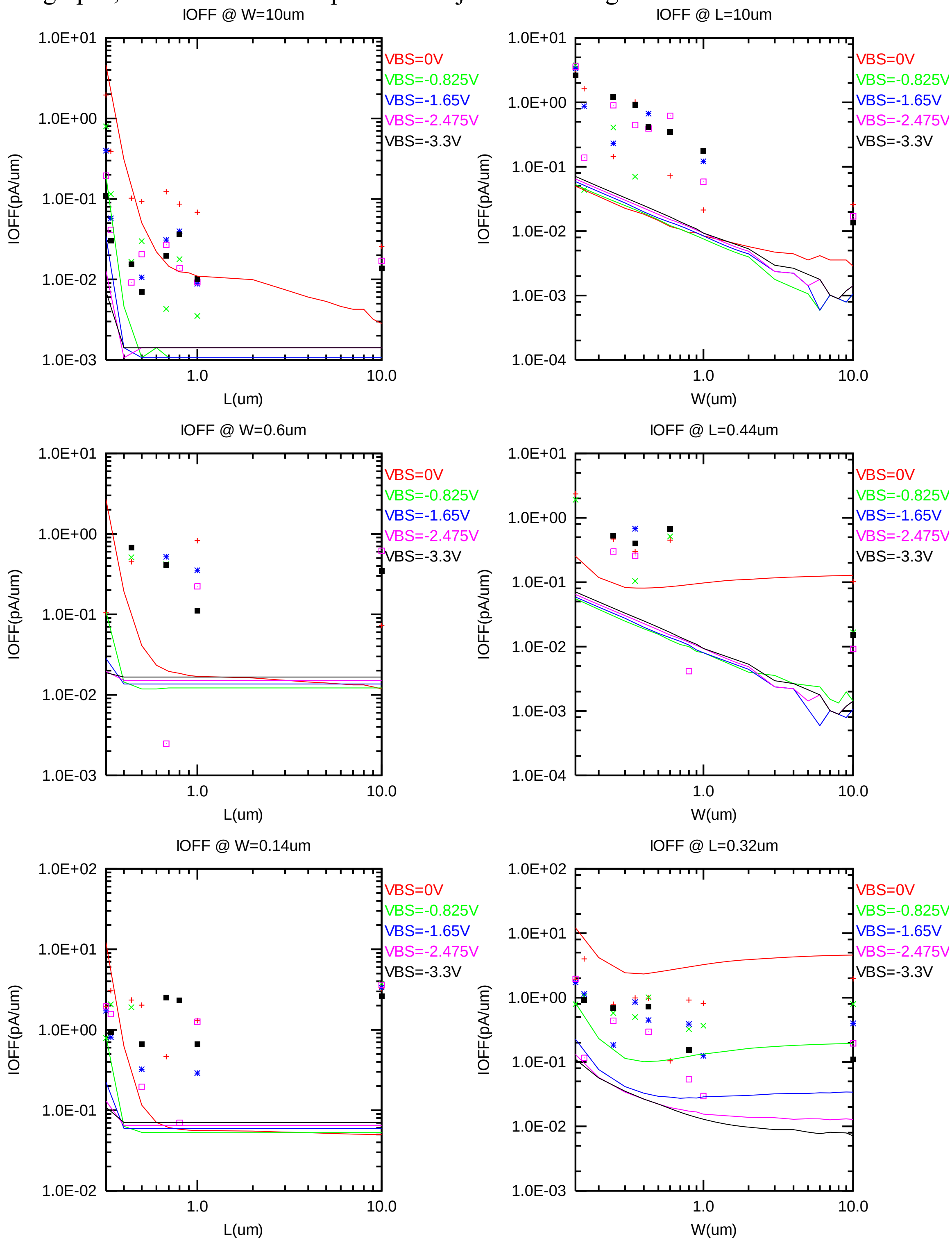

Figure 2-14 I_{OFF} vs. channel length and channel width for NMOS

V_{TLIN} and V_{TSAT} vs. Temperature

The following plots show the temperature dependence of the linear and saturation threshold voltage extracted at I_D = 300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) (The dimensions are after 90% shrink and 12nm sizing channel length back.).

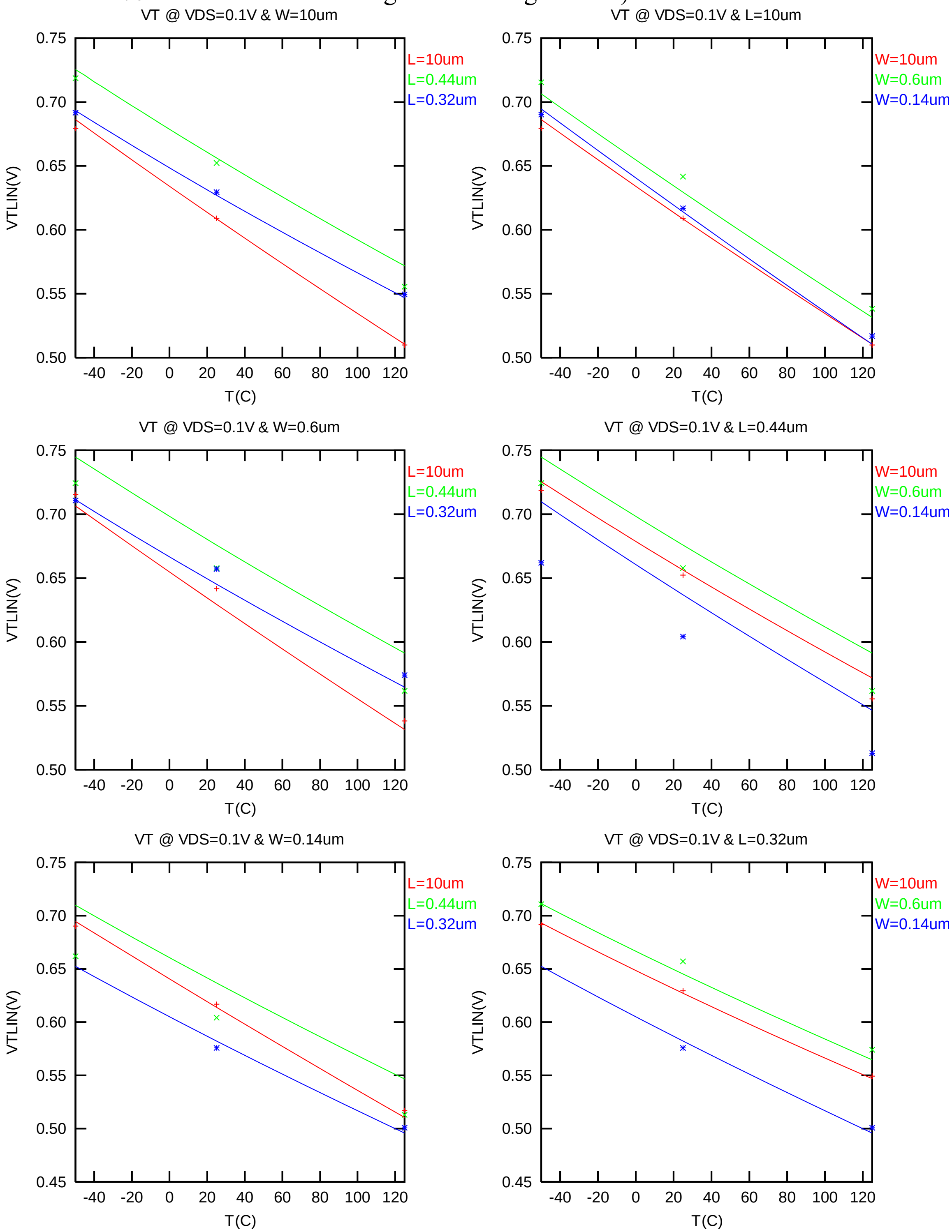

Figure 2-15 V_{TLIN} vs. temperature for NMOS

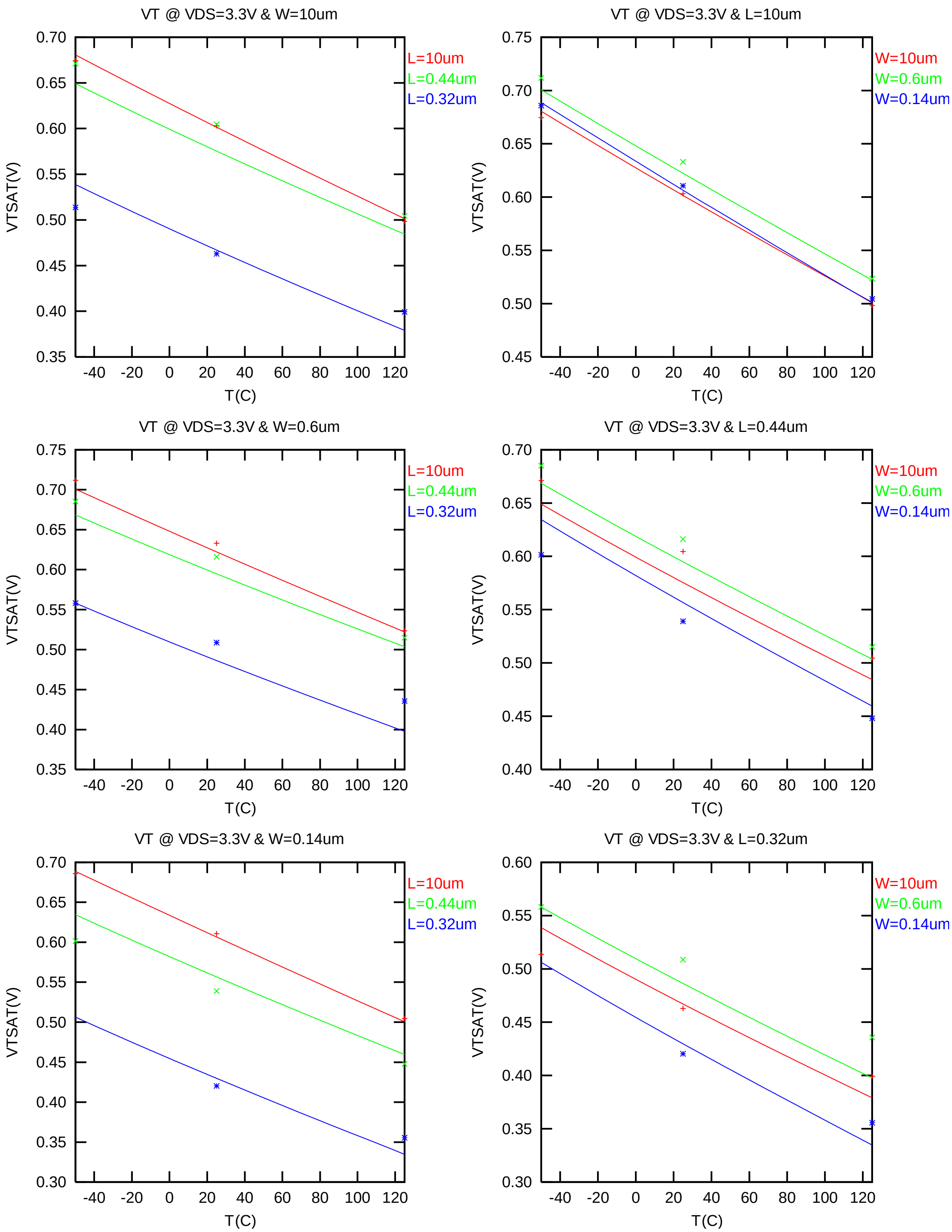

Figure 2-16 V_{TSAT} vs. temperature for NMOS

I_{ON} vs Temperature

The following plots show the temperature dependence of I_{ON} extracted at V_{GS} = V_{DS} = 3.3 V, V_{BS} = 0 V(The dimensions are after 90% shrink and 12nm sizing channel length back.).

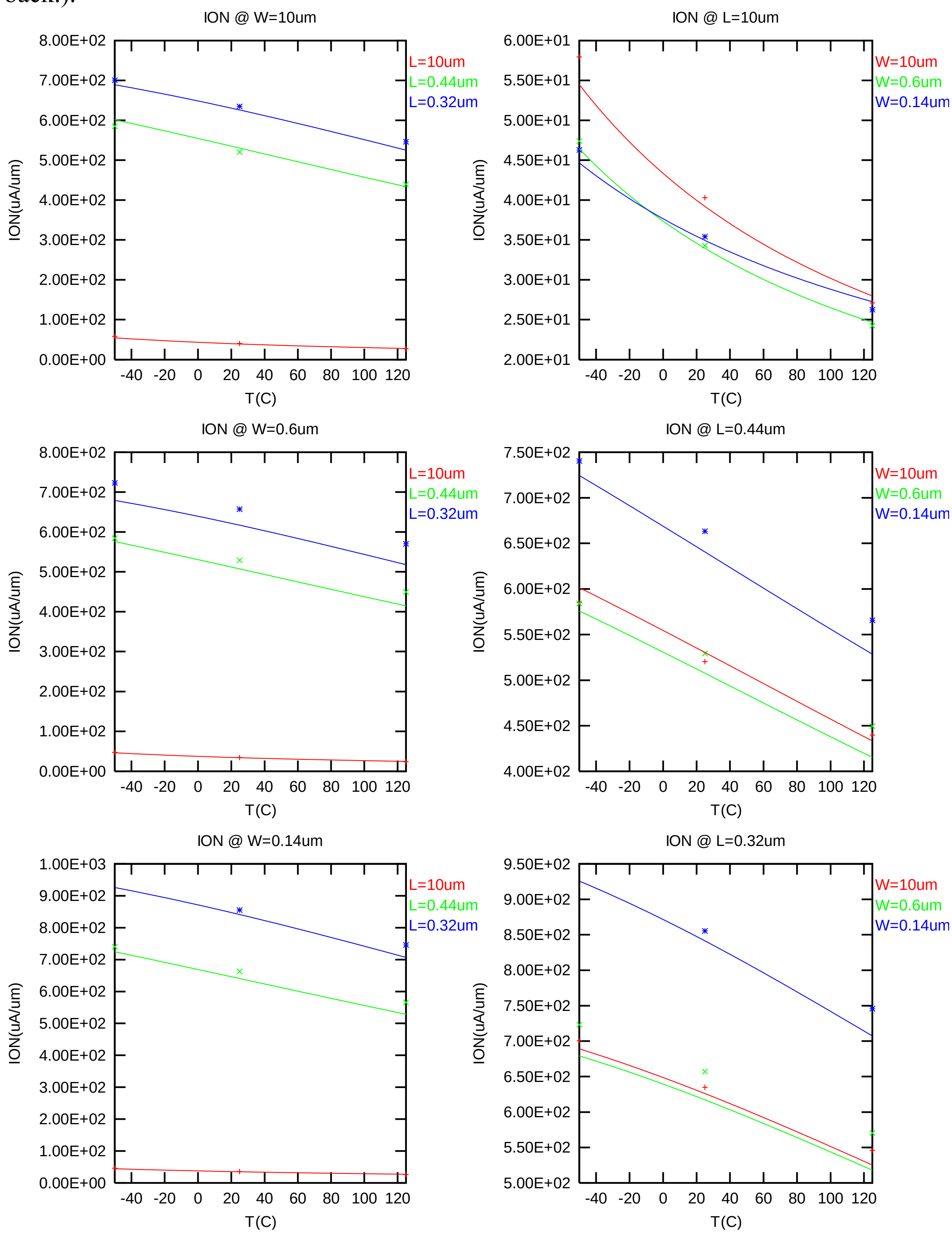

Figure 2-17 Saturation Current vs. temperature for NMOS

I_{OFF} vs Temperature

The following plot shows the temperature dependence of I_{OFF} extracted at V_{DS} = 3.3 V, V_{GS} = V_{BS} = 0 V(The dimensions are after 90% shrink and 12nm sizing channel length back.).

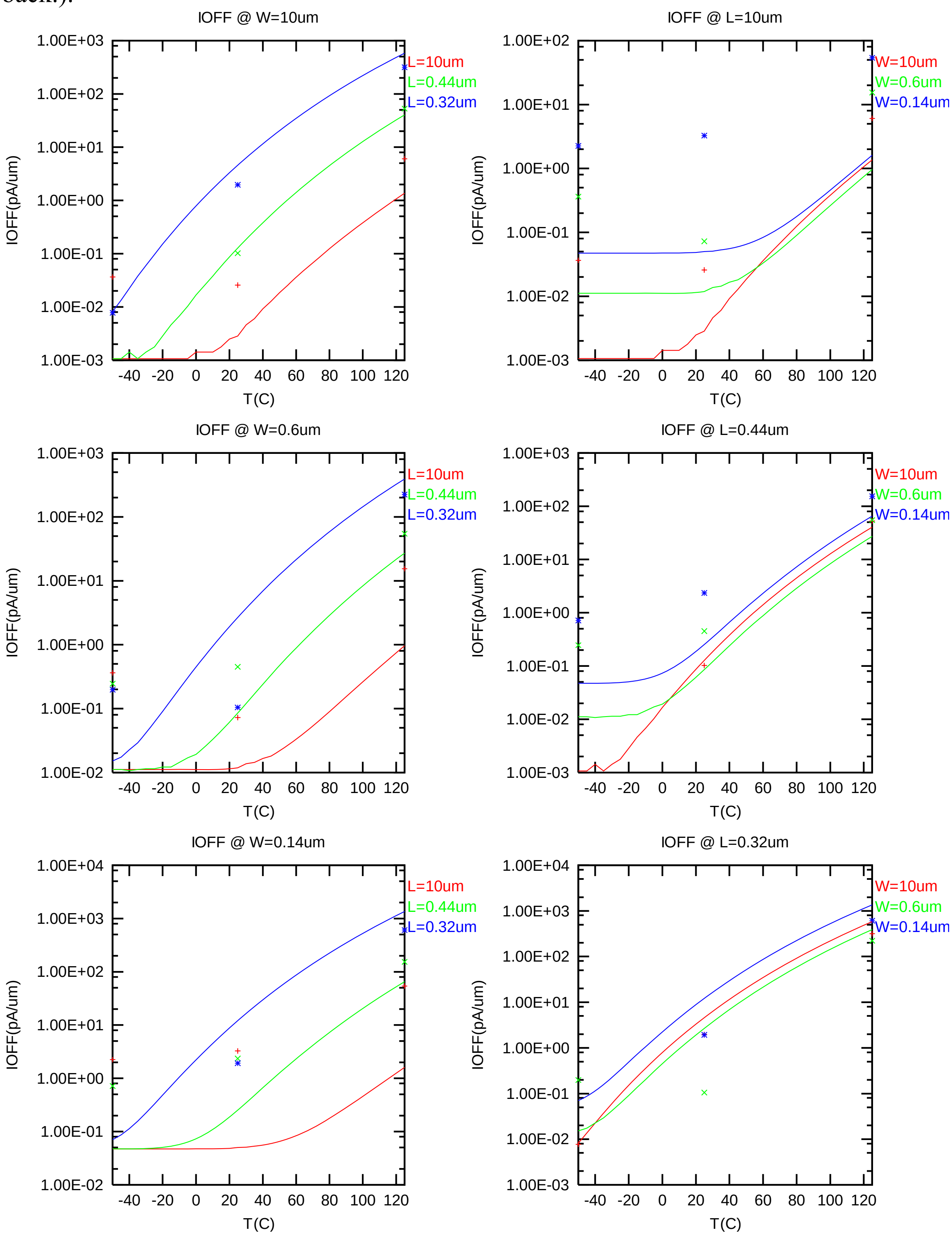

Figure 2-18 I_{OFF} vs. temperature for NMOS

2.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

a. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
b. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
c. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$

2) G_M and G_{DS} accuracy criteria

d. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
e. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 2-5 Requirement for model fit quality

Target of Quality Check	$V_{GS} < V_T$ (sub-threshold)	$V_T \leq V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \leq V_{GS} \leq V_{CC}$ (strong inversion)
I_D	≤75%	≤25%	≤7%
G_M	≤50%	≤20%	≤20%
G_{DS}	≤100%	≤50%	≤50%

I_D *Accuracy*

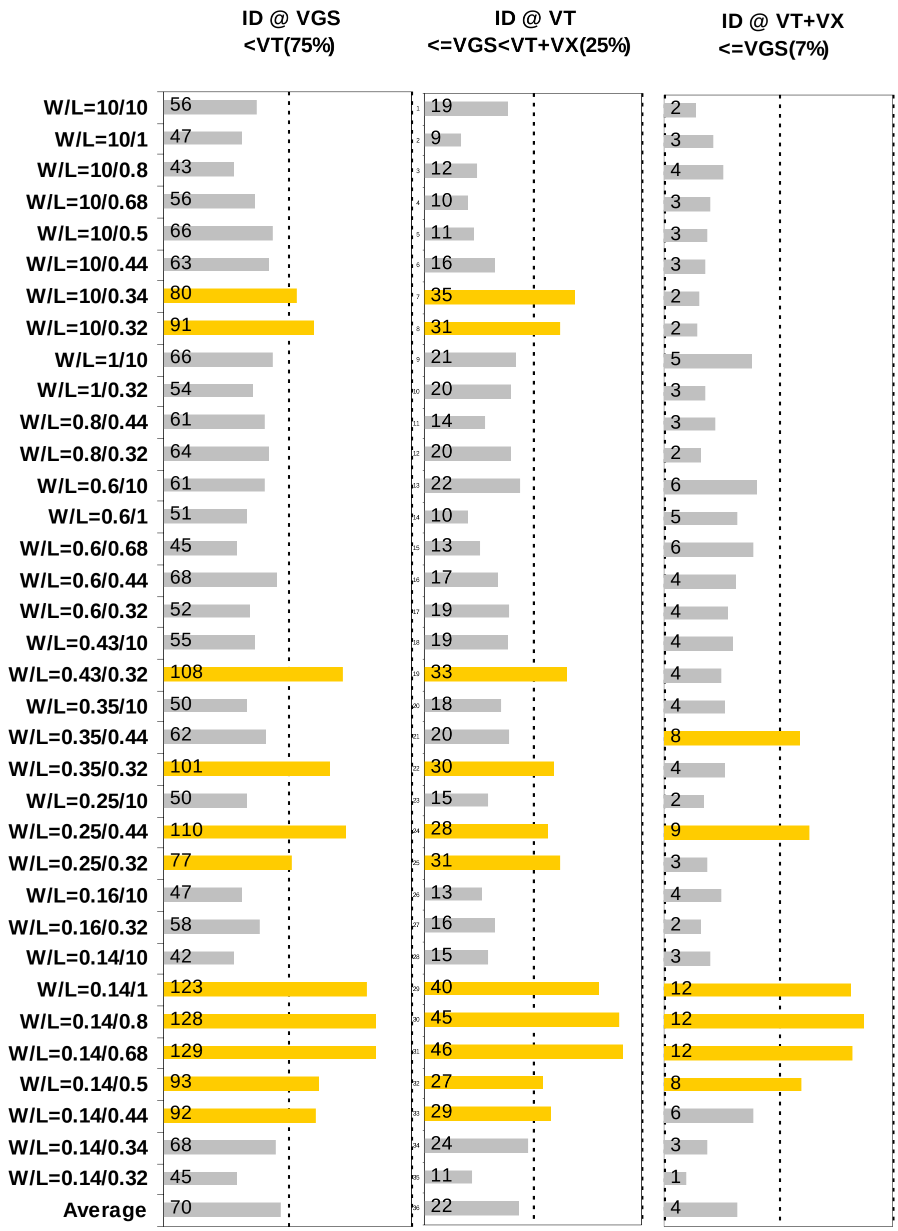

Figure 2-19 I_D accuracy at 25 ℃ for NMOS

G_M and G_{DS} Accuracy

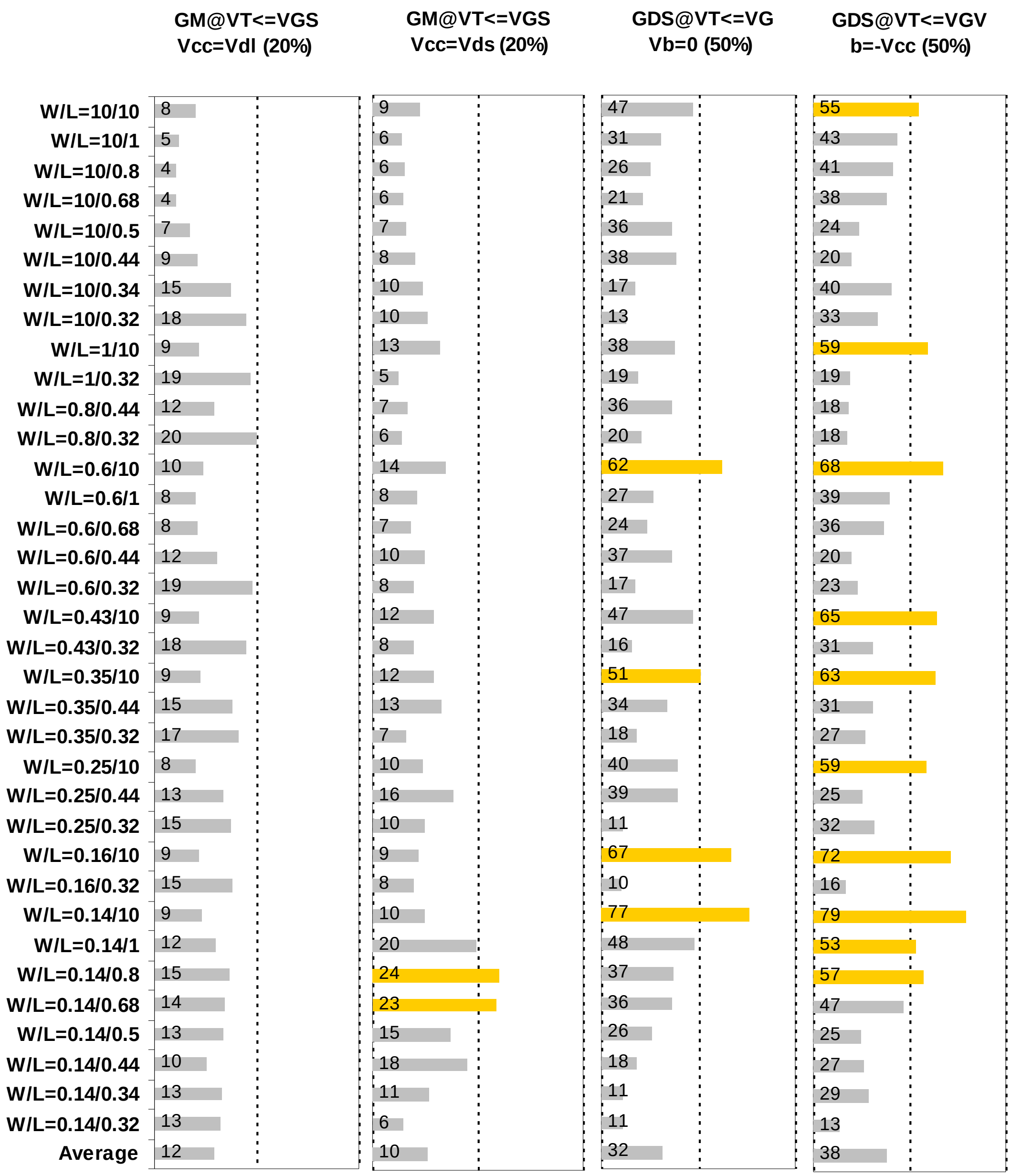

Figure 2-20 G_M and G_{DS} accuracy at 25 ℃ for NMOS

2.4 MOSFET capacitance model

2.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 2-6 Measurement structure for NMOS Cgg capacitance model

Structure	L_{DES}[μm]	W_{DES}[μm]
Gate capacitance components, C_{GDS}	2	9000

Table 2-7 Large area measurement structure for NMOS Cgc capacitance model

Structure	L_{DES}[μm]	W_{DES}[μm]
Gate capacitance components	0.5	100
	0.34	100
	2	100
	1	100
	2	180

2.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized as below.

Table 2-8 Measurement condition for NMOS capacitance model

V_{BIAS} [V]	From -3.65 to 3.65 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

2.4.3 T_{OX} extraction

The oxide thickness model parameter "TOX" was extracted to be 74.9 Å.

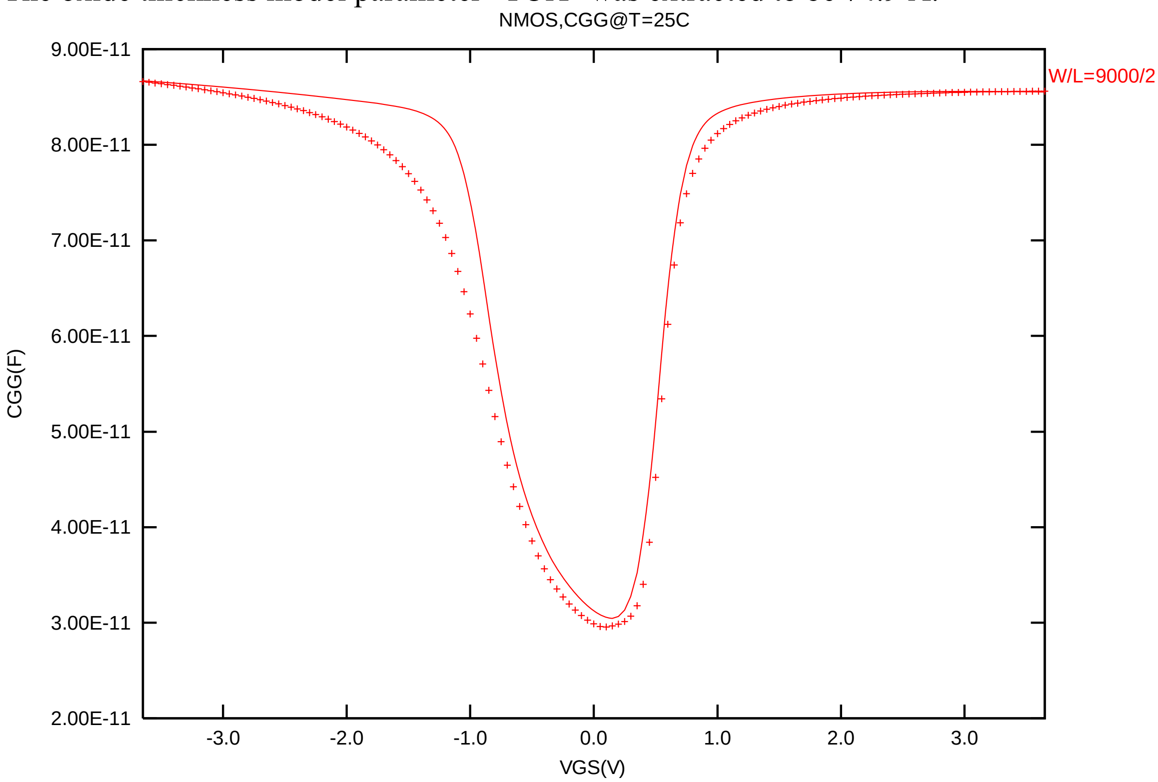

Figure 2-21 Measured and Simulated C_{GG} curves

2.4.4 C_{GC} vs. V_{GS} Plots

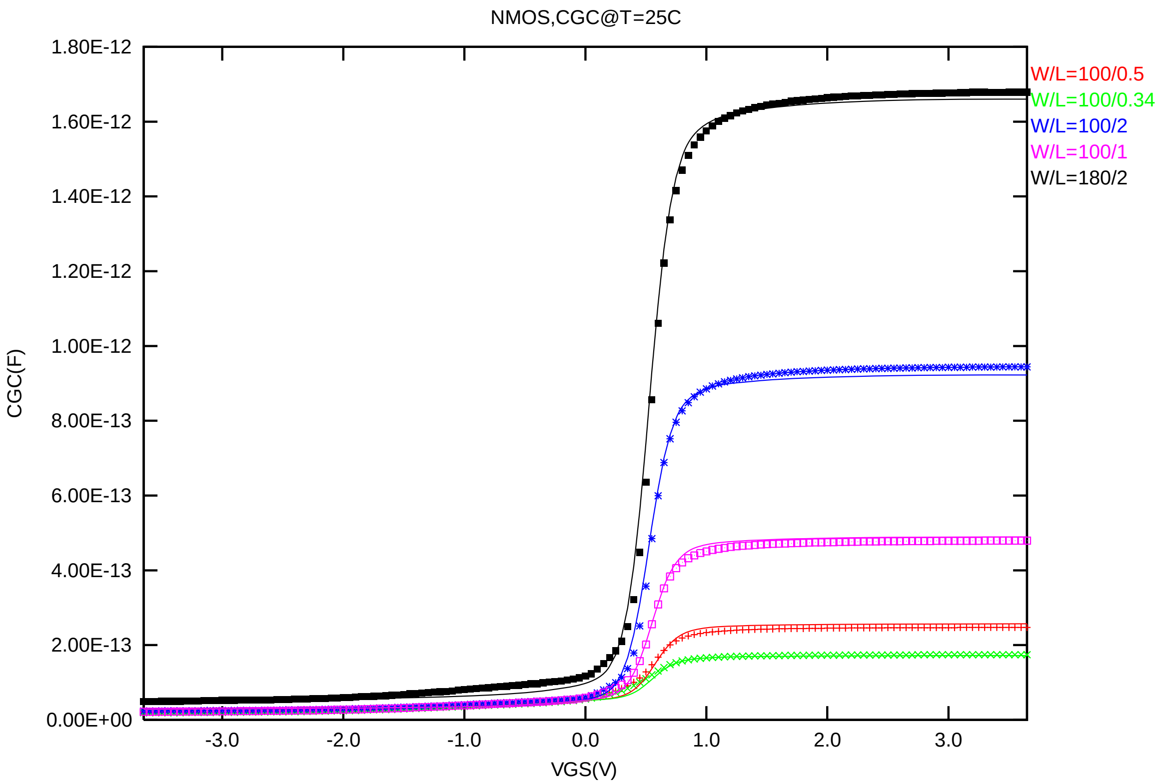

Figure 2-22 Measured and Simulated C_{GC} curves for large area measurement structure for NMOS

2.5 Parasitic source / drain junction capacitance model

2.5.1 Test devices

The following test structures were used for parameter extraction.

Table 2-9 Junction capacitor structures for NMOS

Structures	Area[m^2]	Perimeter[m]	Description
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.830E-02	Poly finger structure

2.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 2-10 Junction capacitor measurement conditions for NMOS

V_{BIAS} [V]	VH: pwell & VL: n+ From 0 to 3.6 V in 0.1 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

2.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 2-11 Extracted parameters from junction capacitor measurement for NMOS

Bottom Area Junction Capacitance Model			STI Bounded Sidewall Junction Capacitance Model			Poly Gate Sidewall Junction Capacitance Model		
Parameter	**Unit**	**Value**	**Parameter**	**Unit**	**Value**	**Parameter**	**Unit**	**Value**
CJ	F/m^2	7.80E-04	CJSW	F/m	1.08E-10	CJSWG	F/m	2.30E-10
PB	V	6.08E-01	PBSW	V	8.00E-01	PBSWG	V	9.05E-01
MJ	--	2.78E-01	MJSW	--	1.20E-01	MJSWG	--	6.24E-01
TCJ	1/K	9.00E-04	TCJSW	1/K	4.00E-04	TCJSWG	1/K	0
TPB	V/K	1.49E-03	TPBSW	V/K	6.00E-04	TPBSWG	V/K	0

CJ, CJSW vs. VBIAS Plots

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

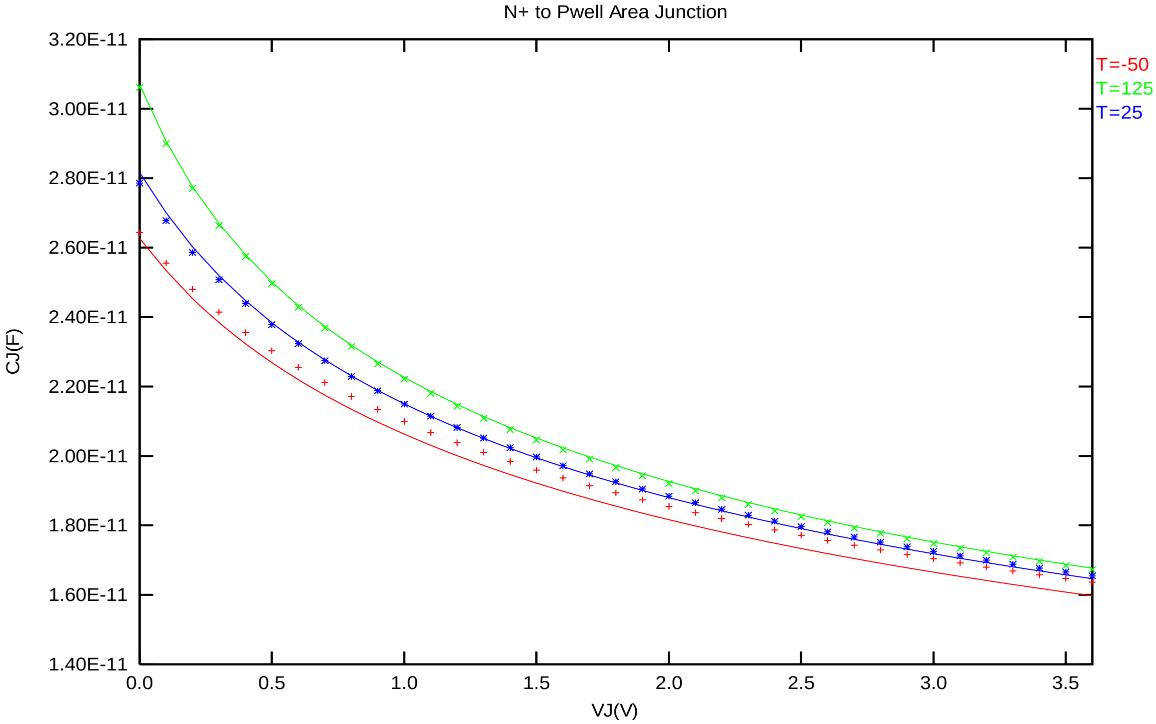

Figure 2-23 Bottom area junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for NMOS

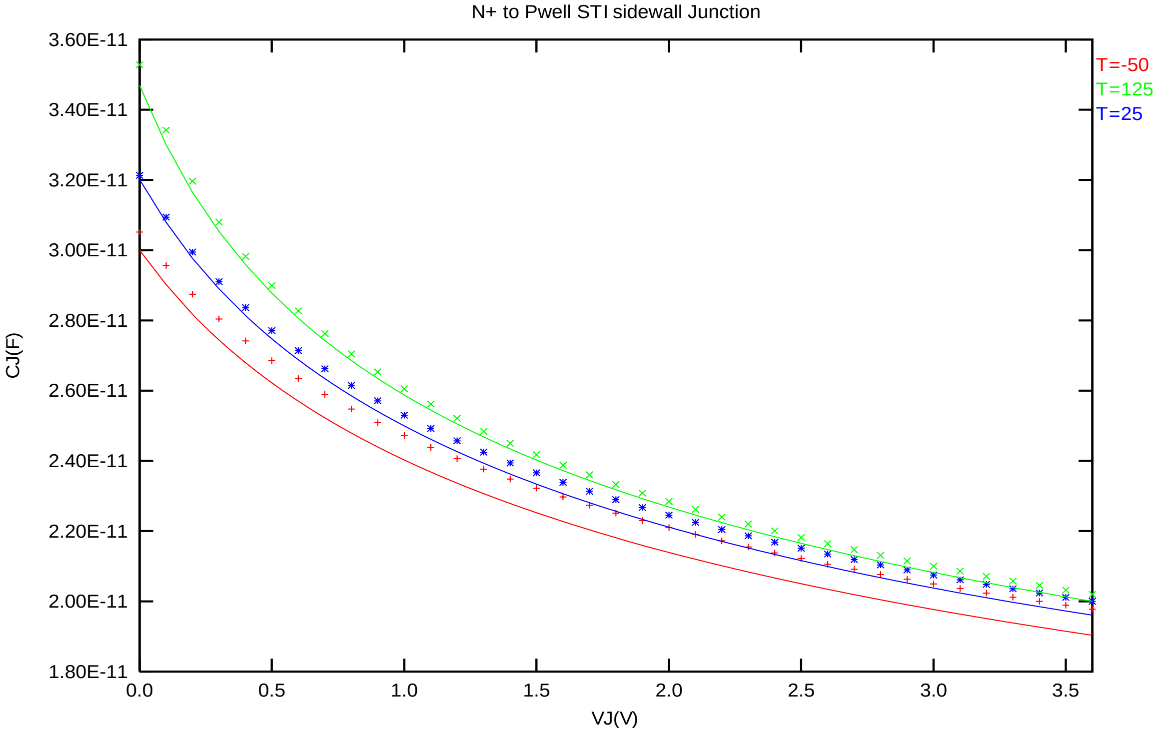

Figure 2-24 STI bounded sidewall junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for NMOS3.3 V p-ch MOSFET

3 3.3 V p-ch MOSFET

3.1 Silicon Wafer for Modeling

The information about the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for PMOS

	MOSFET DC model	MOSFET AC Model	Junction DC Model	Junction AC Model
Mask ID	QGSML	QGSML	QGSML	QGSML
Lot ID	D1A0BFF3	D1A0BFF3	D1A0BFF3	D1A0BFF3
Wafer Number	12	12	12	12

3.2 Electrical parameters

The important electrical parameters for the characterization of this device are summarized in the table below.

Linear and saturation threshold voltage is extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for NMOS and I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for PMOS, respectively. The dimension is after 90% shrink and 12nm sizing channel length back.
All the normalized current are divided with W_{DES}.

All parameters are given for T=25 ℃,V_{BS}=0 V unless specified otherwise.

Table 3-2 Electrical Parameters for PMOS

All parameters are given for T=25 ^oC ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Target	Measured Value	Extracted Value	Centered Value
Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = -0.1 V	V	-0.624	-0.622	-0.624	-0.624
I_{ON}	V_{DS} = -3.3 V	μA/ μm	-8.291	-8.37	-8.32	-8.32
Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.32 μ m)						
V_{TLIN}	V_{DS} = -0.1 V	V	-0.556	-0.553	-0.553	-0.570
V_{TSAT}	V_{DS} = -3.3 V	V	-0.510	-0.504	-0.493	-0.510
Body Effect V_T shift	V_{DS} = -3.3 V V_{BS}= 0 ~ --3.3 V	V	--	-0.561	-0.585	-0.585
DIBL V_T shift	V_{DS}= -0.1~-3.3 V	V	-0.046	-0.049	-0.060	-0.060
I_{ON}	V_{DS}=V_{GS}= -3.3 V	μA/μm	-285	-288.48	-287.27	-285.0
I_{OFF}	V_{DS}= -3.3 V, V_{GS}=0 V	A/μm	-3E-12	-1.2E-12	-1.0E-12	-6.8E-13
Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)						
V_{TLIN}	V_{DS} = -0.1 V	V	-0.639	-0.637	-0.641	-0.641
V_{TSAT}	V_{DS}= -3.3 V	V	--	-0.633	-0.636	-0.636
I_{ON}	V_{DS}=V_{GS}= -3.3 V	μA/μm	-9.293	-9.48	-9.37	-9.36
Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μm / 0.32 μm)						
V_{TLIN}	V_{DS} = -0.1 V	V	-0.588	-0.578	-0.583	-0.596
V_{TSAT}	V_{DS}= -3.3 V	V	-0.539	-0.530	-0.541	-0.554
I_{ON}	V_{DS}=V_{GS}= -3.3 V	μA/μm	-307.14	-309.57	-307.51	-304.96
I_{OFF}	V_{DS}= -3.3 V, V_{GS}=0 V	A/μm	-3.34E-12	-3.5E-12	-5.0E-12	-4.9E-12
Capacitance						
CJ	V_J = 0 V	F/m^2	9.95E-04	1.00E-03	9.93E-04	9.93E-04
CJSW	V_J = 0 V	F/m	1.25E-10	1.17E-10	1.20E-10	1.20E-10
CJSWG	V_J = 0 V	F/m	3.62E-10	3.263E-10	3.59E-10	3.59E-10

3.3 MOSFET DC model

3.3.1 Test devices

The geometry of devices that were used for MOSFET DC model parameter extraction is summarized below(The dimensions are after 90% shrink and 12nm sizing channel length back.).

Table 3-3 Geometry of PMOS devices measured for parameter extraction

W\L	10	1	0.8	0.68	0.5	0.44	0.34	0.32
10	X	X	X	X	X	X	X	X
1	X							X
0.8						X		X
0.6	X	X		X		X		X
0.43	X							X
0.35	X					X		X
0.25	X					X		X
0.16	X							X
0.14	X	X	X	X	X	X	X	X

3.3.2 Measurement conditions

The drain current was measured at various bias conditions summarized in the table below:

Table 3-4 Test conditions for IV curves of PMOS

I_D vs. V_{GS}

	Start	Stop	Step
V_{DS} (V)	-0.1	-3.3	-3.2
V_{GS} (V)	0.5	-3.6	-0.05
V_{BS} (V)	0	3.3	0.825

I_D vs. V_{DS}

	Start	Stop	Step
V_{DS} (V)	0	-3.6	-0.05
V_{GS} (V)	-0.7	-3.3	-0.65
V_{BS} (V)	0	3.3	3.3

3.3.3 Model verification plots

The simulation results (solid lines) obtained from the extracted model were plotted together with the measurement results (dotted lines) for comparison. The L and W in the plots refer to (L_{DES}*0.9+12nm and W_{DES}*0.9) respectively.

I_D *vs.* V_{DS} *Plots*

The following plots show measured and simulated output I-V curves for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

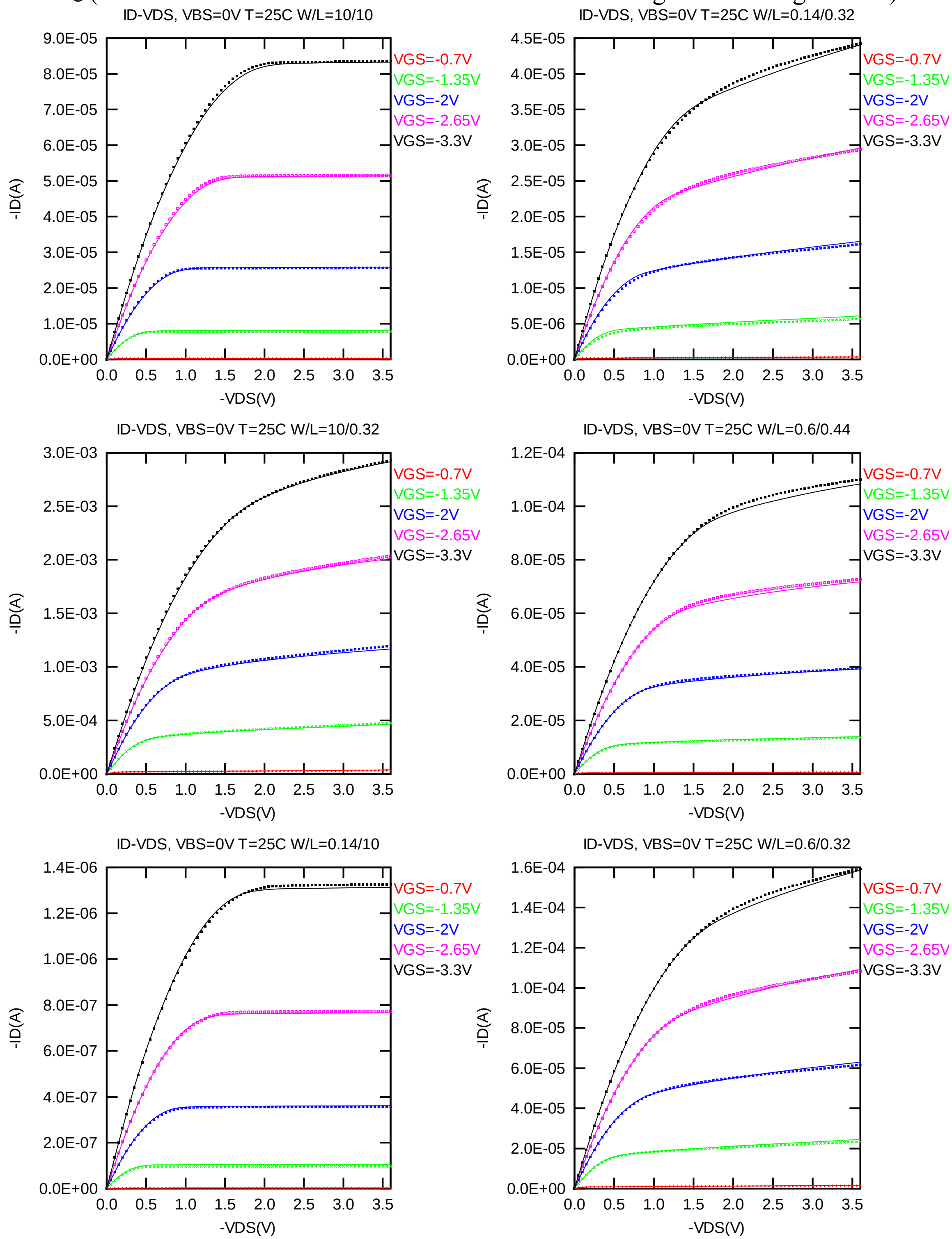

Figure 3-1 I_D vs. V_{DS} at V_{BS} = 0 V for PMOS

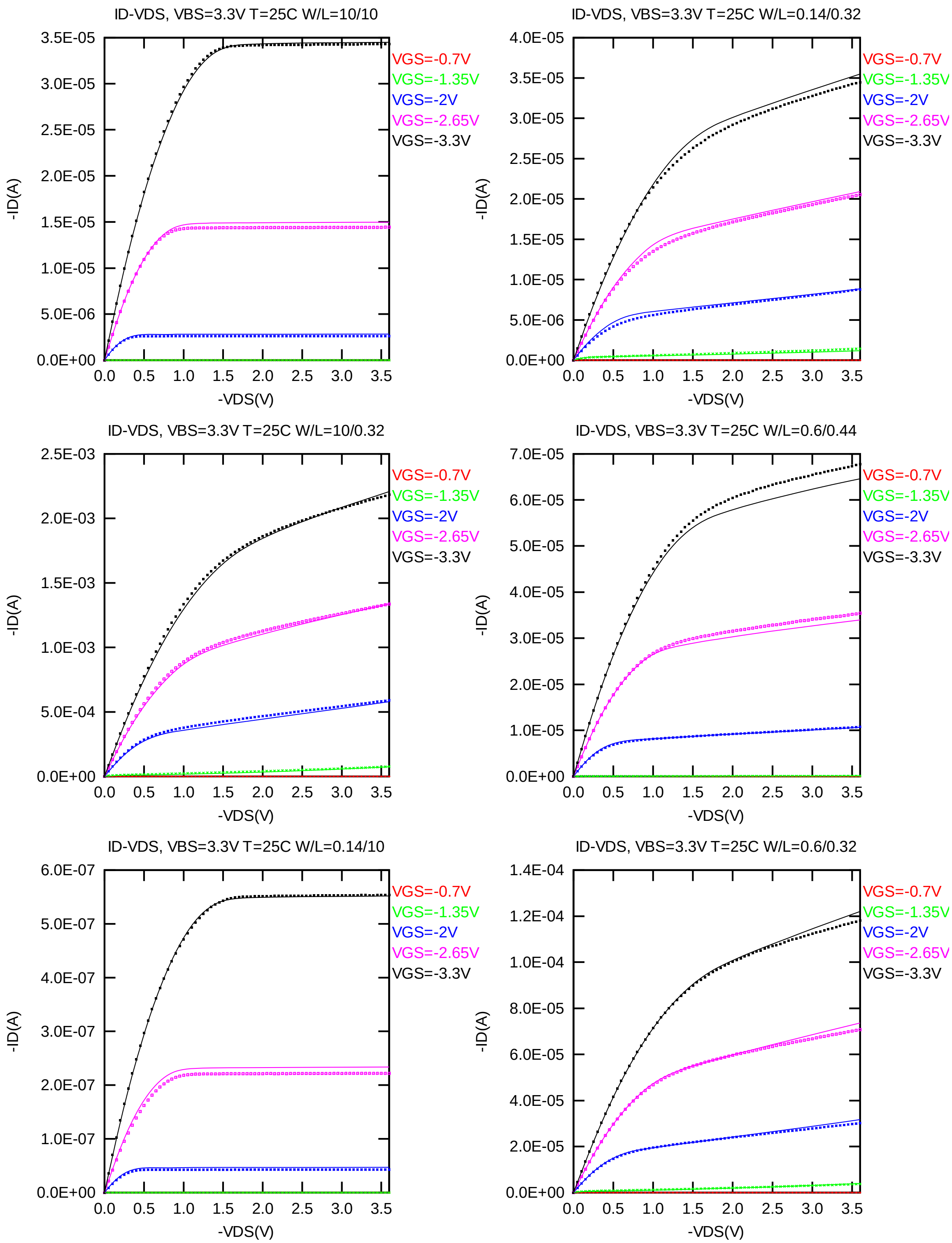

Figure 3-2 I_D vs. V_{DS} at V_{BS}=3.3 V for PMOS

G_{DS} vs. V_{DS} Plots

The following plots show measured and simulated output conductance curves for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

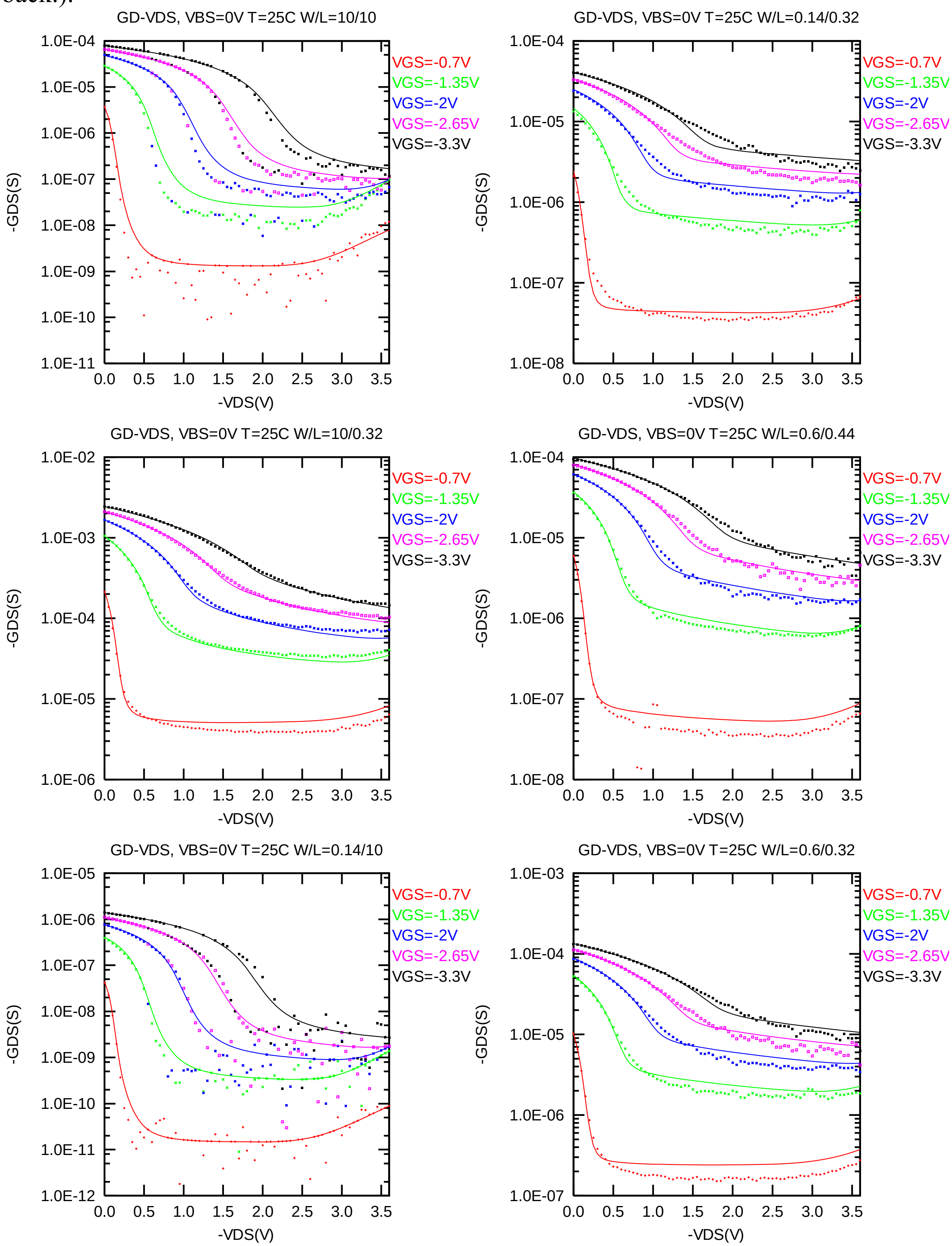

Figure 3-3 G_{DS} vs. V_{DS} at V_{BS} = 0 V for PMOS

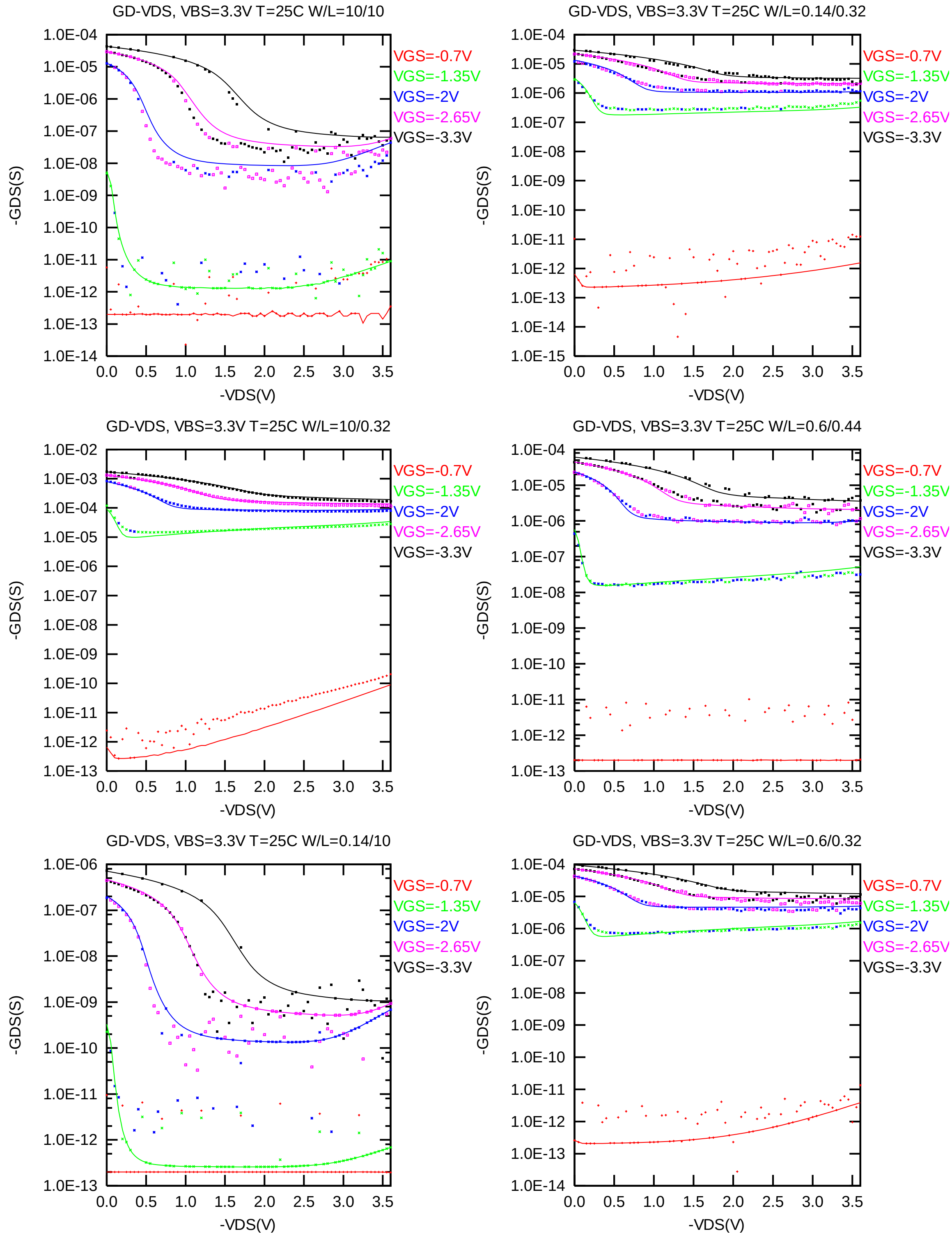

Figure 3-4 G_{DS} vs. V_{DS} at V_{BS} = 3.3V for PMOS

I_D vs. V_{GS} Plots (linear scale)

The following plots show measured and simulated transfer curves in linear scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

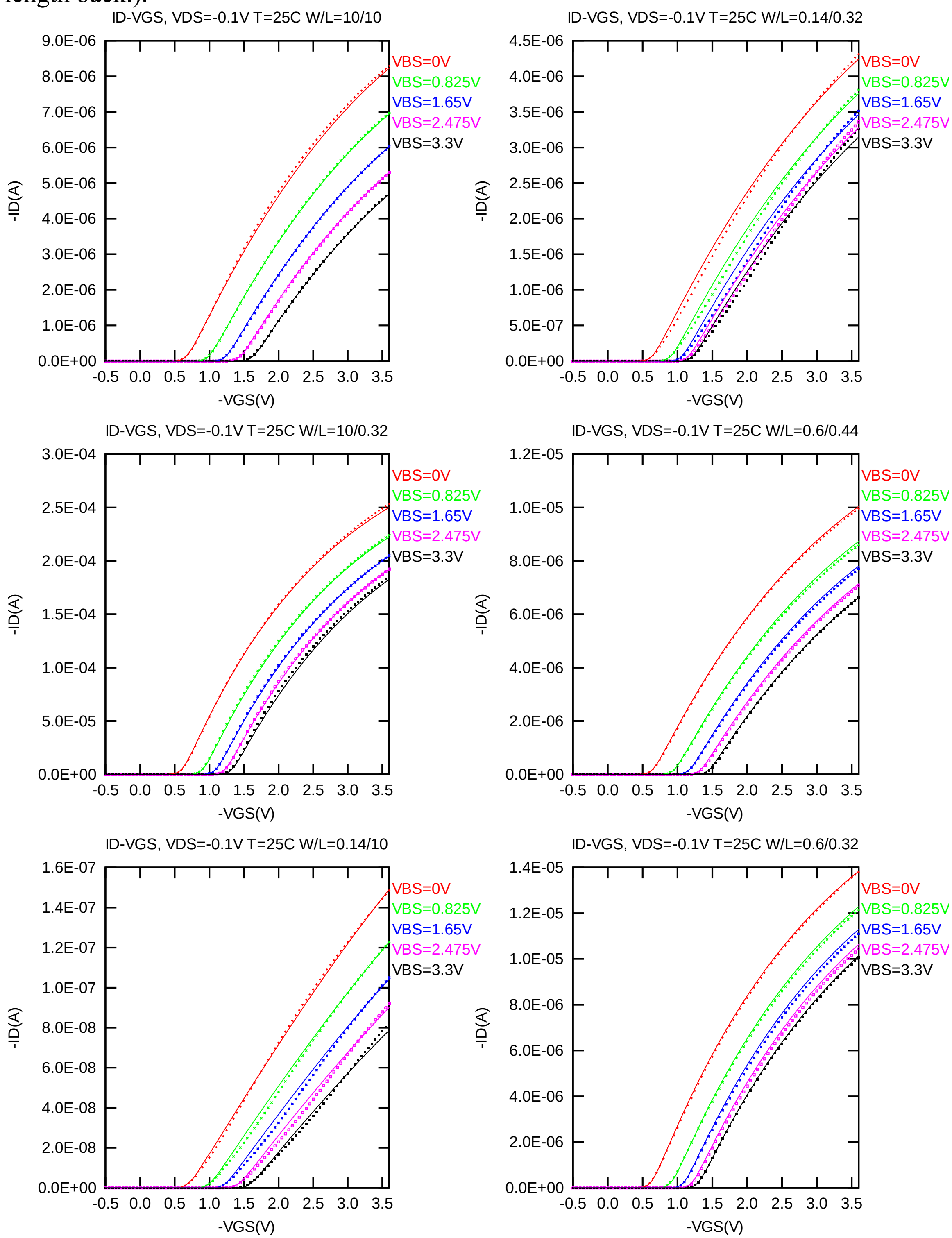

Figure 3-5 I_D vs. V_{GS} at V_{DS} = -0.1 V for PMOS

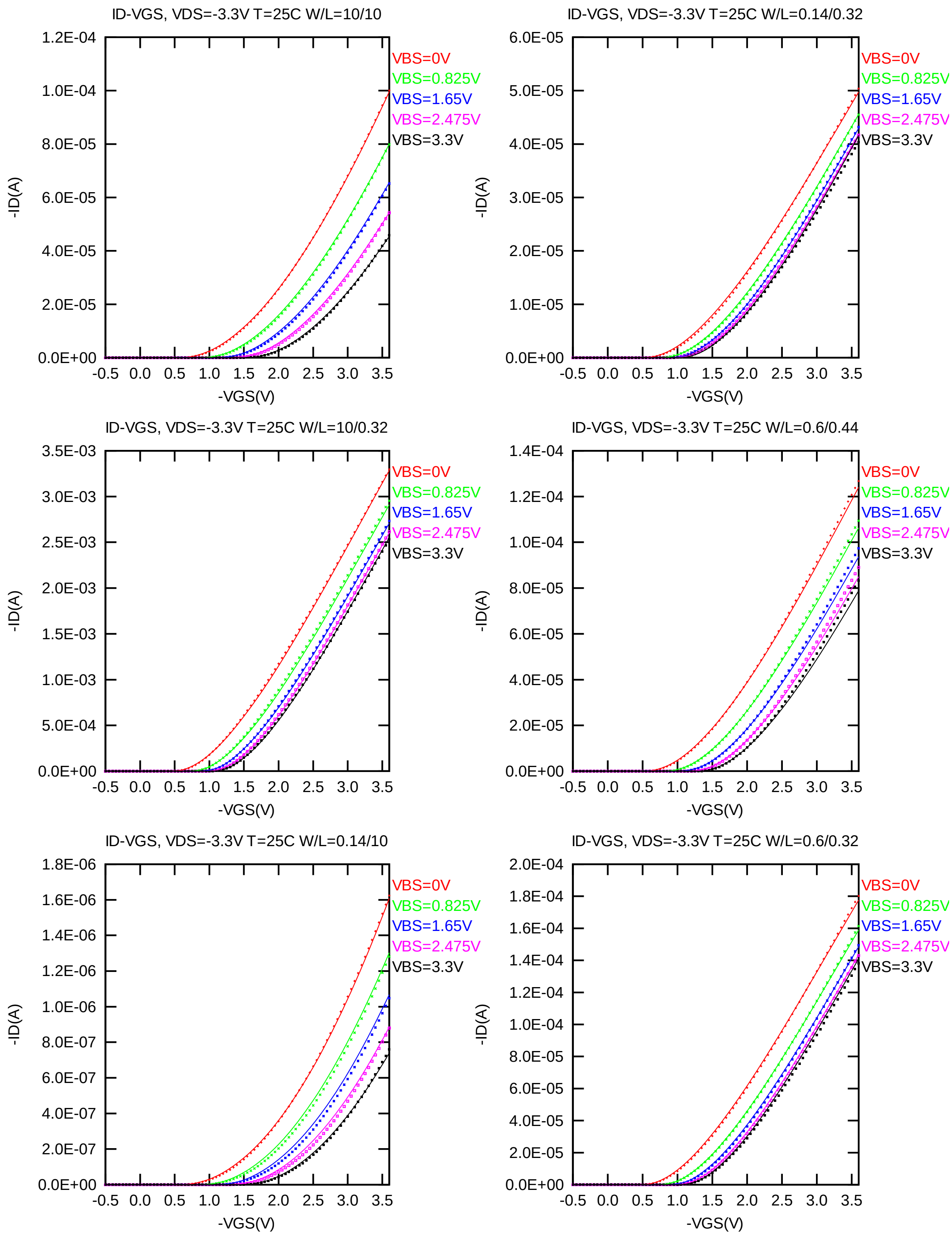

Figure 3-6 I_D vs. V_{GS} at V_{DS} = -3.3 V for PMOS

I_D vs. V_{GS} Plots (logarithmic scale)

The following plots show measured and simulated transfer curves in logarithmic scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

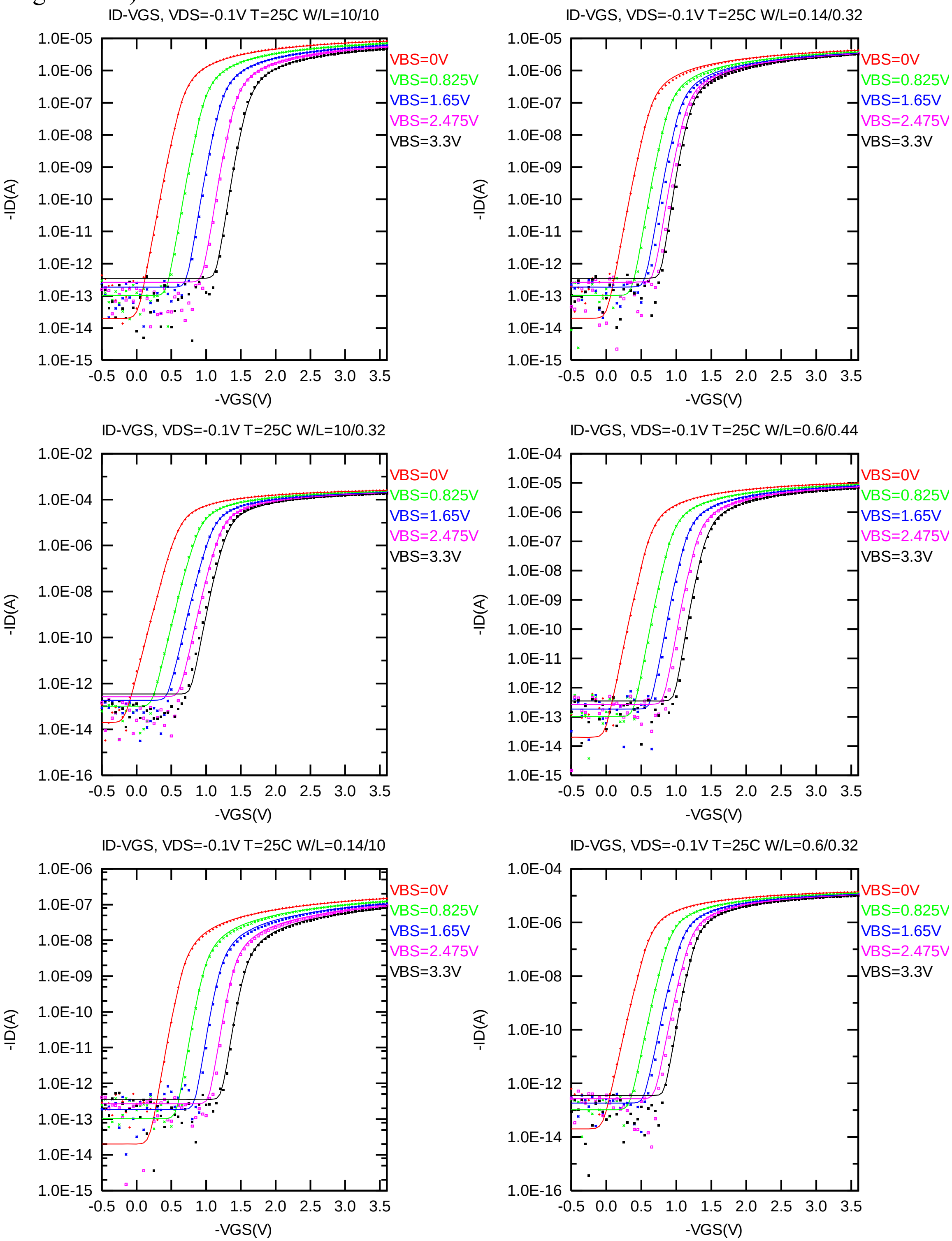

Figure 3-7 Log scaled I_D vs. V_{GS} at V_{DS} = -0.1 V for PMOS

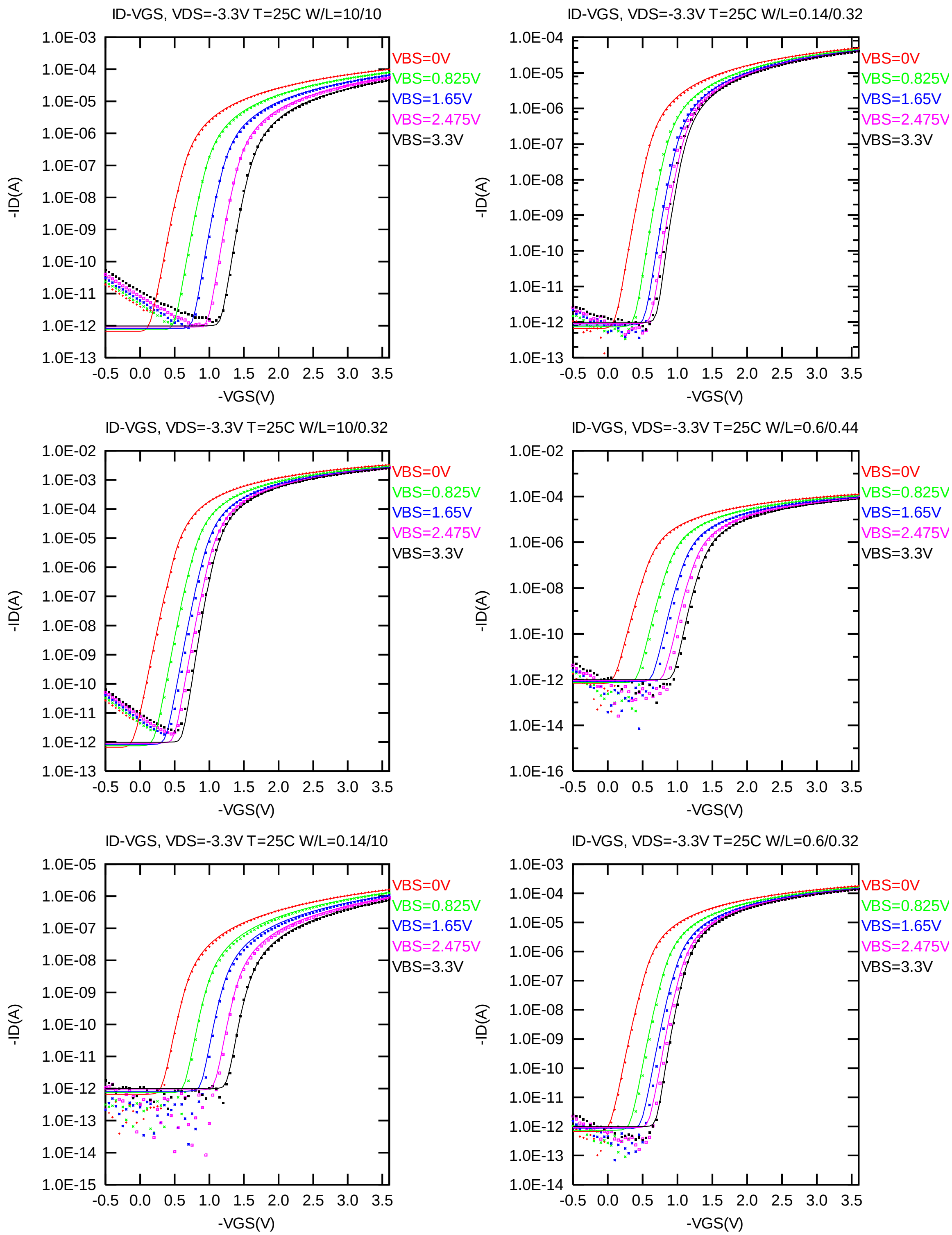

Figure 3-8 Log scaled I_D vs. V_{GS} at V_{DS} = -3.3 V for PMOS

G_M vs. V_{GS} Plots

The following plots show the measured and simulated transfer conductance curves in linear scale for selected devices at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

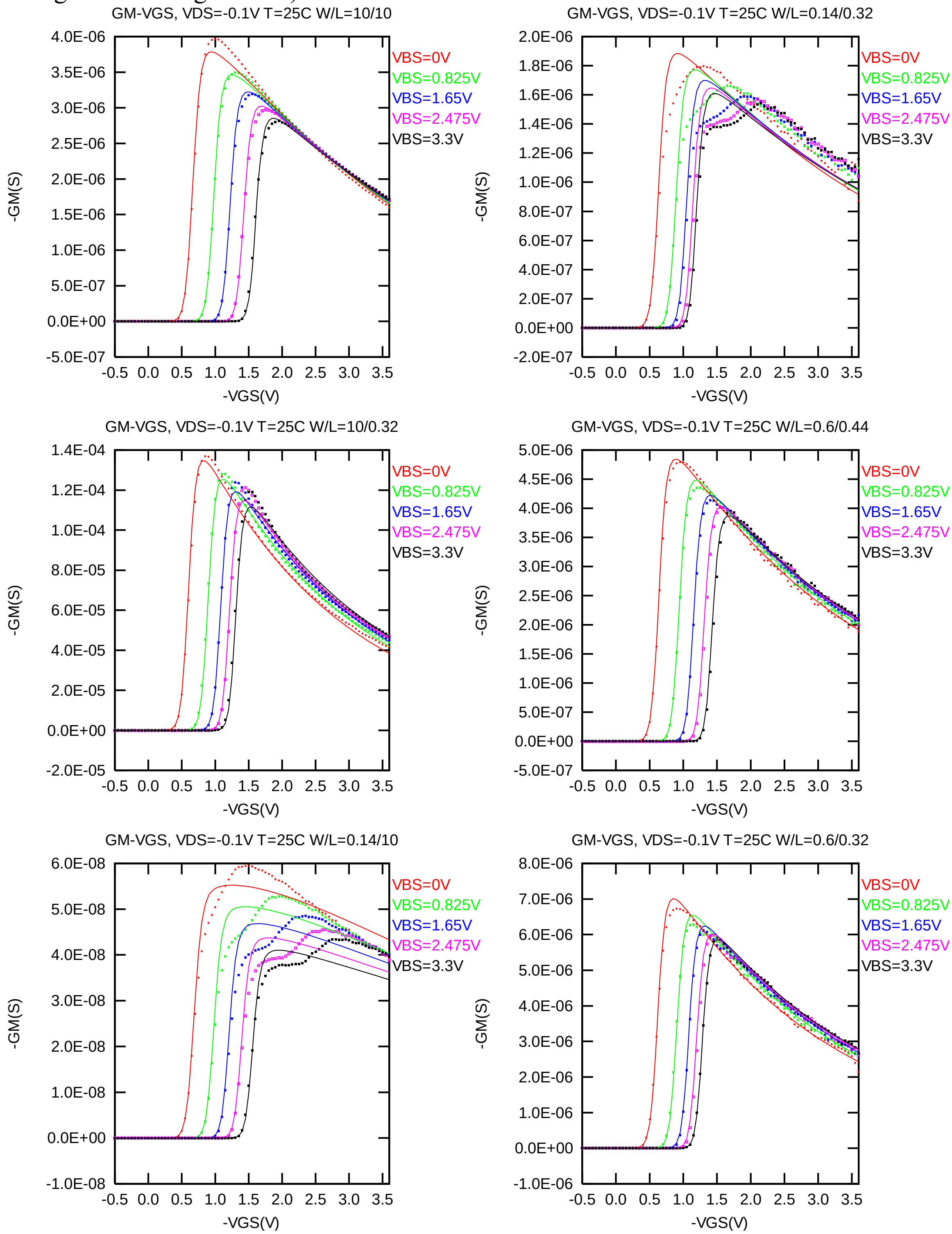

Figure 3-9 G_M vs. V_{GS} at V_{DS} = -0.1 V for PMOS

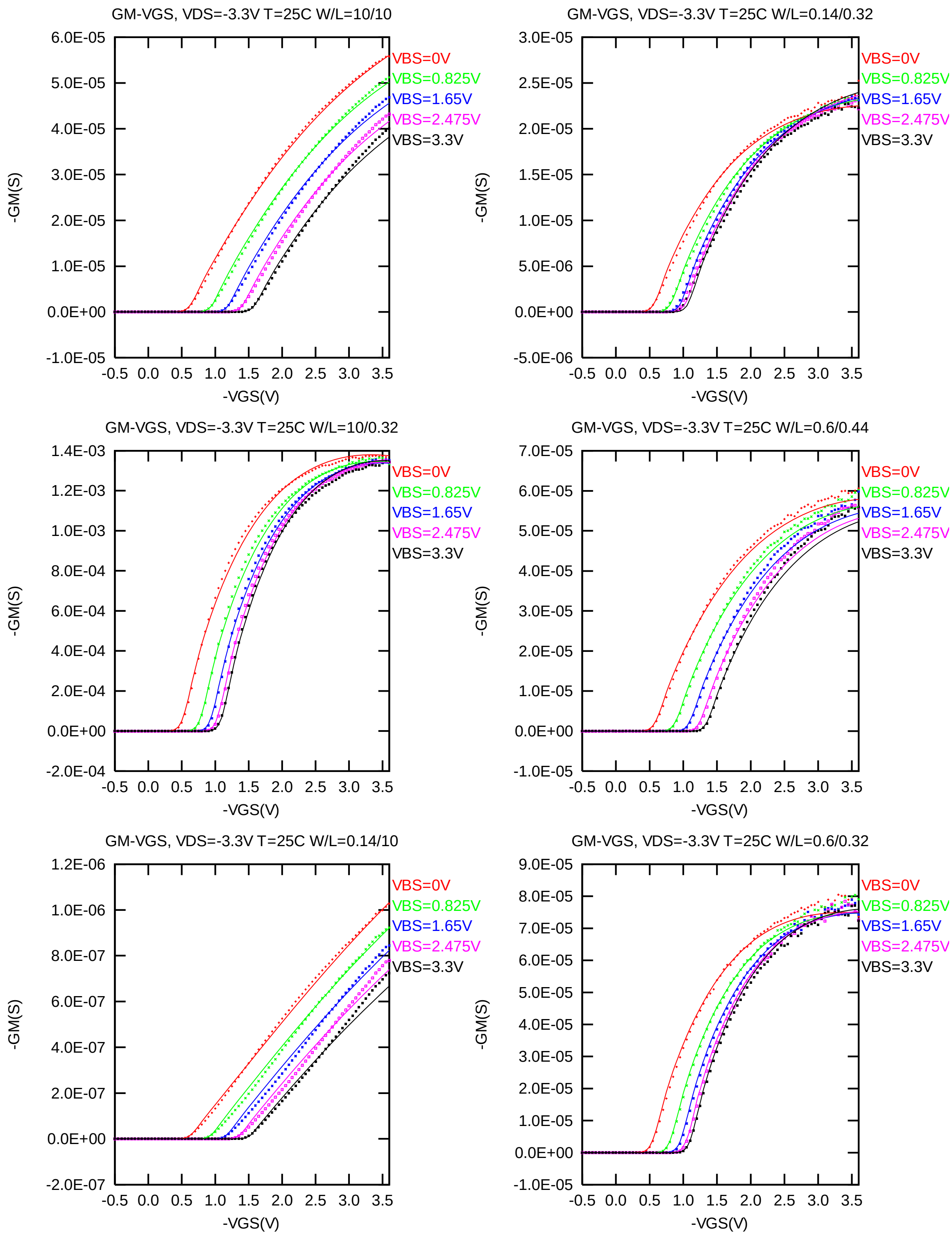

Figure 3-10 G_M vs. V_{GS} at V_{DS} = - 3.3 V for PMOS

V_{TLIN} & V_{TSAT} vs. L_{DES} & W_{DES}

The following figures show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

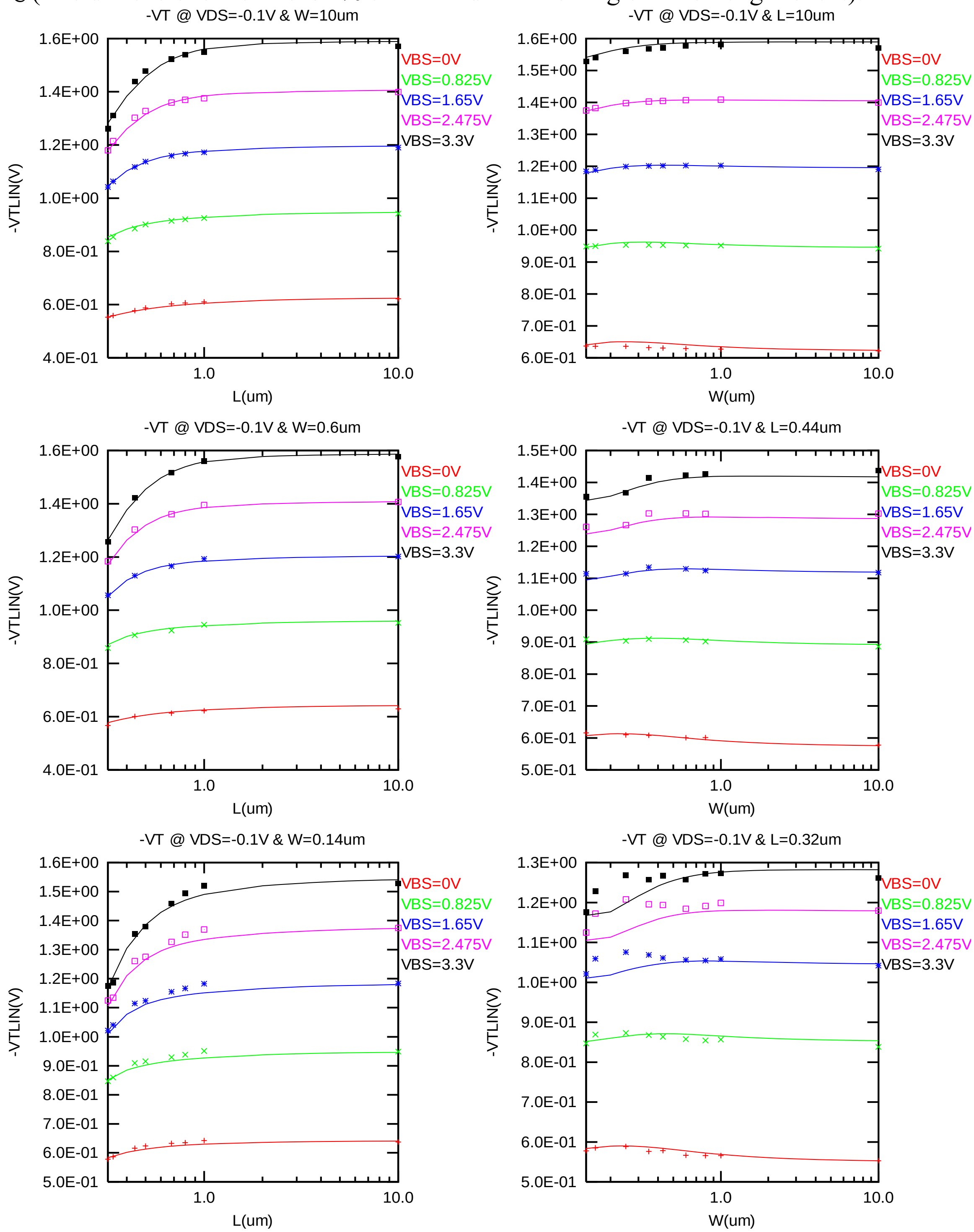

Figure 3-11 V_{TLIN} vs. channel length & channel width for low drain bias (V_{DS} = -0.1 V) for PMOS

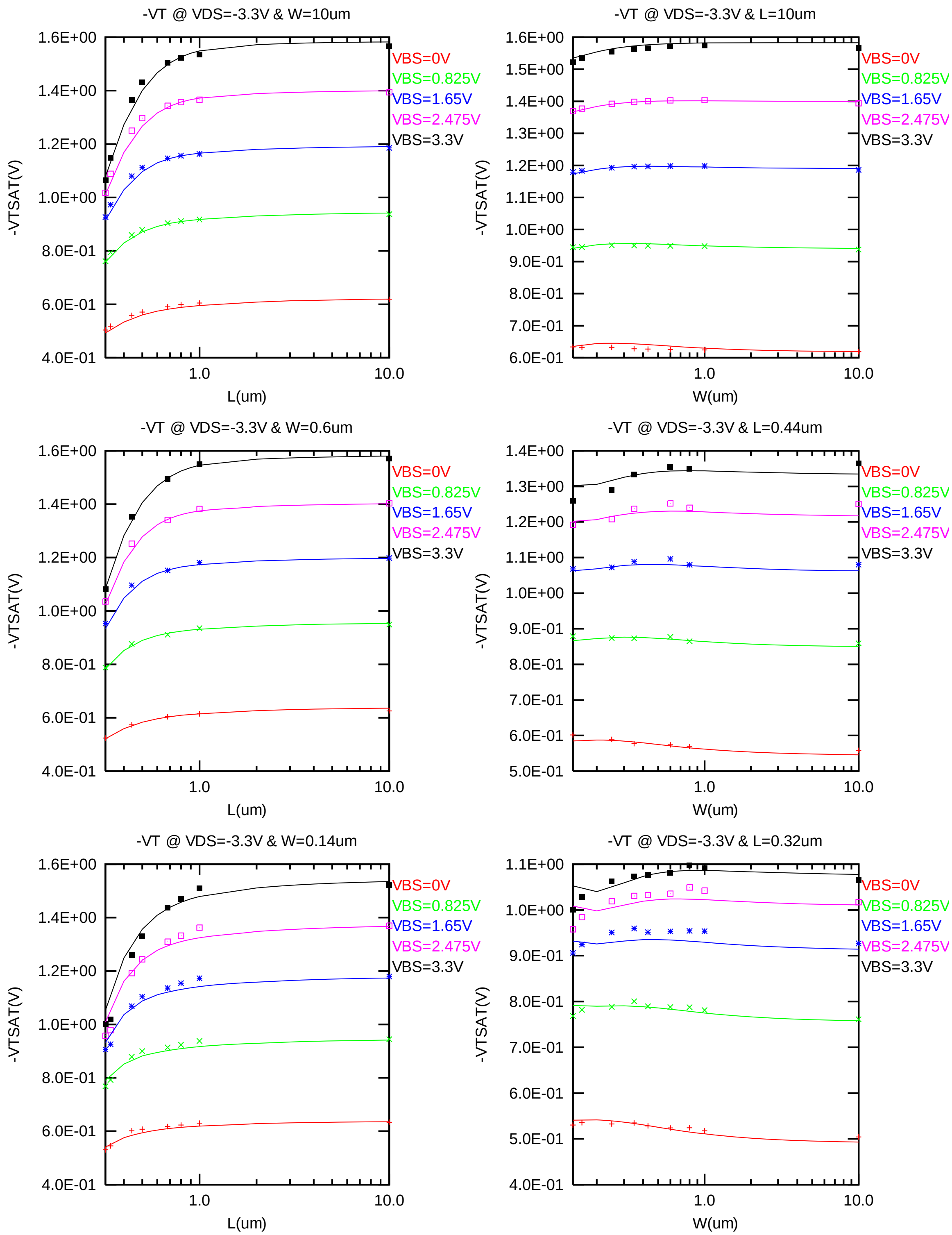

Figure 3-12 V_{TSAT} vs. channel length & channel width for high drain bias (V_{DS} = -3.3 V) for PMOS

I_{ON} vs. L_{DES} & I_{ON} vs. W_{DES}

The following plots show the channel length and width dependence of I_{ON} extracted at $V_{DS} = V_{GS} = -V_{CC}$ for $V_{BS} = 0$ V and $V_{BS} = 3.3$V at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.).

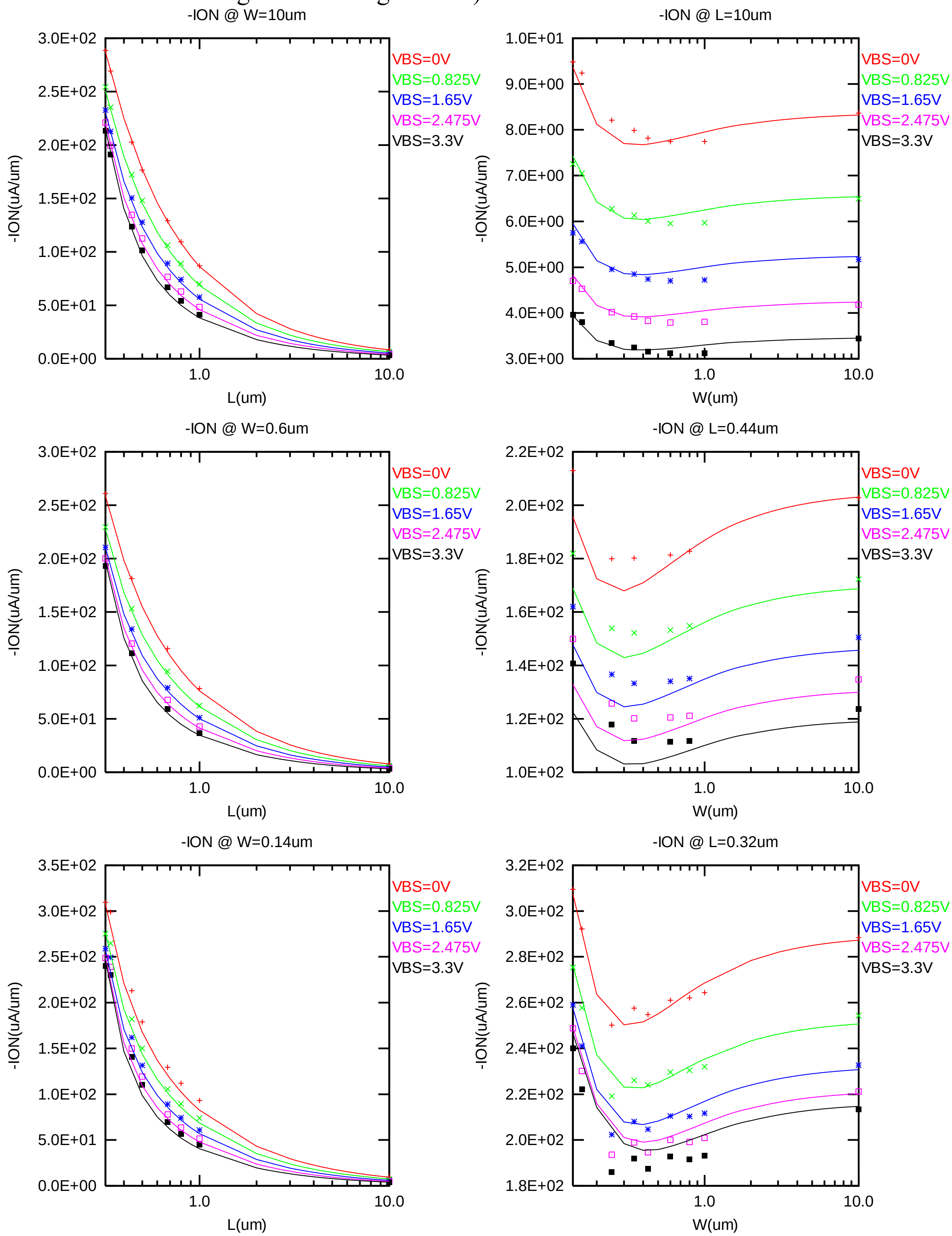

Figure 3-13 Saturation current I_{ON} vs. channel length and channel width for PMOS

I_{OFF} vs. L_{DES} & I_{OFF} vs. W_{DES}

The following plots show the channel length and width dependence of I_{OFF} extracted at V_{DS} = -V_{CC} and V_{GS} = 0V for V_{BS} = 0 V to 3.3V at 25 ℃(The dimensions are after 90% shrink and 12nm sizing channel length back.). A large discrepancy of I_{OFF} can be seen in the graphs, which is due to the presence of junction leakage current.

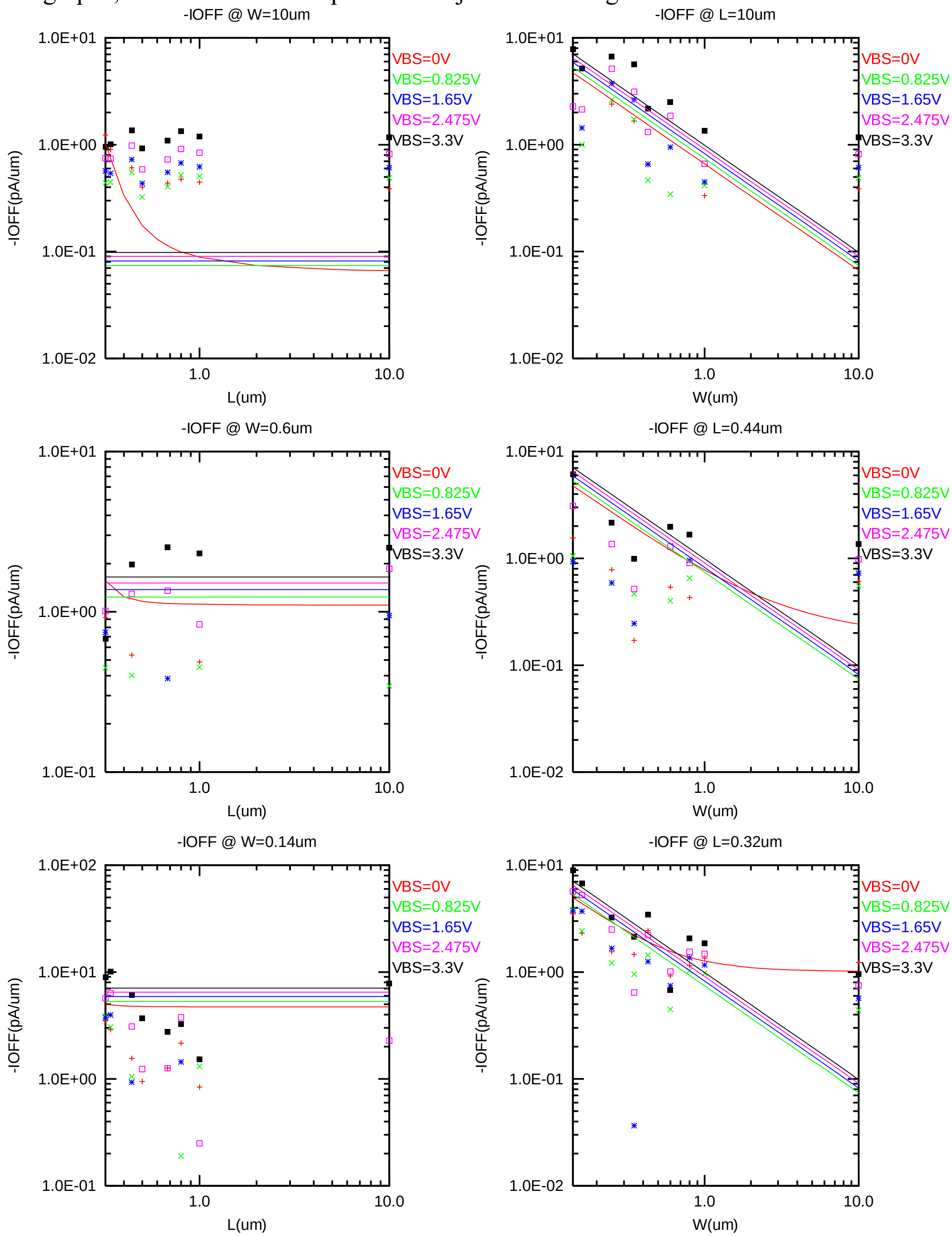

Figure 3-14 I_{OFF} vs. channel length and channel width for PMOS

V_{TLIN} & V_{TSAT} vs. Temperature

The following plots give the temperature dependence of the linear and saturation threshold voltage extracted at I_D = -70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm)(The dimensions are after 90% shrink and 12nm sizing channel length back.).

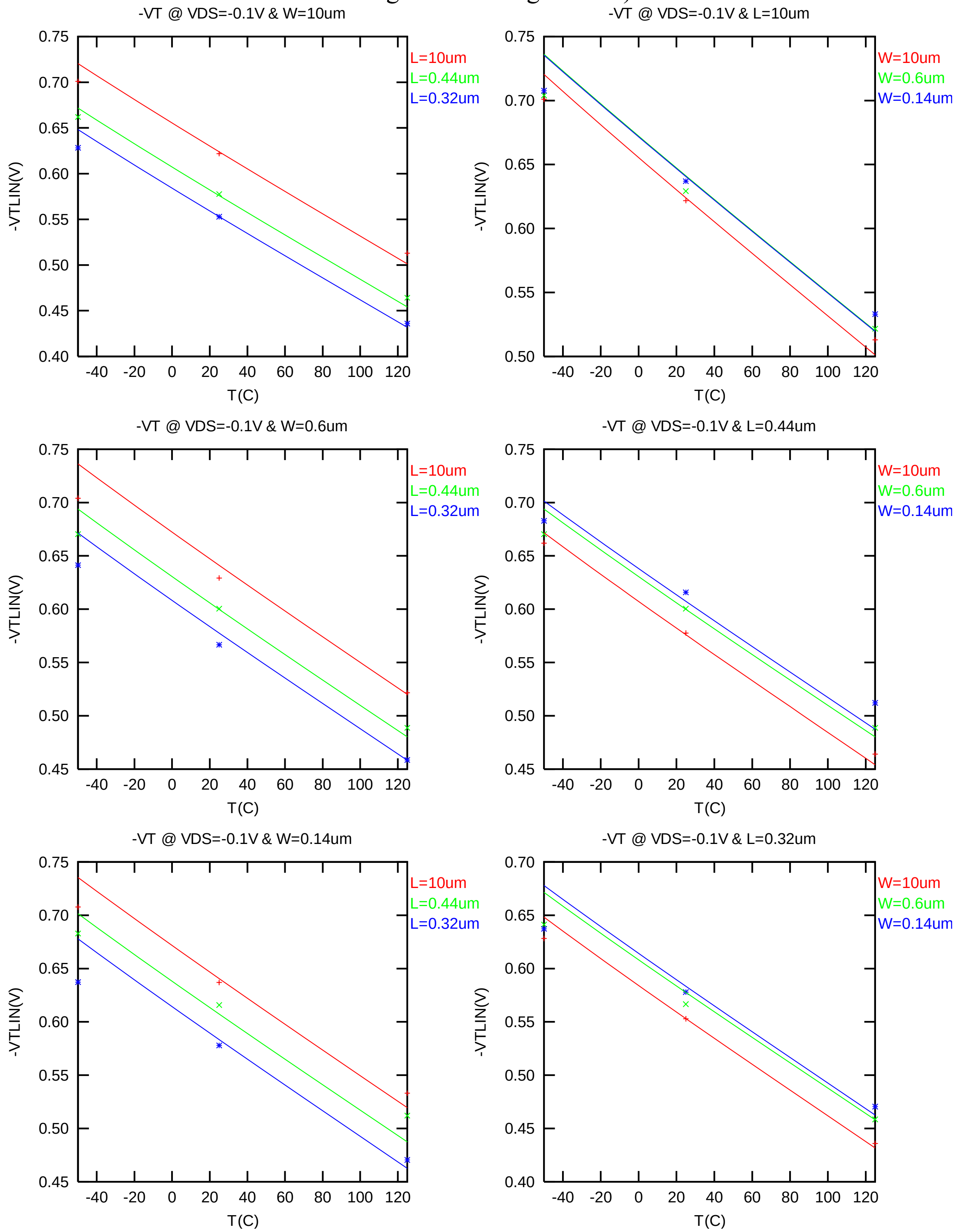

Figure 3-15 V_{TLIN} vs. temperature for PMOS

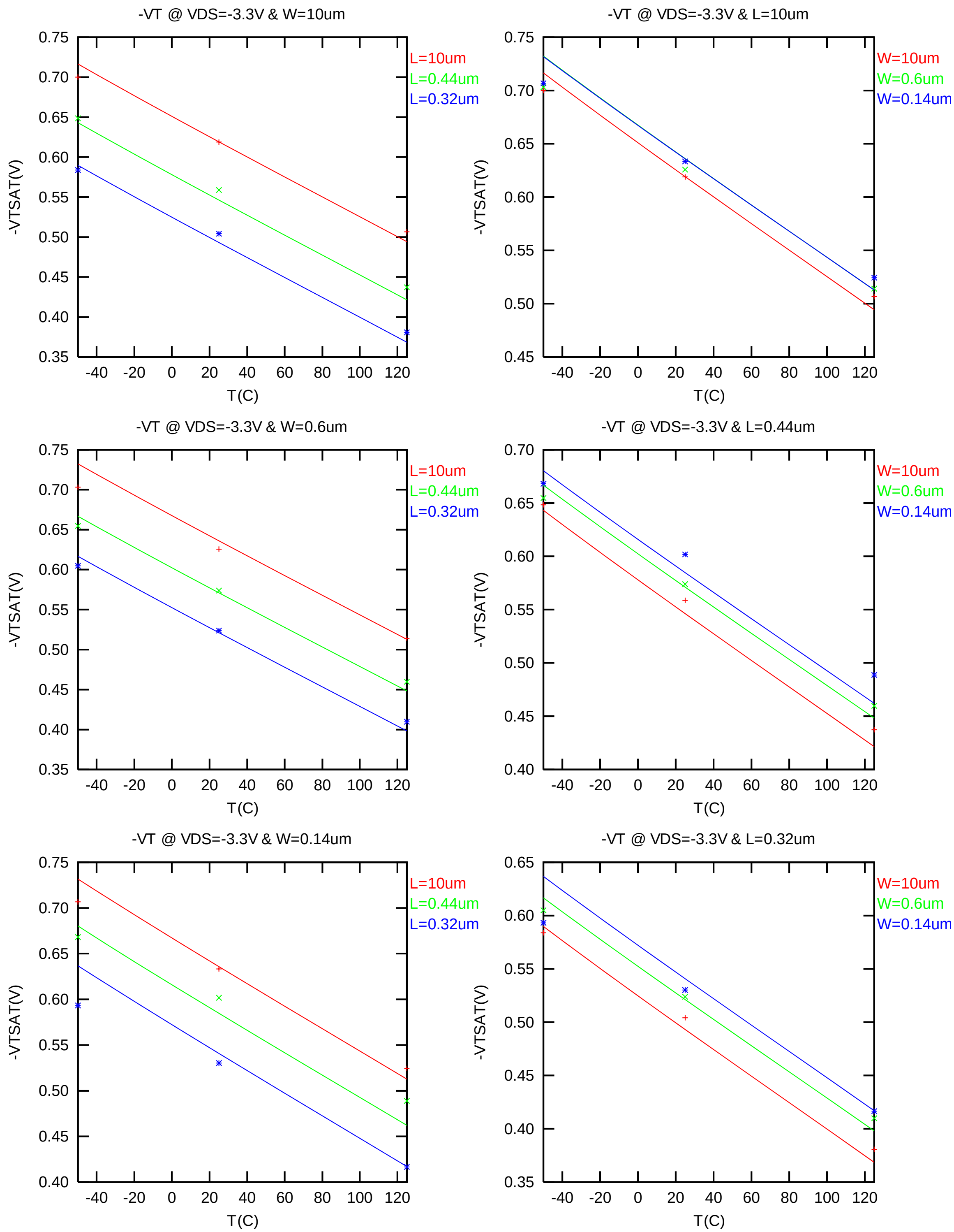

Figure 3-16 V_{TSAT} vs. temperature for PMOS

I_{ON} vs Temperature

The following plot shows the temperature dependence of I_{ON} extracted at $V_{GS} = V_{DS} = -3.3$ V, $V_{BS} = 0$ V(The dimensions are after 90% shrink and 12nm sizing channel length back.).

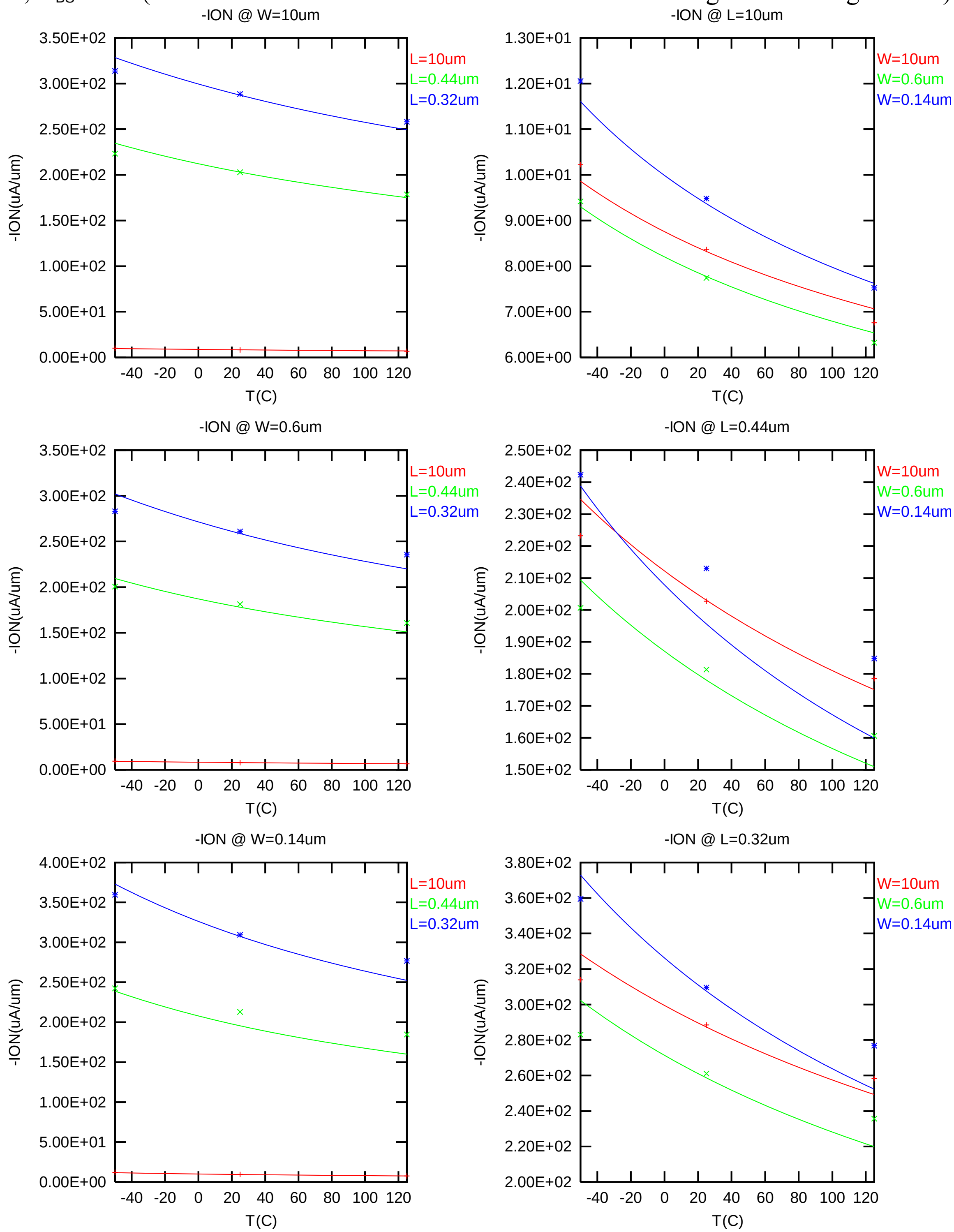

Figure 3-17 Saturation Current vs. temperature for PMOS

I_{OFF} vs Temperature

The following plot shows the temperature dependence of I_{OFF} extracted at V_{DS}= -3.3 V, V_{GS} = V_{BS} = 0 V(The dimensions are after 90% shrink and 12nm sizing channel length back.).

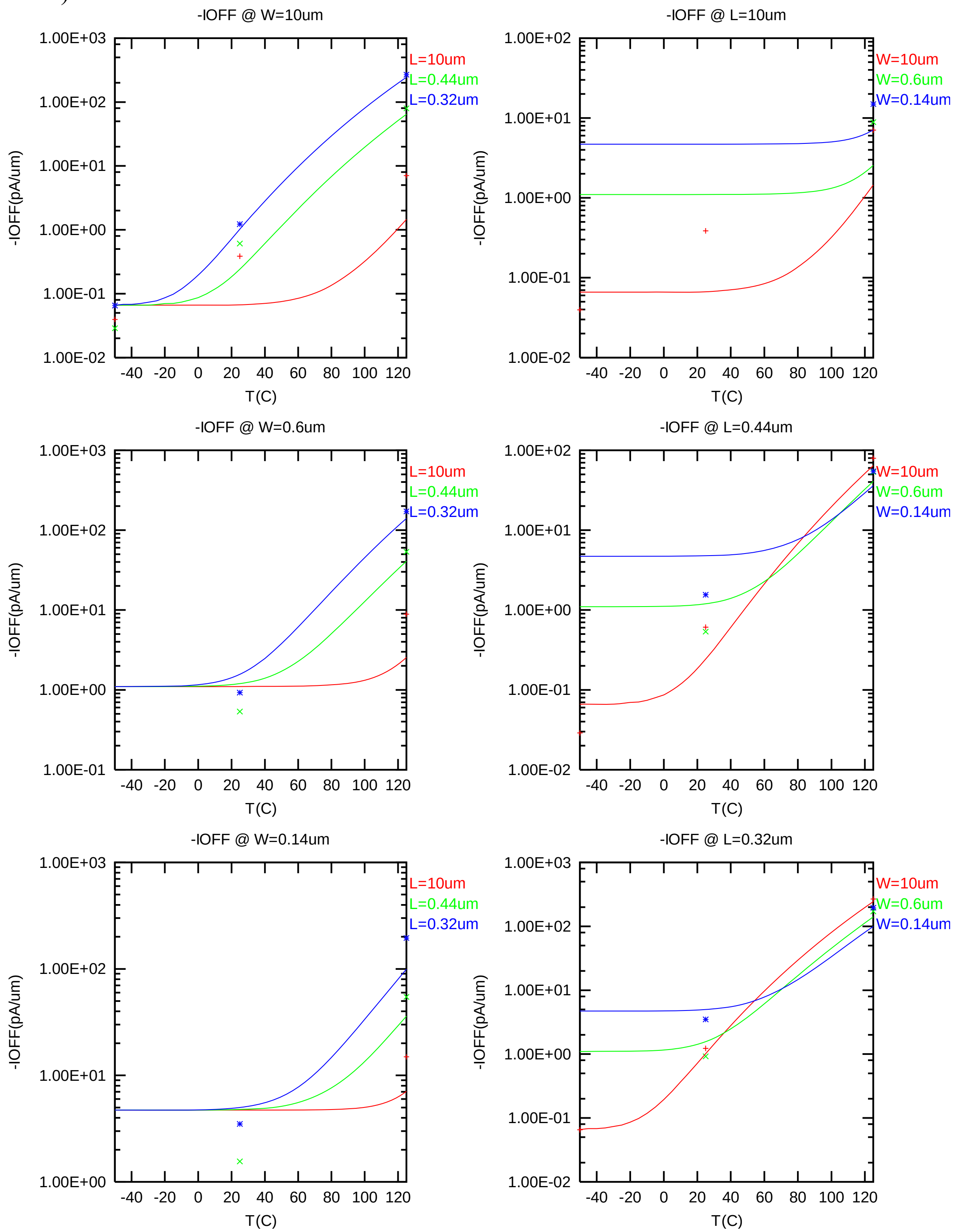

Figure 3-18 I_{OFF} vs. temperature for PMOS

3.3.4 Model fitting accuracy

The model fitting accuracy of parameters I_D, G_M and G_{DS} is quantified using the following equation:

error % = 100 * | measurement - simulation | / [(measurement + simulation) / 2]

Different error criteria are defined for different parameters (I_D, G_M and G_{DS}) as well as for different operation region. These criteria are explained below.

1) I_D accuracy criteria

f. I_D in sub-threshold region, i.e. $V_{GS} < V_T$
g. I_D in moderate inversion region, i.e. $V_T \leqq V_{GS} < V_T + V_X$
h. I_D in strong inversion region, i.e. $V_T + V_X \leqq V_{GS} \leqq V_{CC}$

, where $V_X = (V_{CC} - V_T) / 2$, $V_{CC} = -3.3$ V

2) G_M and G_{DS} accuracy criteria

i. G_M and G_{DS} in sub-threshold region, i.e. $V_{GS} < V_T$
j. G_M and G_{DS} in inversion region, i.e. $V_T \leqq V_{GS} \leqq V_{CC}$

Table 3-5 Requirement for model fit quality

Target of Quality Check	$V_{GS}< V_T$ (sub-threshold)	$V_T\leq V_{GS} < V_T+V_X$ (moderate inversion)	$V_T+V_X \leq V_{GS}\leq V_{CC}$ (strong inversion)
I_D	≤75%	≤25%	≤7%
G_M	≤50%	≤20%	≤20%
G_{DS}	≤100%	≤50%	≤50%

I_D Accuracy

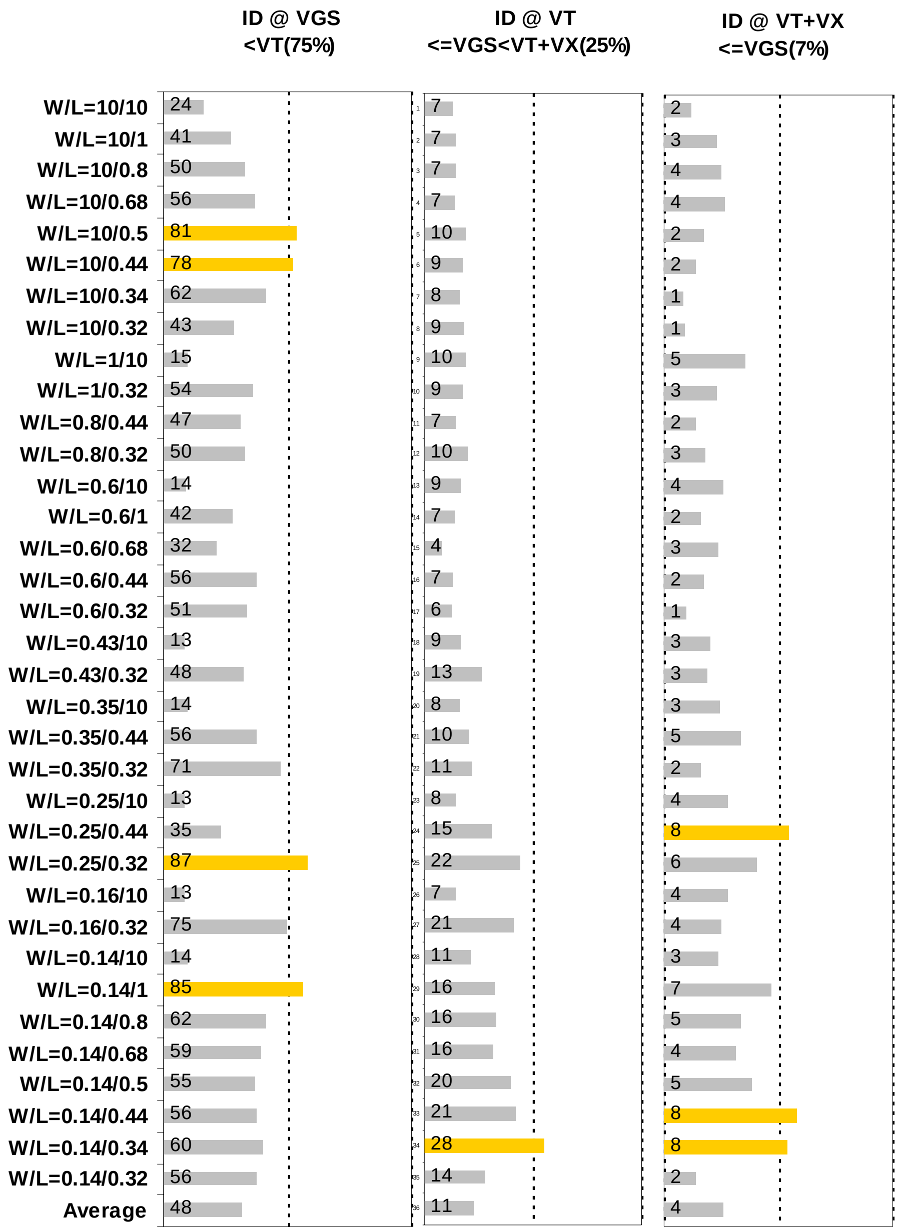

Figure 3-19 I_D accuracy at 25 ℃ for PMOS

G_M and G_{DS} Accuracy

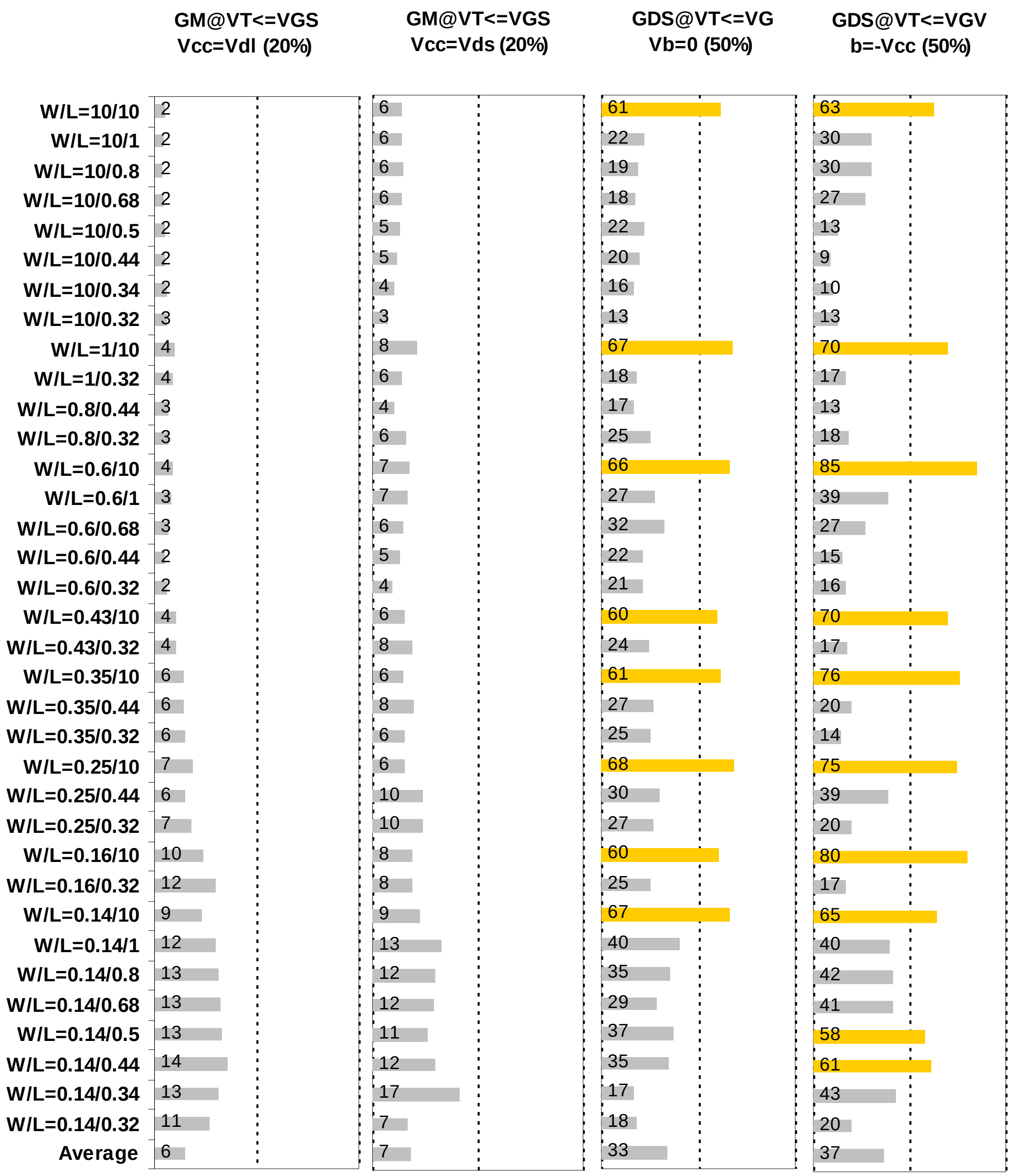

Figure 3-20 G_M and G_{DS} accuracy at 25 ℃ for PMOS

3.4 MOSFET capacitance model

3.4.1 Test devices

The following test structures were used for MOSFET capacitance model parameter extraction.

Table 3-6 Measurement structure for PMOS Cgg capacitance model

Structure	L_{DES}[μm]	W_{DES}[μm]
Gate capacitance components, C_{GDS}	2	9000

Table 3-7 Large area measurement structure for PMOS Cgc capacitance model

Structure	L_{DES}[μm]	W_{DES}[μm]
Gate capacitance components	0.5	100
	0.34	100
	2	100
	1	100
	2	180

3.4.2 Measurement conditions

The measurement conditions for MOSFET capacitance are summarized as below.

Table 3-8 Meaurement condition for PMOS capacitance model

V_{BIAS} [V]	From -3.65 to 3.65 V in 0.05 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	25

3.4.3 T_{OX} extraction

The oxide thickness model parameter "TOX" was extracted to be 76.8 Å.

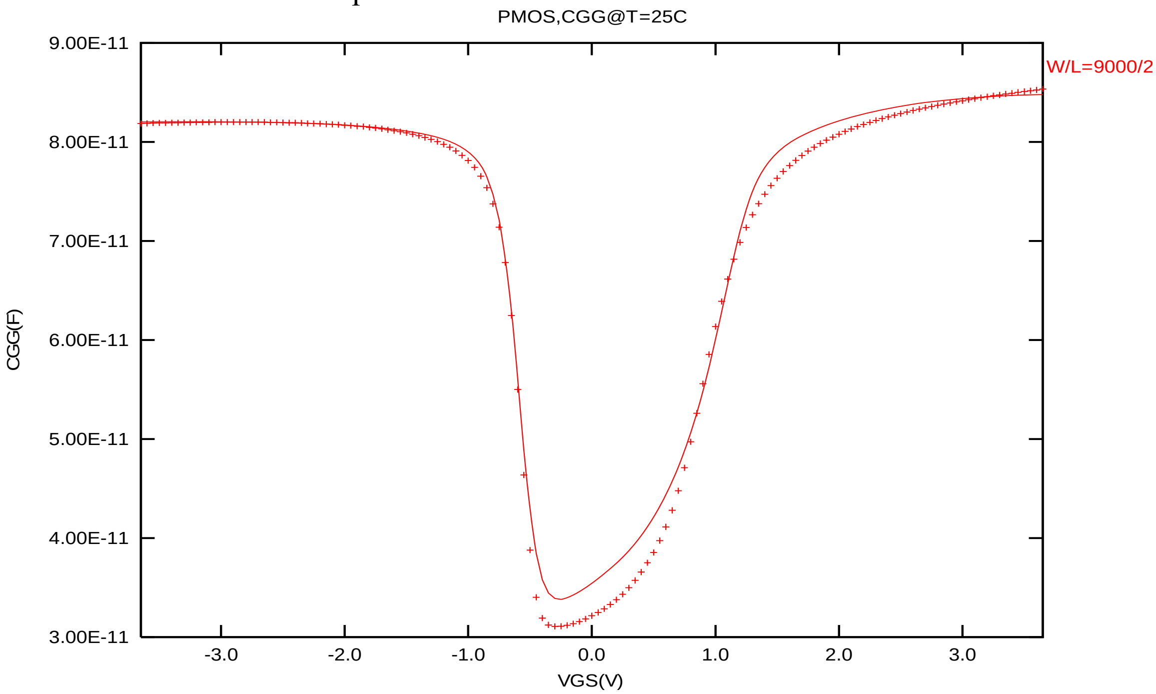

Figure 3-21 Measured and Simulated C_{GG} curves for PMOS

3.4.4 C_{GC} vs. V_{GS} Plots

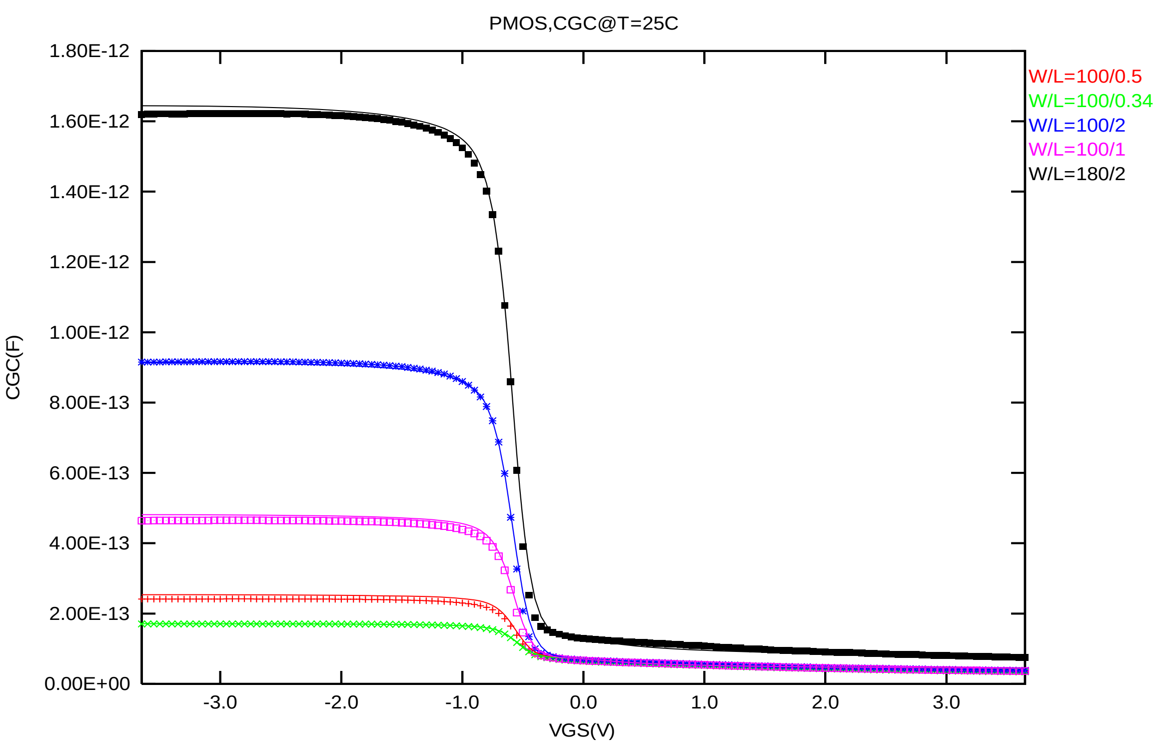

Figure 3-22 Measured and Simulated C_{GC} curves for large area measurement structure for PMOS

3.5 Parasitic source / drain junction capacitance model

3.5.1 Test devices

The following test structures were used for parameter extraction.

Table 3-9 Junction capacitor structures for PMOS

Structures	Area[m^2]	Perimeter[m]	Description
Bottom area junction capacitance	3.60E-08	7.60E-04	Plate junction capacitor
STI bounded sidewall junction capacitance	3.60E-08	3.64E-02	Finger diode
Inner gate sidewall junction	2.70E-08	1.830E-02	Poly finger structure

3.5.2 Measurement conditions

All junction capacitance was measured under voltage sweep conditions summarized in the following table.

Table 3-10 Junction capacitor measurement conditions for PMOS

V_{BIAS} [V]	VH: nwell & VL: p+ From 0 to 3.6 V in 0.1 V step
Frequency [kHz]	100
V_{OSC} [mV]	50
Temp [°C]	-50,25,125

3.5.3 Extracted model parameters and verification plots

The following junction capacitance parameters were extracted from hardware.

Table 3-11 Extracted parameter from junction capacitor measurement for PMOS

Bottom Area Junction Capacitance Model			**STI Bounded Sidewall Junction Capacitance Model**			**Poly Gate Sidewall Junction Capacitance Model**		
Parameter	**Unit**	**Value**	**Parameter**	**Unit**	**Value**	**Parameter**	**Unit**	**Value**
CJ	F/m^2	9.93E-04	CJSW	F/m	1.20E-10	CJSWG	F/m	3.59E-10
PB	V	7.90E-01	PBSW	V	7.80E-01	PBSWG	V	7.00E-01
MJ	--	3.30E-01	MJSW	--	1.30E-01	MJSWG	--	4.50E-01
TCJ	1/K	7.20E-04	TCJSW	1/K	1.00E-04	TCJSWG	1/K	1.42E-03
TPB	V/K	1.50E-03	TPBSW	V/K	5.00E-04	TPBSWG	V/K	2.58E-03

CJ, CJSW vs. VBIAS Plots

The following plot shows the variation of bottom area junction and STI bounded sidewall at different bias voltage and temperatures.

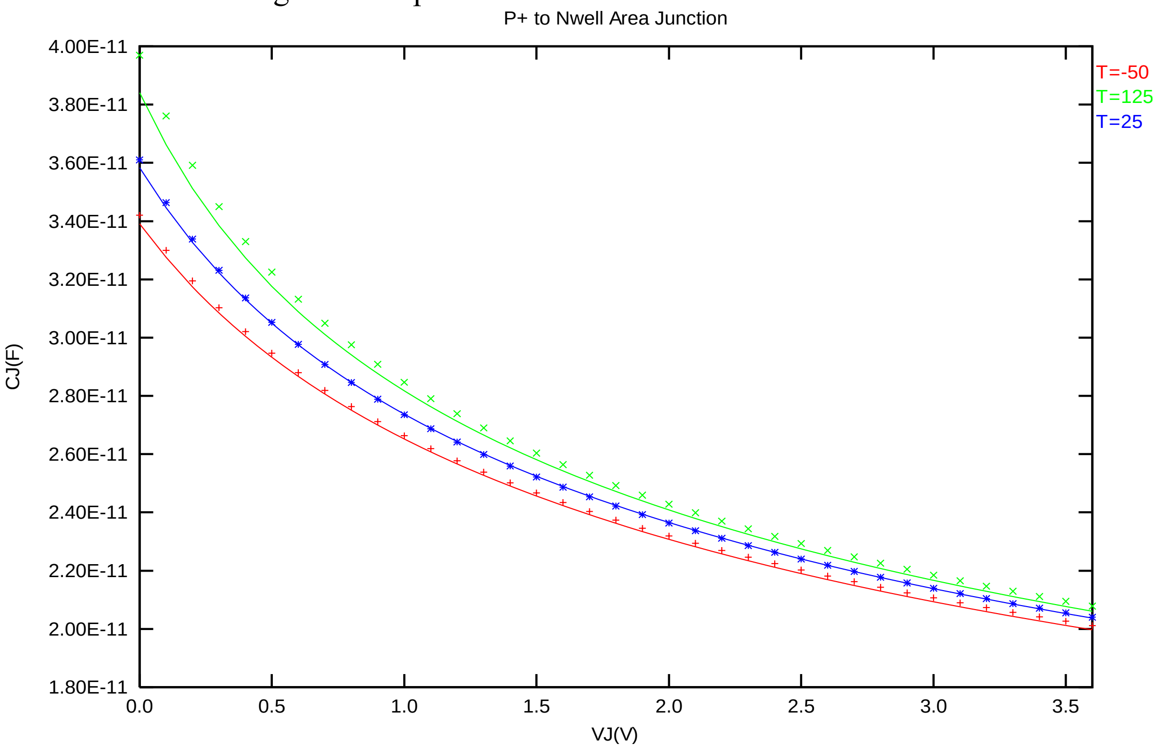

Figure 3-23 Bottom area junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for PMOS

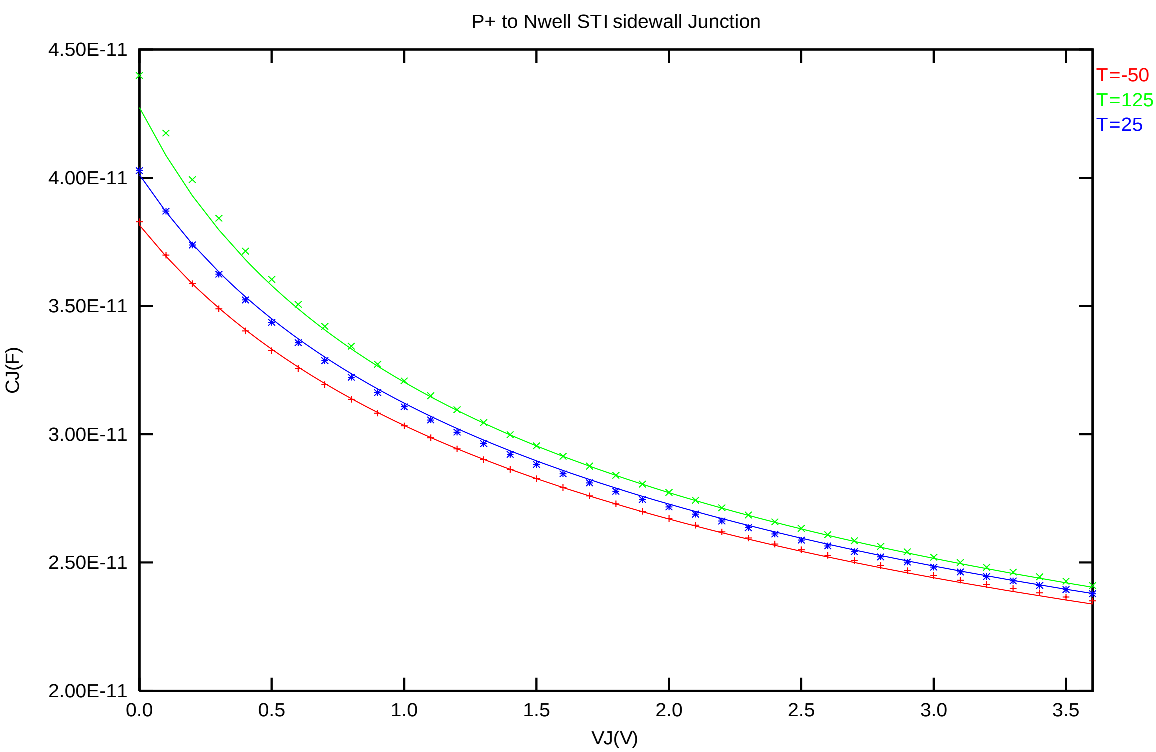

Figure 3-24 STI bounded sidewall junction capacitance at -50 ℃, 25 ℃ and 125 ℃ for PMOS

4 Worst case model

4.1 Parameter Variations

Table 4-1 Corner parameters

	n_33_l110e	TT	SS	FF	SNFP	FNSP
1	xl	-9.00E-09	+1.00E-08	-1.00E-08	+1.00E-08	-1.00E-08
2	xw	1.00E-09	-3.04E-09	+3.00E-10	-9.85E-10	+5.58E-10
3	tox	7.49E-09	+1.50E-10	-1.50E-10	+0.00E+00	+0.00E+00
4	wwl	-2.19E-22	+1.95E-22	-1.35E-22	+7.05E-23	-5.05E-23
5	cgso	5.00E-11	-10.0%	+10.0%	+0.0%	+0.0%
6	cgdo	5.00E-11	-10.0%	+10.0%	+0.0%	+0.0%
7	vth0	5.80E-01	+1.93E-02	-2.03E-02	+1.68E-02	-1.71E-02
8	pvth0	5.29E-16	+6.32E-16	-4.20E-16	+1.67E-16	-1.55E-16
9	k3	1.39E+00	+6.94E-01	-7.55E-01	+3.59E-01	-3.54E-01
10	nlx	8.83E-10	+6.37E-08	-5.95E-08	+2.40E-08	-2.21E-08
11	u0	3.92E-02	-2.11E-03	+1.86E-03	-1.20E-03	+1.20E-03
12	rdsw	3.24E+02	+2.10E+02	-1.35E+02	+9.08E+01	-7.56E+01
13	cgdl	1.80E-10	-10.0%	+10.0%	+0.0%	+0.0%
14	cgsl	1.80E-10	-10.0%	+10.0%	+0.0%	+0.0%
15	cj	7.80E-04	+10.0%	-10.0%	+10.0%	-10.0%
16	cjsw	1.08E-10	+10.0%	-10.0%	+10.0%	-10.0%
17	cjswg	2.30E-10	+10.0%	-10.0%	+10.0%	-10.0%

	p_33_l110e	TT	SS	FF	SNFP	FNSP
1	xl	-9.00E-09	+5.00E-09	-5.00E-09	-5.00E-09	+5.00E-09
2	xw	4.00E-09	-2.20E-08	+1.26E-08	+7.65E-09	-1.00E-08
3	tox	7.68E-09	+1.50E-10	-1.50E-10	+0.00E+00	+0.00E+00
4	wwl	-7.98E-22	+4.21E-22	-2.77E-22	-1.15E-22	+1.52E-22
5	cgso	9.11E-11	-10.0%	+10.0%	+0.0%	+0.0%
6	cgdo	9.11E-11	-10.0%	+10.0%	+0.0%	+0.0%
7	vth0	-5.96E-01	-2.74E-02	+2.86E-02	+2.29E-02	-2.24E-02
8	pvth0	-4.60E-16	-1.28E-15	+1.65E-15	+8.23E-16	-7.60E-16
9	k3	1.13E+00	+9.58E-02	-1.20E-01	-3.45E-02	+3.17E-02
10	nlx	6.42E-09	+1.82E-08	-1.71E-08	-6.89E-09	+7.85E-09
11	u0	1.06E-02	-4.89E-04	+4.06E-04	+2.72E-04	-2.93E-04
12	rdsw	8.28E+02	+4.20E+02	-2.36E+02	-9.00E+01	+1.26E+02
13	cgdl	2.02E-10	-10.0%	+10.0%	+0.0%	+0.0%
14	cgsl	2.02E-10	-10.0%	+10.0%	+0.0%	+0.0%
15	cj	9.93E-04	+10.0%	-10.0%	-10.0%	+10.0%
16	cjsw	1.20E-10	+10.0%	-10.0%	-10.0%	+10.0%
17	cjswg	3.59E-10	+10.0%	-10.0%	-10.0%	+10.0%

4.2 Wafer Data vs Corner graph

Threshold voltage in linear region V_{TLIN}

The threshold voltage extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for NMOS and I_D =-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) for PMOS at 25 ℃. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

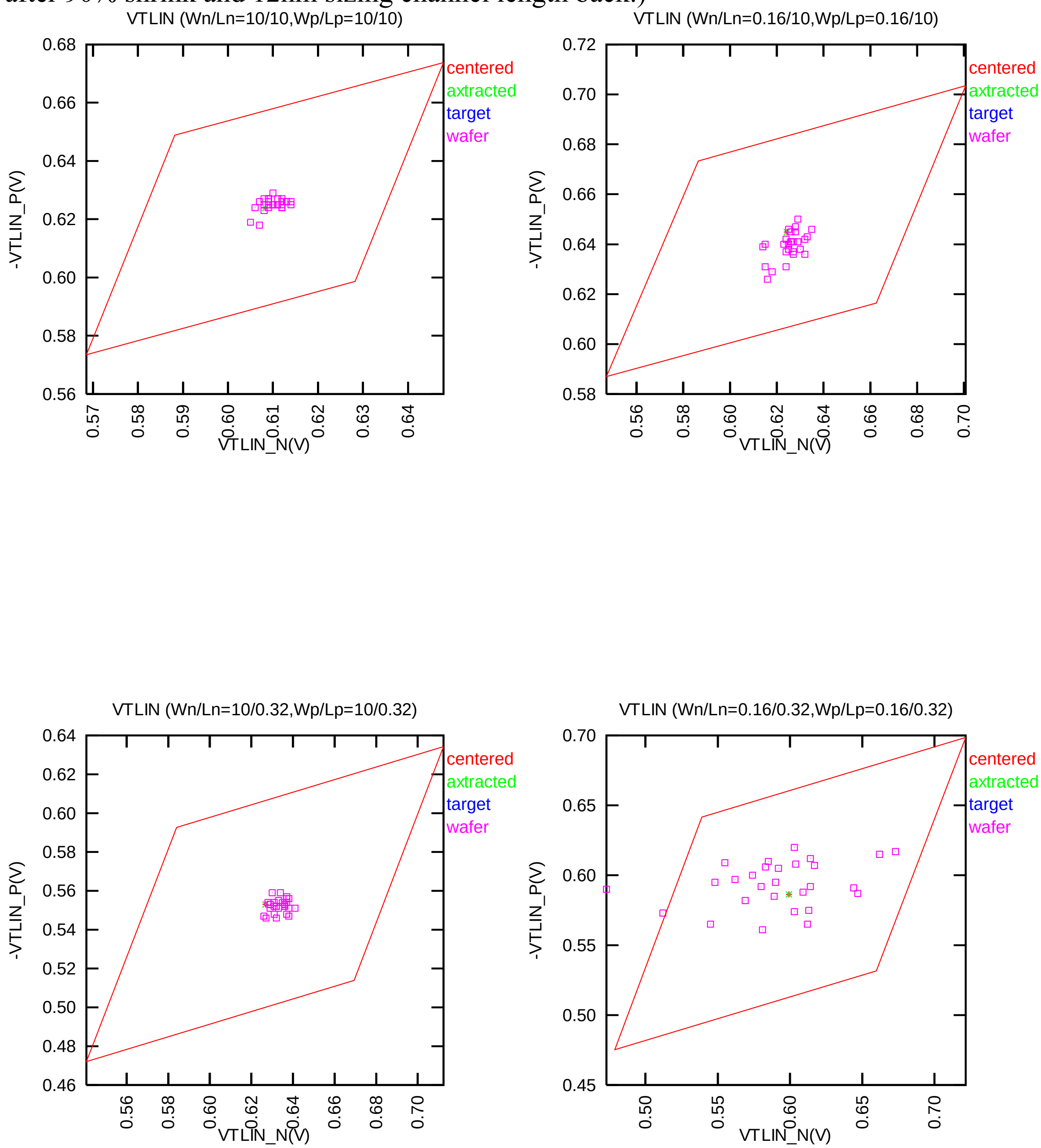

Figure 4-1 Corner graph of threshold voltage at low drain bias

Threshold voltage in saturation region V_{TSAT}

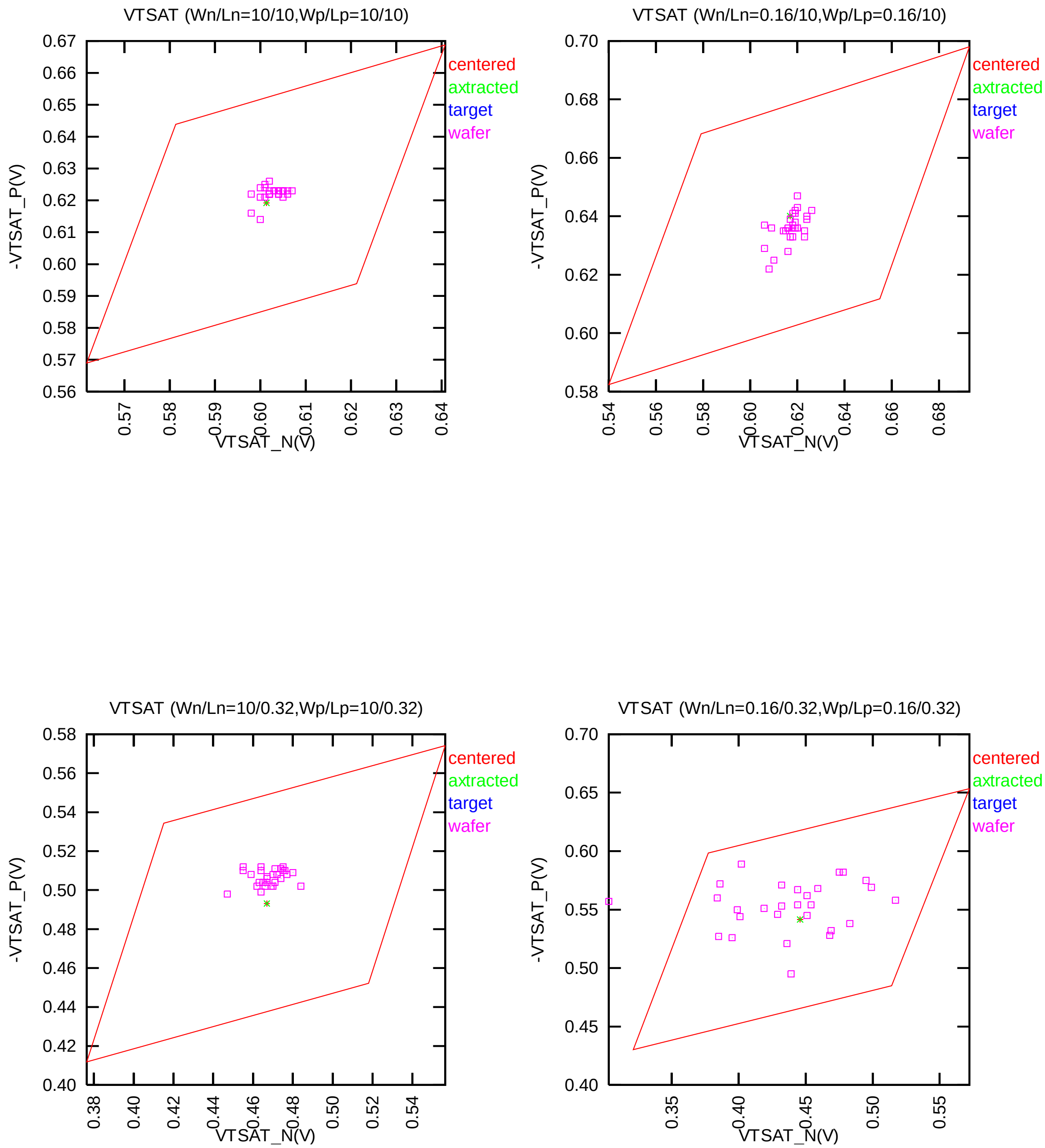

Figure 4-2 Corner graph of threshold voltage at high drain bias

Saturation Current I_{ON}

The dimensions are after 90% shrink and 12nm sizing channel length back.

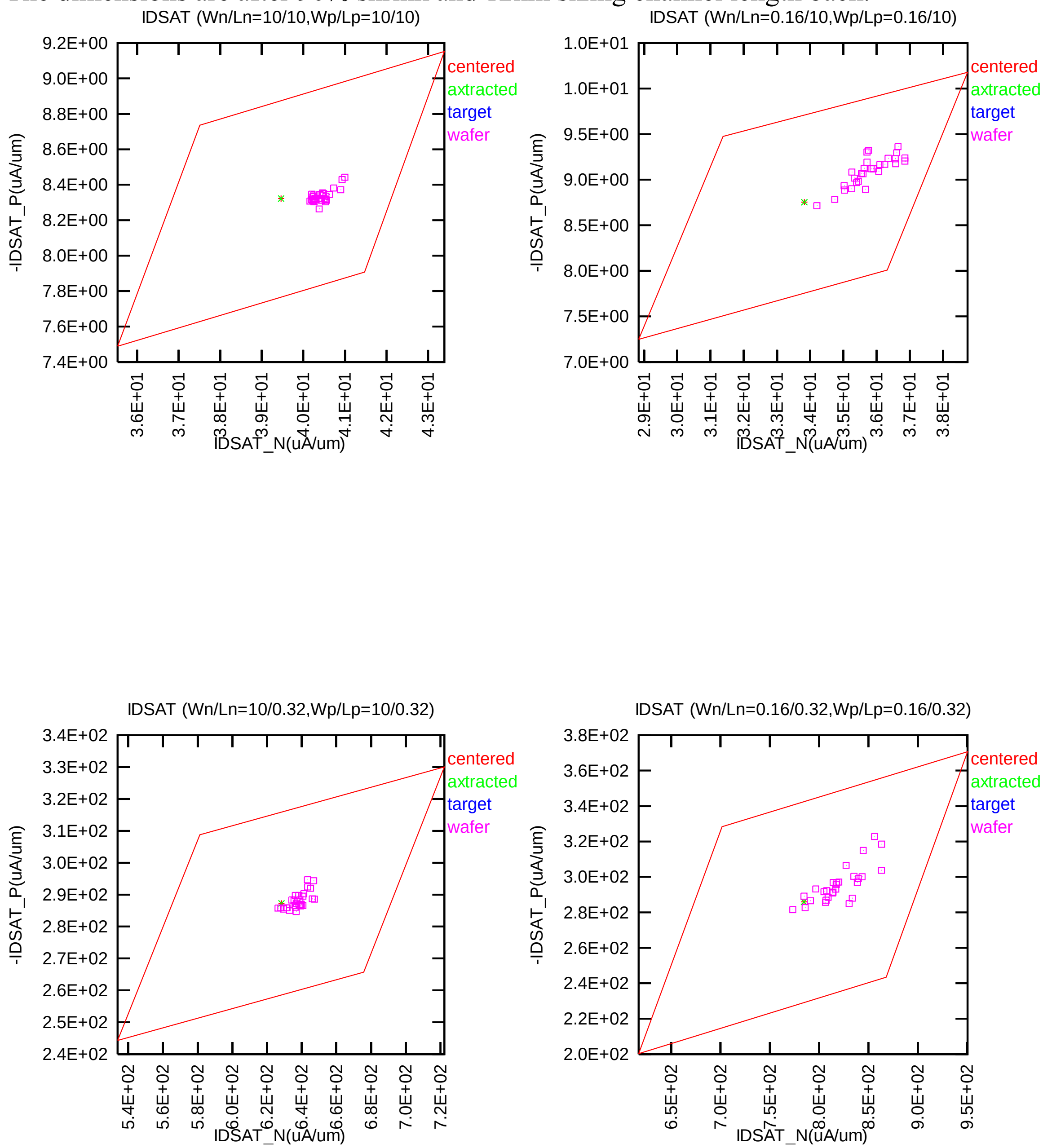

Figure 4-3 Corner graph of on-current

4.3 Device performance reference tables

Note: The model simulation results in the following tables have already reflected silicon data that is sizing channel length back for 12nm. It means: The real silicon size of W/L=9μm/0.318μm would be the same as the model simulation results in size of W/L=9μm/0.306μm as below table (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 4-2 NMOS ID current at VGS=VDS= Vcc/2 and VGS=VDS =Vcc.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (μA)	VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	87.99	544.59	67.39	392.41	52.73	280.24
SS	9/9	75.31	489.52	58.03	353.04	45.70	252.35
FF	9/9	101.75	599.25	77.49	431.73	60.26	308.21
SNFP	9/9	81.87	516.55	62.88	372.83	49.34	266.69
FNSP	9/9	94.44	573.38	72.14	412.52	56.28	294.15
TT	9/0.306	2050.90	6293.90	1816.90	5728.40	1536.00	4823.70
SS	9/0.306	1650.90	5335.20	1464.20	4861.10	1246.40	4116.50
FF	9/0.306	2479.10	7239.60	2198.40	6592.60	1850.40	5532.70
SNFP	9/0.306	1841.00	5824.30	1628.60	5294.90	1379.20	4463.10
FNSP	9/0.306	2269.60	6765.40	2014.50	6166.20	1700.90	5189.00
TT	0.144/9	1.08	7.12	0.91	5.56	0.79	4.33
SS	0.144/9	0.83	6.02	0.71	4.72	0.62	3.69
FF	0.144/9	1.37	8.21	1.14	6.40	0.97	4.97
SNFP	0.144/9	0.96	6.57	0.81	5.15	0.70	4.02
FNSP	0.144/9	1.21	7.68	1.02	5.99	0.87	4.65
TT	0.144/0.306	41.59	133.46	37.22	121.51	32.02	102.49
SS	0.144/0.306	29.53	104.13	26.50	94.67	23.03	80.16
FF	0.144/0.306	54.82	162.42	49.11	148.24	41.98	124.80
SNFP	0.144/0.306	35.30	119.10	31.57	108.17	27.25	91.27
FNSP	0.144/0.306	48.21	147.74	43.22	134.85	37.08	113.75

Table 4-3 NMOS ID current density at VGS=VDS= Vcc/2 and VGS=VDS =Vcc.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IDSAT (μA/um)	VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	9.78	60.51	7.49	43.60	5.86	31.14
SS	9/9	8.37	54.39	6.45	39.23	5.08	28.04
FF	9/9	11.31	66.58	8.61	47.97	6.70	34.25
SNFP	9/9	9.10	57.39	6.99	41.43	5.48	29.63
FNSP	9/9	10.49	63.71	8.02	45.84	6.25	32.68
TT	9/0.306	227.88	699.32	201.88	636.49	170.67	535.97
SS	9/0.306	183.43	592.80	162.69	540.12	138.49	457.39
FF	9/0.306	275.46	804.40	244.27	732.51	205.60	614.74
SNFP	9/0.306	204.56	647.14	180.96	588.32	153.24	495.90
FNSP	9/0.306	252.18	751.71	223.83	685.13	188.99	576.56
TT	0.144/9	7.51	49.44	6.32	38.63	5.46	30.07
SS	0.144/9	5.76	41.78	4.91	32.79	4.31	25.65
FF	0.144/9	9.49	57.00	7.90	44.43	6.75	34.48
SNFP	0.144/9	6.64	45.61	5.62	35.75	4.89	27.92
FNSP	0.144/9	8.43	53.33	7.06	41.57	6.06	32.26
TT	0.144/0.306	288.78	926.81	258.48	843.82	222.35	711.74
SS	0.144/0.306	205.09	723.13	184.01	657.42	159.96	556.64
FF	0.144/0.306	380.72	1127.92	341.05	1029.44	291.55	866.67
SNFP	0.144/0.306	245.14	827.08	219.22	751.18	189.26	633.81
FNSP	0.144/0.306	334.77	1025.97	300.14	936.46	257.51	789.93

Table 4-4 NMOS VTLIN and VTSAT.

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
VTH (V)	VDS	0.1V	3.3V	0.1V	3.3V	0.1V	3.3V
TT	9/9	0.687	0.681	0.609	0.602	0.511	0.502
SS	9/9	0.725	0.720	0.649	0.642	0.552	0.543
FF	9/9	0.648	0.642	0.569	0.562	0.470	0.461
SNFP	9/9	0.706	0.700	0.629	0.622	0.532	0.522
FNSP	9/9	0.667	0.661	0.589	0.582	0.490	0.481
TT	9/0.306	0.695	0.539	0.629	0.467	0.549	0.380
SS	9/0.306	0.780	0.628	0.715	0.557	0.636	0.470
FF	9/0.306	0.610	0.449	0.543	0.377	0.462	0.288
SNFP	9/0.306	0.737	0.590	0.672	0.519	0.592	0.431
FNSP	9/0.306	0.653	0.487	0.586	0.415	0.506	0.327
TT	0.144/9	0.697	0.691	0.617	0.609	0.514	0.504
SS	0.144/9	0.775	0.769	0.696	0.689	0.595	0.585
FF	0.144/9	0.618	0.612	0.536	0.529	0.431	0.422
SNFP	0.144/9	0.736	0.730	0.657	0.649	0.555	0.544
FNSP	0.144/9	0.658	0.652	0.577	0.570	0.473	0.463
TT	0.144/0.306	0.657	0.509	0.587	0.433	0.501	0.339
SS	0.144/0.306	0.777	0.634	0.709	0.559	0.626	0.466
FF	0.144/0.306	0.538	0.385	0.466	0.308	0.379	0.212
SNFP	0.144/0.306	0.717	0.577	0.648	0.502	0.563	0.408
FNSP	0.144/0.306	0.597	0.441	0.526	0.364	0.440	0.269

Table 4-5 NMOS ID current at VGS=0 and VDS=0.1V or VDS=Vcc(off current).

NMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
IOFF (A)	VDS	0.1V	3.3V	0.1V	3.3V	0.1V	3.3V
TT	9/9	2.01E-14	6.61E-13	4.30E-14	6.82E-13	1.20E-11	1.40E-11
SS	9/9	2.01E-14	6.61E-13	2.79E-14	6.71E-13	5.51E-12	6.77E-12
FF	9/9	2.01E-14	6.61E-13	8.88E-14	7.39E-13	2.71E-11	3.10E-11
SNFP	9/9	2.01E-14	6.64E-13	3.29E-14	6.75E-13	7.84E-12	9.38E-12
FNSP	9/9	2.01E-14	6.61E-13	6.15E-14	7.11E-13	1.85E-11	2.13E-11
TT	9/0.306	2.02E-14	7.25E-13	4.69E-13	4.17E-11	1.62E-10	5.20E-09
SS	9/0.306	2.01E-14	6.64E-13	6.10E-14	3.87E-12	2.70E-11	7.67E-10
FF	9/0.306	2.45E-14	2.71E-12	5.20E-12	5.50E-10	1.02E-09	3.55E-08
SNFP	9/0.306	2.01E-14	6.68E-13	1.51E-13	9.87E-12	6.45E-11	1.71E-09
FNSP	9/0.306	2.10E-14	1.15E-12	1.57E-12	1.87E-10	4.12E-10	1.59E-08
TT	0.144/9	2.00E-14	6.60E-13	2.03E-14	6.60E-13	2.27E-13	8.91E-13
SS	0.144/9	2.00E-14	6.60E-13	2.00E-14	6.60E-13	7.48E-14	7.20E-13
FF	0.144/9	2.00E-14	6.60E-13	2.36E-14	6.64E-13	1.12E-12	1.90E-12
SNFP	0.144/9	2.00E-14	6.60E-13	2.01E-14	6.60E-13	1.17E-13	7.68E-13
FNSP	0.144/9	2.00E-14	6.60E-13	2.11E-14	6.61E-13	4.91E-13	1.19E-12
TT	0.144/0.306	2.00E-14	6.63E-13	4.35E-14	2.25E-12	6.87E-12	1.82E-10
SS	0.144/0.306	2.00E-14	6.60E-13	2.08E-14	7.04E-13	5.58E-13	1.29E-11
FF	0.144/0.306	2.12E-14	1.01E-12	7.51E-13	5.80E-11	9.07E-11	2.53E-09
SNFP	0.144/0.306	2.00E-14	6.60E-13	2.41E-14	8.77E-13	1.87E-12	4.15E-11
FNSP	0.144/0.306	2.01E-14	7.03E-13	1.56E-13	1.25E-11	2.56E-11	7.99E-10

Table 4-6 PMOS ID current at VGS=VDS= -Vcc/2 and VGS=VDS = -Vcc.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IDSAT (μA)	-VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	16.26	98.45	15.10	83.07	14.41	70.51
SS	9/9	13.68	88.51	12.82	74.72	12.35	63.47
FF	9/9	19.09	108.33	17.57	91.39	16.61	77.52
SNFP	9/9	17.59	103.50	16.26	87.21	15.44	73.91
FNSP	9/9	14.99	93.39	13.98	78.92	13.40	67.09
TT	9/0.306	645.63	2952.40	591.77	2580.60	552.34	2238.90
SS	9/0.306	505.28	2491.00	469.56	2192.60	444.62	1915.40
FF	9/0.306	804.38	3410.10	729.16	2967.40	672.45	2562.90
SNFP	9/0.306	723.10	3179.10	659.41	2775.00	611.95	2403.20
FNSP	9/0.306	572.28	2723.80	527.63	2385.50	495.65	2074.70
TT	0.144/9	0.25	1.82	0.23	1.47	0.22	1.20
SS	0.144/9	0.19	1.53	0.18	1.23	0.17	1.00
FF	0.144/9	0.31	2.11	0.28	1.70	0.27	1.39
SNFP	0.144/9	0.28	1.96	0.25	1.59	0.24	1.29
FNSP	0.144/9	0.22	1.67	0.20	1.35	0.19	1.10
TT	0.144/0.306	9.73	52.89	8.67	43.62	7.94	35.81
SS	0.144/0.306	6.03	36.96	5.43	30.46	5.04	24.97
FF	0.144/0.306	14.17	68.88	12.48	56.77	11.28	46.58
SNFP	0.144/0.306	11.86	60.80	10.51	50.17	9.55	41.19
FNSP	0.144/0.306	7.79	45.01	6.98	37.10	6.44	30.45

Table 4-7 PMOS ID current density at VGS=VDS= -Vcc/2 and VGS=VDS = -Vcc.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IDSAT (μA/um)	-VDS	1.65V	3.3V	1.65V	3.3V	1.65V	3.3V
TT	9/9	1.81	10.94	1.68	9.23	1.60	7.83
SS	9/9	1.52	9.83	1.42	8.30	1.37	7.05
FF	9/9	2.12	12.04	1.95	10.15	1.85	8.61
SNFP	9/9	1.96	11.50	1.81	9.69	1.72	8.21
FNSP	9/9	1.67	10.38	1.55	8.77	1.49	7.45
TT	9/0.306	71.74	328.04	65.75	286.73	61.37	248.77
SS	9/0.306	56.14	276.78	52.17	243.62	49.40	212.82
FF	9/0.306	89.38	378.90	81.02	329.71	74.72	284.77
SNFP	9/0.306	80.34	353.23	73.27	308.33	67.99	267.02
FNSP	9/0.306	63.59	302.64	58.63	265.06	55.07	230.52
TT	0.144/9	1.73	12.62	1.58	10.20	1.50	8.30
SS	0.144/9	1.34	10.59	1.24	8.54	1.18	6.94
FF	0.144/9	2.16	14.62	1.97	11.83	1.85	9.64
SNFP	0.144/9	1.93	13.63	1.77	11.02	1.67	8.97
FNSP	0.144/9	1.53	11.60	1.41	9.37	1.34	7.62
TT	0.144/0.306	67.60	367.27	60.21	302.92	55.13	248.71
SS	0.144/0.306	41.88	256.65	37.73	211.53	35.00	173.39
FF	0.144/0.306	98.38	478.35	86.68	394.21	78.36	323.47
SNFP	0.144/0.306	82.37	422.22	72.95	348.38	66.35	286.05
FNSP	0.144/0.306	54.12	312.57	48.50	257.65	44.72	211.44

Table 4-8 PMOS VTLIN and VTSAT.

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-VTH (V)	-VDS	0.1V	3.3V	0.1V	3.3V	0.1V	3.3V
TT	9/9	0.721	0.717	0.624	0.620	0.502	0.495
SS	9/9	0.770	0.766	0.674	0.669	0.553	0.545
FF	9/9	0.671	0.667	0.574	0.569	0.450	0.444
SNFP	9/9	0.696	0.692	0.599	0.594	0.476	0.469
FNSP	9/9	0.745	0.741	0.649	0.644	0.527	0.520
TT	9/0.306	0.664	0.605	0.569	0.509	0.448	0.384
SS	9/0.306	0.745	0.685	0.651	0.590	0.531	0.467
FF	9/0.306	0.584	0.525	0.488	0.427	0.366	0.302
SNFP	9/0.306	0.626	0.565	0.530	0.468	0.408	0.343
FNSP	9/0.306	0.703	0.646	0.609	0.550	0.488	0.426
TT	0.144/9	0.737	0.733	0.642	0.637	0.521	0.514
SS	0.144/9	0.793	0.789	0.700	0.694	0.581	0.574
FF	0.144/9	0.680	0.676	0.584	0.580	0.461	0.455
SNFP	0.144/9	0.709	0.705	0.614	0.609	0.492	0.485
FNSP	0.144/9	0.764	0.760	0.670	0.665	0.550	0.543
TT	0.144/0.306	0.691	0.649	0.596	0.553	0.476	0.429
SS	0.144/0.306	0.799	0.758	0.708	0.664	0.591	0.545
FF	0.144/0.306	0.582	0.539	0.485	0.441	0.362	0.315
SNFP	0.144/0.306	0.637	0.593	0.542	0.496	0.420	0.371
FNSP	0.144/0.306	0.744	0.704	0.651	0.609	0.533	0.488

Table 4-9 PMOS ID current at VGS=0 and VDS= -0.1V or VDS= -Vcc(off current).

PMOS	Temp. (℃)	-50 ℃		25 ℃		125 ℃	
-IOFF (A)	-VDS	0.1V	3.3V	0.1V	3.3V	0.1V	3.3V
TT	9/9	1.99E-14	6.57E-13	3.19E-14	6.68E-13	1.23E-11	1.38E-11
SS	9/9	1.99E-14	6.54E-13	2.35E-14	6.57E-13	8.30E-12	9.47E-12
FF	9/9	1.99E-14	6.54E-13	6.13E-14	7.03E-13	2.30E-11	2.51E-11
SNFP	9/9	1.99E-14	6.54E-13	4.32E-14	6.82E-13	1.67E-11	1.84E-11
FNSP	9/9	1.99E-14	6.57E-13	2.62E-14	6.64E-13	9.74E-12	1.10E-11
TT	9/0.306	2.05E-14	6.61E-13	1.34E-12	6.33E-12	5.14E-10	1.62E-09
SS	9/0.306	1.99E-14	6.54E-13	1.89E-13	1.38E-12	1.12E-10	3.40E-10
FF	9/0.306	2.74E-14	7.11E-13	1.05E-11	4.68E-11	2.46E-09	7.86E-09
SNFP	9/0.306	2.20E-14	6.71E-13	3.70E-12	1.74E-11	1.11E-09	3.64E-09
FNSP	9/0.306	2.00E-14	6.61E-13	4.85E-13	2.57E-12	2.37E-10	7.12E-10
TT	0.144/9	2.00E-14	6.60E-13	2.01E-14	6.60E-13	3.47E-13	1.01E-12
SS	0.144/9	2.00E-14	6.60E-13	2.00E-14	6.60E-13	2.84E-13	9.40E-13
FF	0.144/9	2.00E-14	6.60E-13	2.05E-14	6.61E-13	5.04E-13	1.17E-12
SNFP	0.144/9	2.00E-14	6.60E-13	2.03E-14	6.60E-13	4.08E-13	1.07E-12
FNSP	0.144/9	2.00E-14	6.60E-13	2.01E-14	6.60E-13	3.09E-13	9.67E-13
TT	0.144/0.306	2.00E-14	6.60E-13	3.07E-14	6.88E-13	5.09E-12	1.15E-11
SS	0.144/0.306	2.00E-14	6.60E-13	2.07E-14	6.62E-13	8.31E-13	2.11E-12
FF	0.144/0.306	2.01E-14	6.60E-13	2.01E-13	1.16E-12	4.22E-11	9.62E-11
SNFP	0.144/0.306	2.00E-14	6.60E-13	6.42E-14	7.83E-13	1.45E-11	3.37E-11
FNSP	0.144/0.306	2.00E-14	6.60E-13	2.26E-14	6.66E-13	1.89E-12	4.32E-12

5 Shallow Trench Isolation (STI) or LOD Stress Effect

5.1 Simulation results for NMOS at SA=0.36µm (SAREF=2.2µm)

Note: The LOD stress effect model was based on very limit and early silicon data. SB is equal to SA, SD=0 in all following plots and tables. The following table show the channel length and width dependence of linear and saturation threshold voltage extracted at I_D =300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm). The dimension is after 90% shrink and 12nm sizing channel length back.
The simulation results for the NMOS characterization are summarized in the table below.

Table 5-1 Simulation results for NMOS

All parameters are given for T=25 °C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
Long channel device (W_{DES}/L_{DES}= 10 μ m / 10 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.610
I_{ON}	V_{DS} = 3.3 V	μA/ μm	38.98
Wide and short channel device (W_{DES}/L_{DES}= 10 μ m / 0.32 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.649
V_{TSAT}	V_{DS} = 3.3 V	V	0.489
Body Effect V_T shift	V_{DS} = 3.3 V V_{BS}= 0 ~ -3.3 V	V	0.158
DIBL V_T shift	V_{DS}= 0.1~3.3 V	V	0.160
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	610.05
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	2.5E-12
Narrow and long channel device (W_{DES}/L_{DES}= 0.14 μ m / 10 μ m)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.615
V_{TSAT}	V_{DS}= 3.3 V	V	0.608
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	34.77
Narrow and short channel device (W_{DES}/L_{DES}= 0.14 μm / 0.32 μm)			
V_{TLIN}	V_{DS} = 0.1 V	V	0.604
V_{TSAT}	V_{DS}= 3.3 V	V	0.452
I_{ON}	V_{DS}=V_{GS}= 3.3 V	μA/μm	819.43
I_{OFF}	V_{DS}= 3.3 V, V_{GS}=0 V	A/μm	1.1E-11

5.1.1 Model verification plots for NMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at I_D=300nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) at 25℃(The dimension is after 90% shrink and 12nm sizing channel length back.).

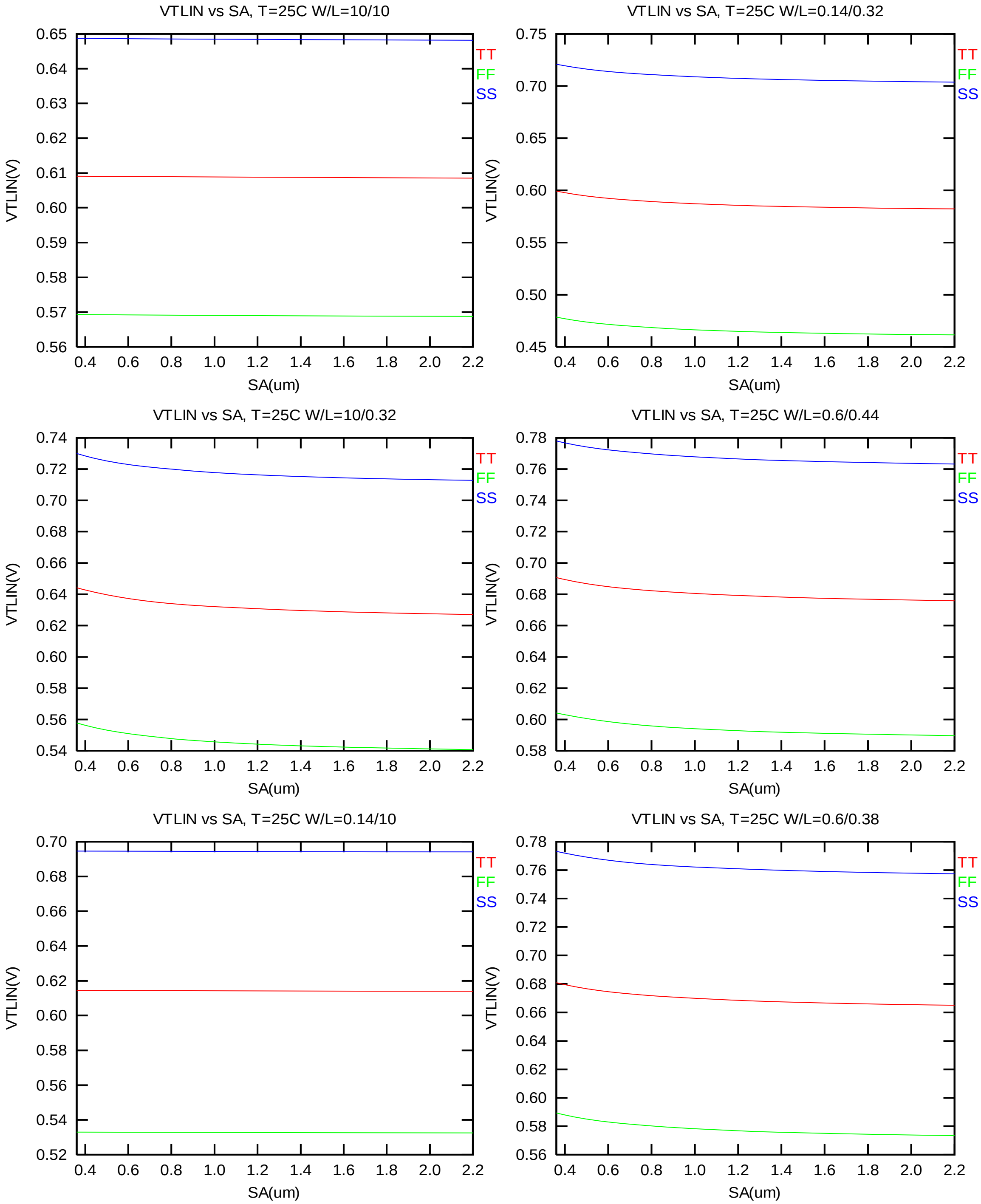

Figure 5-1 V_{TLIN} vs. SA for low drain bias (V_{DS} = 0.1 V) for NMOS

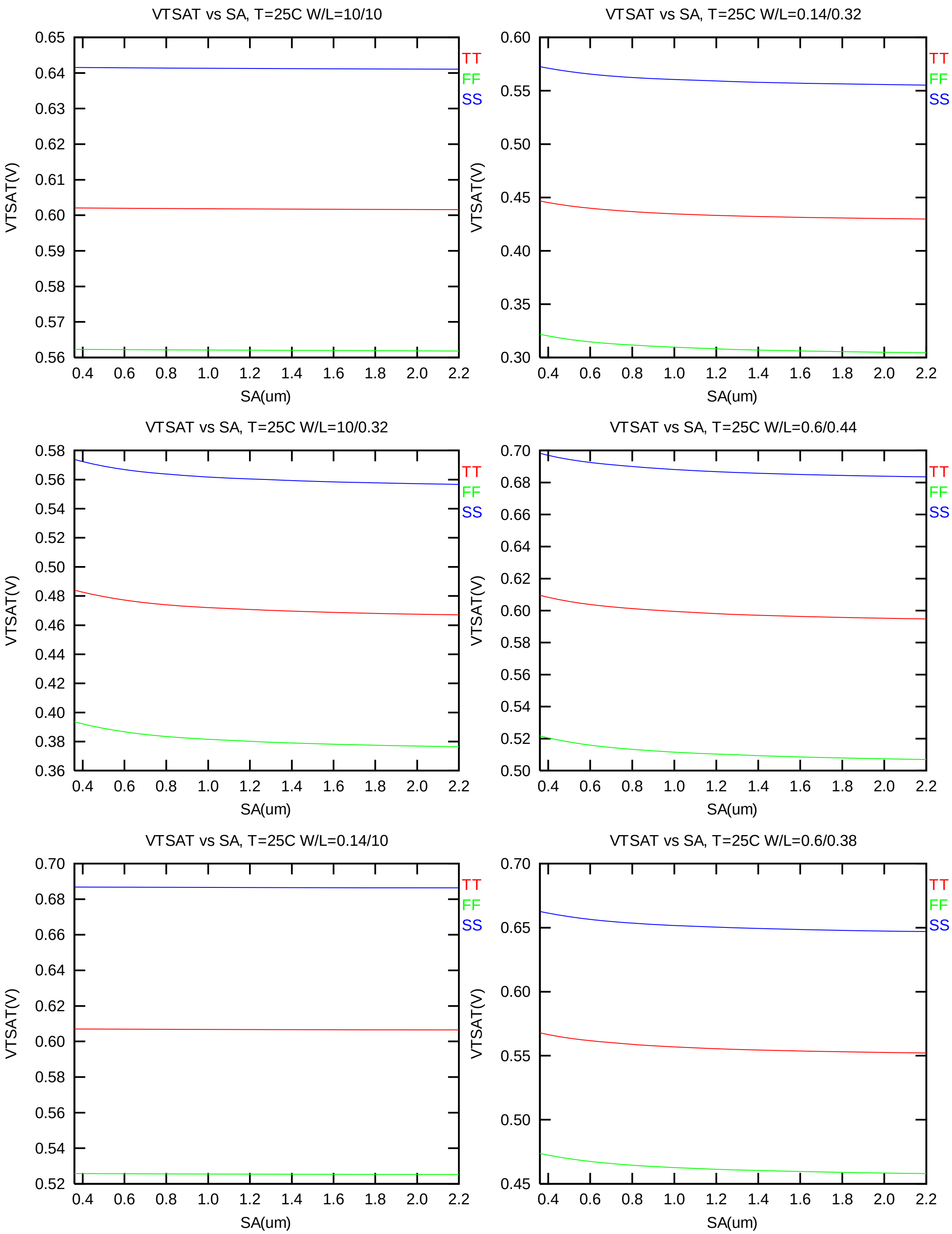

Figure 5-2 V_{TSAT} vs. SA for high drain bias (V_{DS} = 3.3 V) for NMOS

I_{ON} vs. SA

The following plots show the SA dependence of I_{ON} extracted at V_{DS}=V_{GS}=3.3 V for V_{BS}=0 at 25℃(The dimension is after 90% shrink and 12nm sizing channel length back.).

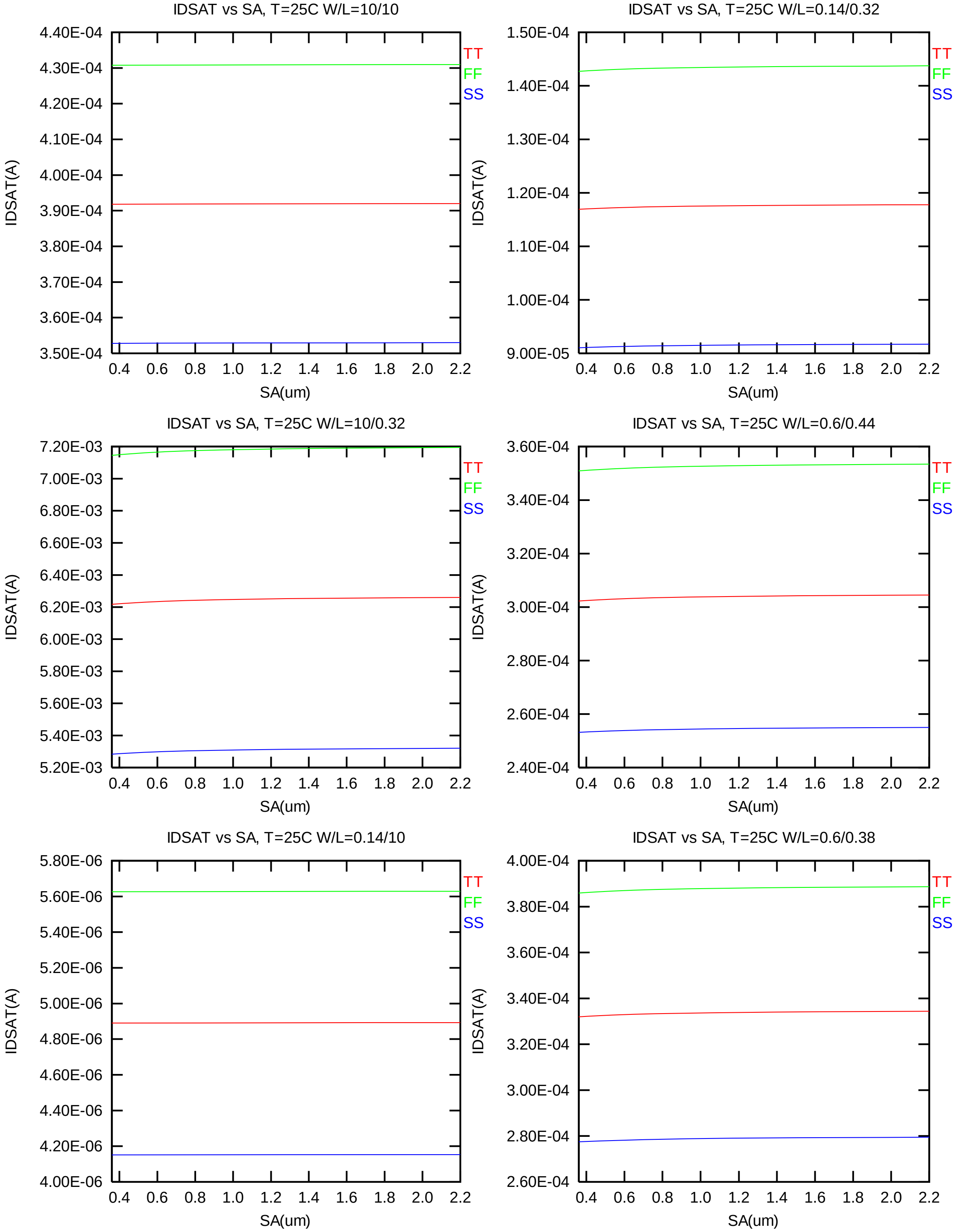

Figure 5-3 Saturation current I_{ON} vs. SA for NMOS

5.2 Simulation results for PMOS at SA=0.36µm (SAREF=2.2µm)

The simulation results for the PMOS characterization are summarized in the table below.

Table 5-2 Simulation results for PMOS

All parameters are given for T=25 °C ,V_{BS}=0 V unless specified otherwise.

Parameter	Condition	Unit	Simulation Value
Long channel device (W_{DES}/L_{DES}= 10 µ m / 10 µ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.624
I_{ON}	V_{DS} = -3.3 V	µA/ µm	-8.39
Wide and short channel device (W_{DES}/L_{DES}= 10 µ m / 0.32 µ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.570
V_{TSAT}	V_{DS} = -3.3 V	V	-0.510
Body Effect V_T shift	V_{DS} = -3.3 V V_{BS}= 0 ~ --3.3 V	V	-0.596
DIBL V_T shift	V_{DS}= -0.1~-3.3 V	V	-0.060
I_{ON}	V_{DS}=V_{GS}= -3.3 V	µA/µm	-300.76
I_{OFF}	V_{DS}= -3.3 V, V_{GS}=0 V	A/µm	-6.6E-13
Narrow and long channel device (W_{DES}/L_{DES}= 0.14 µ m / 10 µ m)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.641
V_{TSAT}	V_{DS}= -3.3 V	V	-0.636
I_{ON}	V_{DS}=V_{GS}= -3.3 V	µA/µm	-9.45
Narrow and short channel device (W_{DES}/L_{DES}= 0.14 µm / 0.32 µm)			
V_{TLIN}	V_{DS} = -0.1 V	V	-0.596
V_{TSAT}	V_{DS}= -3.3 V	V	-0.553
I_{ON}	V_{DS}=V_{GS}= -3.3 V	µA/µm	-324.96
I_{OFF}	V_{DS}= -3.3 V, V_{GS}=0 V	A/µm	-4.9E-12

5.2.1 Model verification plots for PMOS

The L and W in the plots refer to L_{DES} and W_{DES} respectively.

V_T vs. SA

The following figures show the SA dependence of linear and saturation threshold voltage extracted at I_D=-70nA*(W_{DES}*0.9)/(L_{DES}*0.9+12nm) at 25℃(The dimension is after 90% shrink and 12nm sizing channel length back).

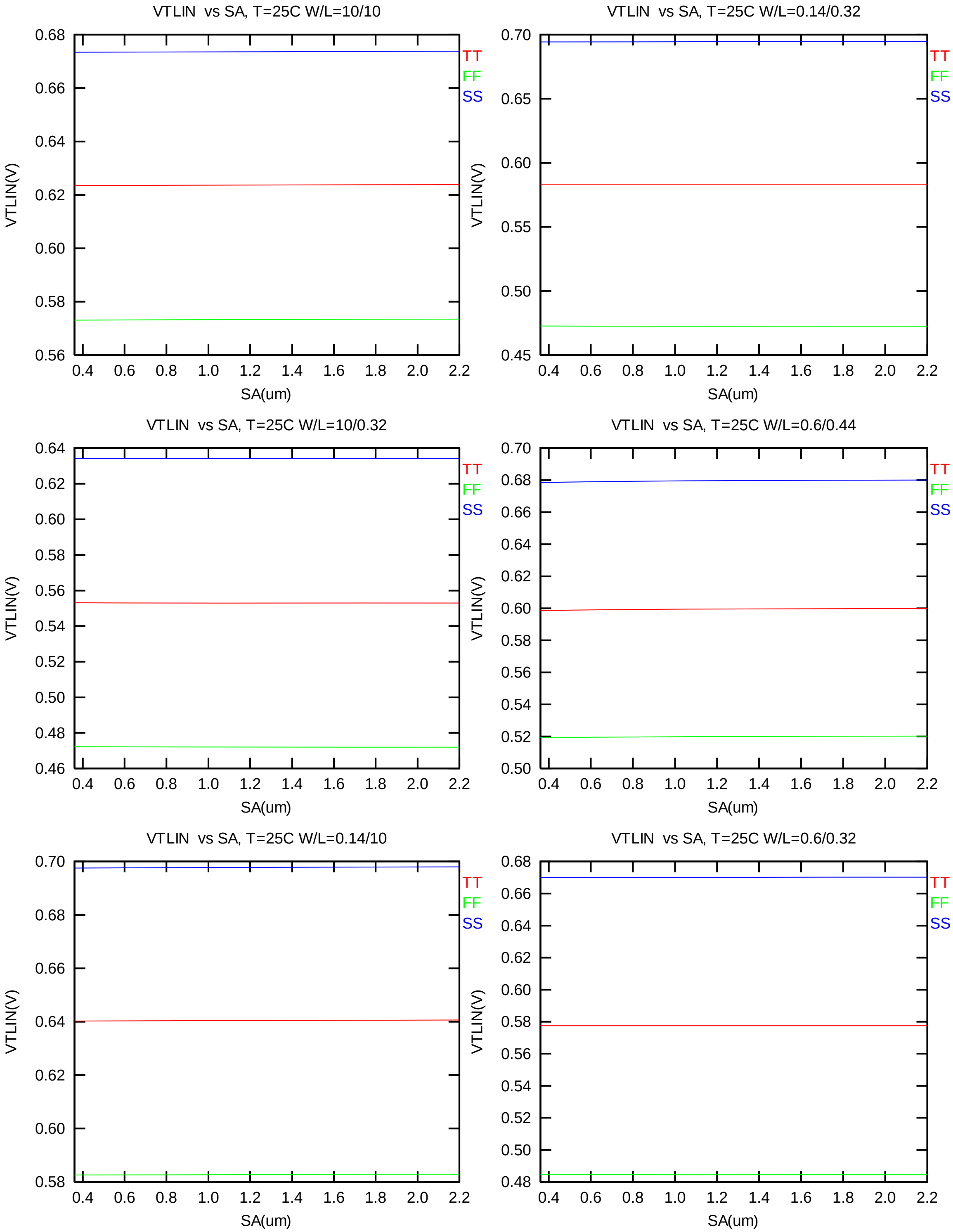

Figure 5-4 V_{TLIN} vs. SA for low drain bias (V_{DS}= -0.1 V) for PMOS

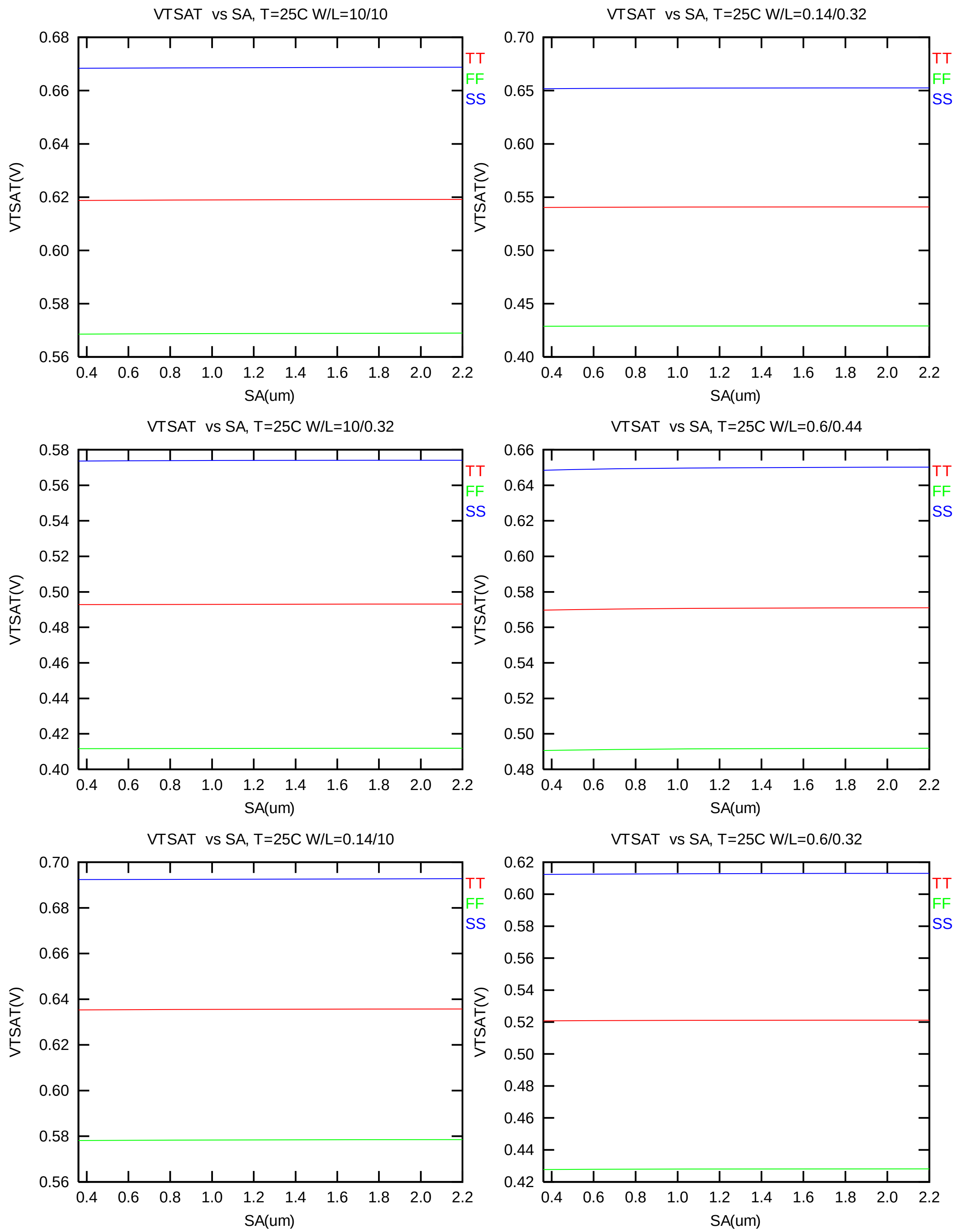

Figure 5-5 V_{TSAT} vs. SA for high drain bias (V_{DS}= -3.3 V) for PMOS

The following plots show the SA dependence of I_{ON} extracted at V_{DS}=V_{GS}= -3.3 V for V_{BS}=0 at 25℃(The dimension is after 90% shrink and 12nm sizing channel length back).

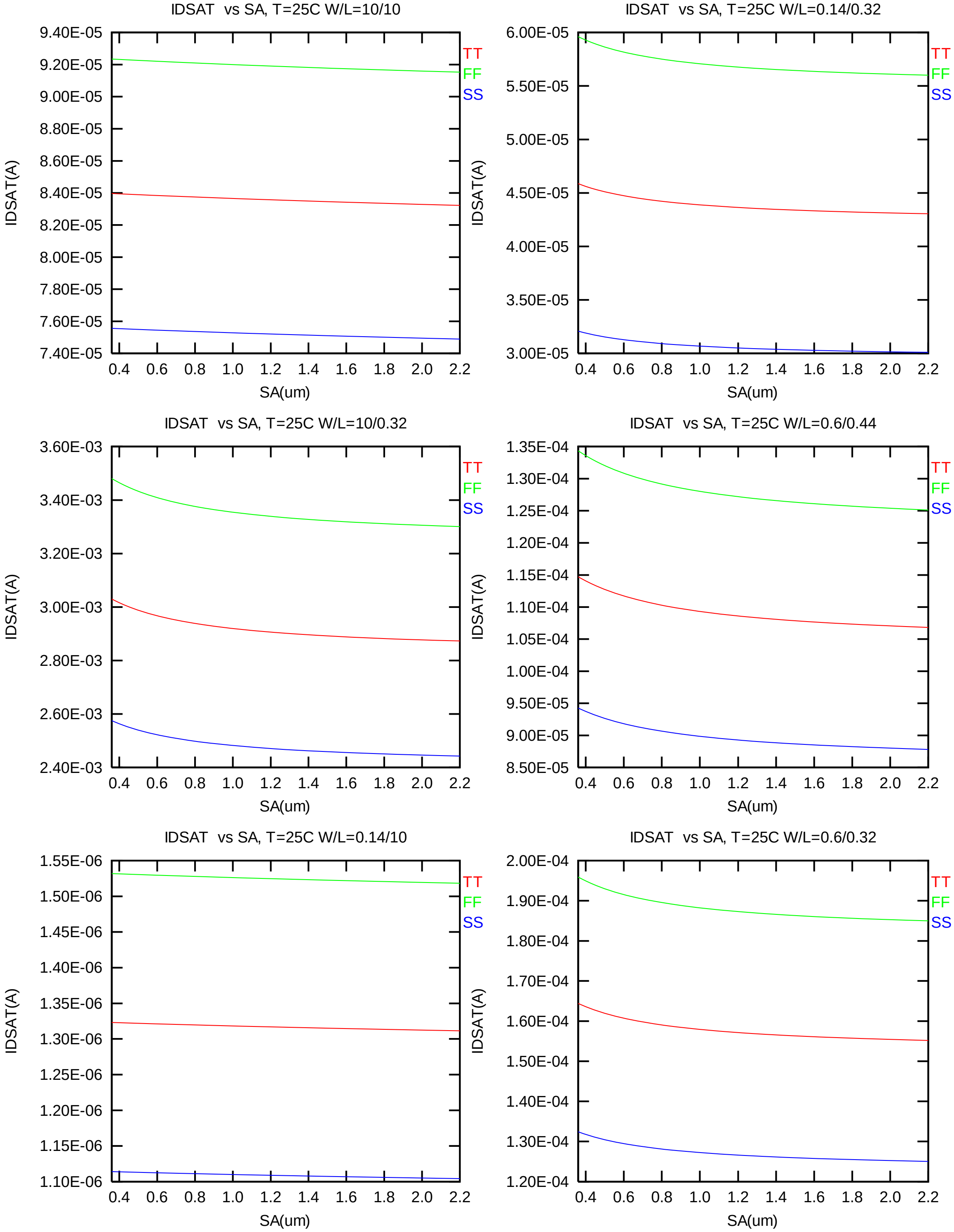

Figure 5-6 Saturation current I_{ON} vs. SA for PMOS

5.3 V_T and I_{ON} variation v. SA for both N/PMOS

Delta V_T is defined as V_T (SA)-V_T (0.36μm) for NMOS and -[V_T (SA)-V_T (0.36μm)] for PMOS. Delta I_{ON} is defined as [I_{ON} (SA)-I_{ON} (0.36μm)]/I_{ON} (0.36μm)*100%.

Delta V_T and I_{ON} vs. SA

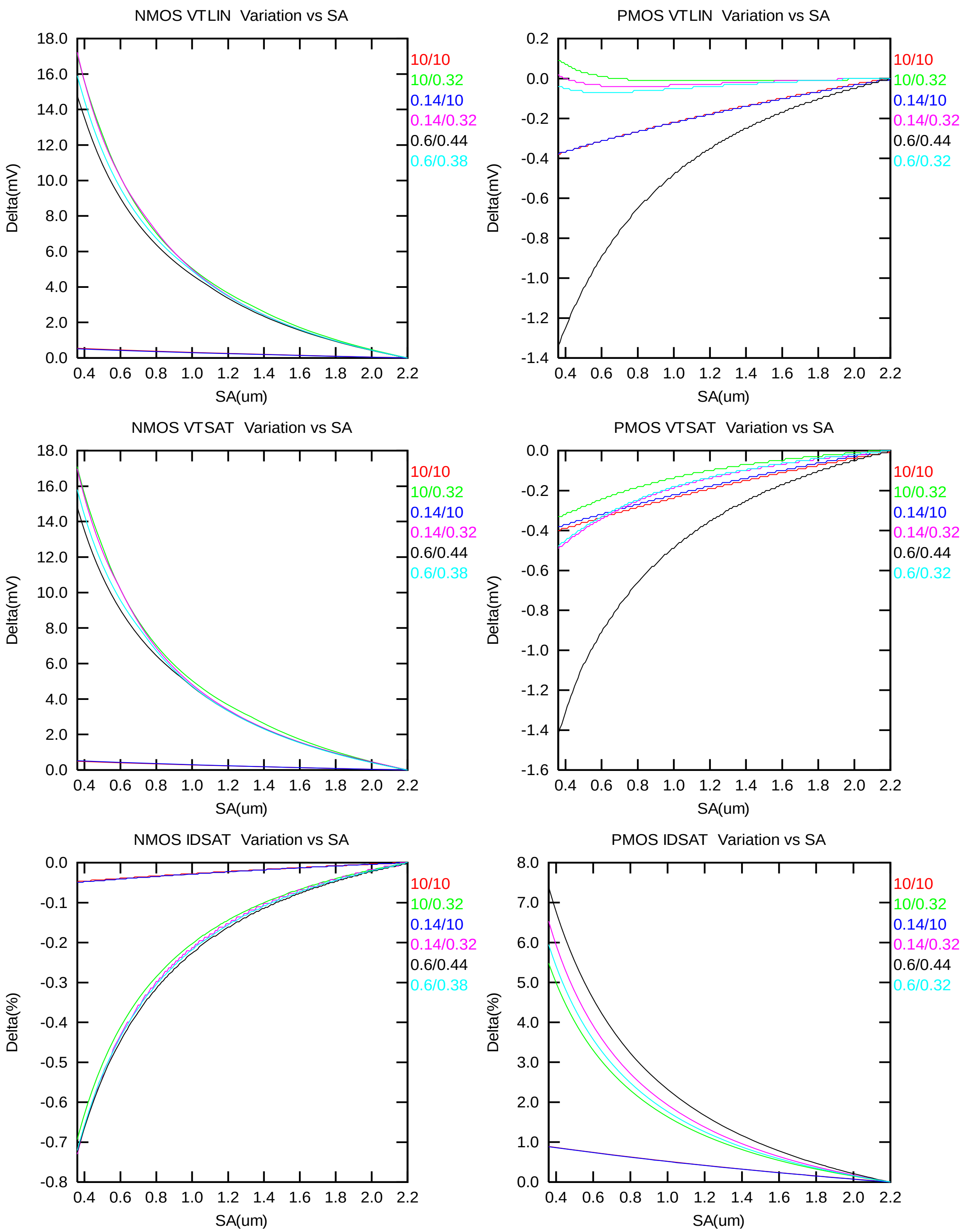

Figure 5-7 Delta V_T and I_{ON} vs. SA for both N/PMOS

6 Dynamic performance

6.1 Ring Oscillator Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configuration is adopted.

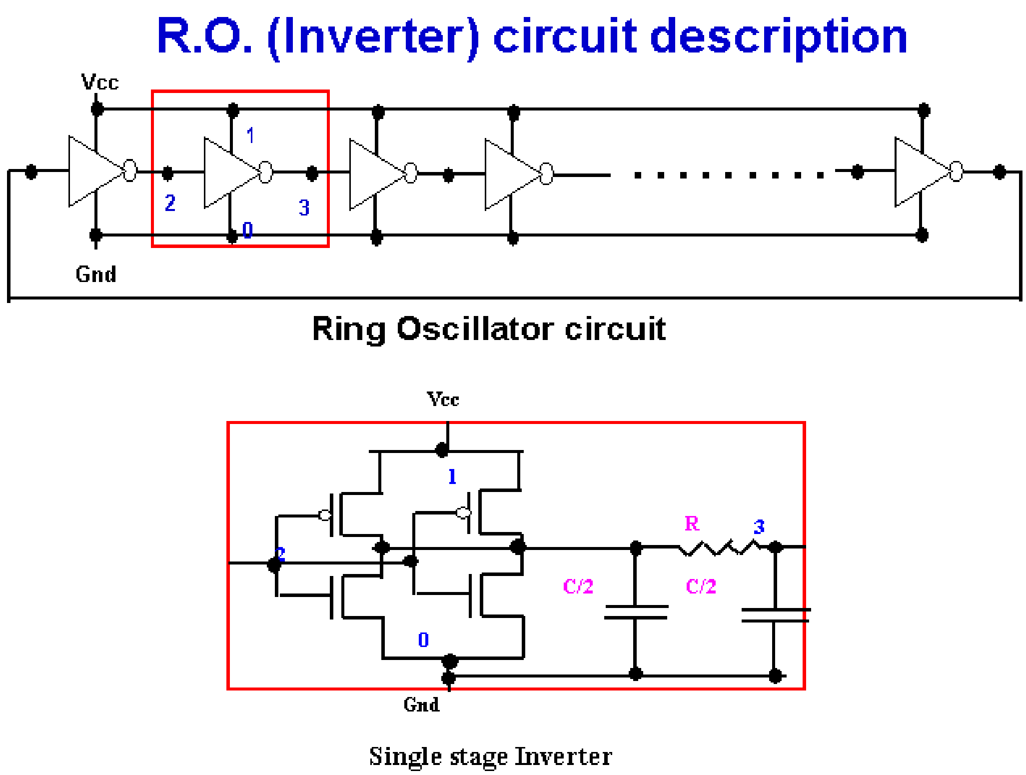

Figure 6-1 Ring Oscillator Netlist Diagram

6.2 Dynamic Performance for SA=0.36µm

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted. (The dimensions are after 90% shrink and 12nm sizing channel length back.)

Table 6-1 Ring oscillator (inverter) structure description for SA=0.36µm

R.O. (Inverter) Structure description		
Contact size	0.16 um X 0.16 um	
Polysilicon gate to contact spacing	0.14 um	
Contact to diffusion edge spacing	0.06 um	
Without RC loading	R = 0ohm	C = 0 F
Inverter		
Logic gate type	Inverter	
Channel width for n-ch MOSFET	3.75 um X 2	
Channel length for n-ch MOSFET	0.32 um(with 12nm sizing back)	
Channel width for p-ch MOSFET	2.5 um X 2	
Channel Length for p-ch MOSFET	0.32 um(with 12nm sizing back)	

The simulation results at different temperatures and using different worst-case models are listed as below.

Table 6-2 Ring Oscillator simulation results (ps/stage) for SA=0.36µm

Inverter, R=0, C=0f, Fanout=1, Vcc=3.3V			
Temp(C)	TT	FF	SS
-50	27.966	24.41	32.907
0	30.793	26.904	36.135
25	32.262	28.207	37.807
85	35.958	31.521	41.972
125	38.579	33.896	44.885

Inverter, R=0, C=0f, Fanout=1, T=25C			
Vcc(V)	TT	FF	SS
2.4	42.88	36.603	51.357
2.7	38.08	32.854	45.161
3	34.72	30.182	40.89
3.3	32.262	28.207	37.807
3.63	30.233	26.553	35.286

7 Noise Characteristic

7.1 Verification plots

NMOS, W=10µm L=0.34µm

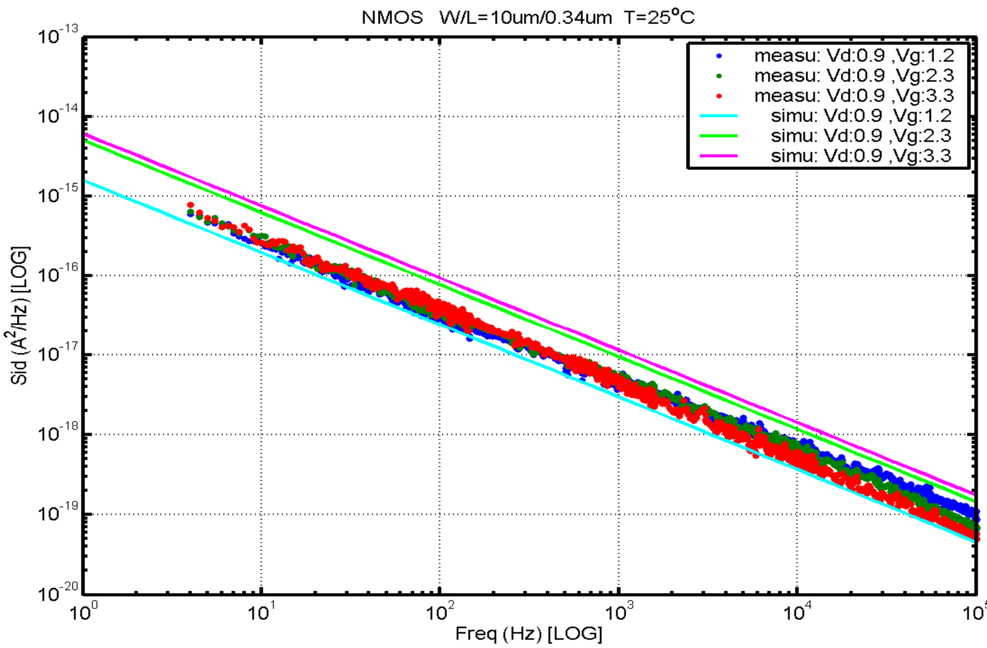

Figure 7-1 NMOS W/L=10µm/0.34µm @ Vd=0.9V

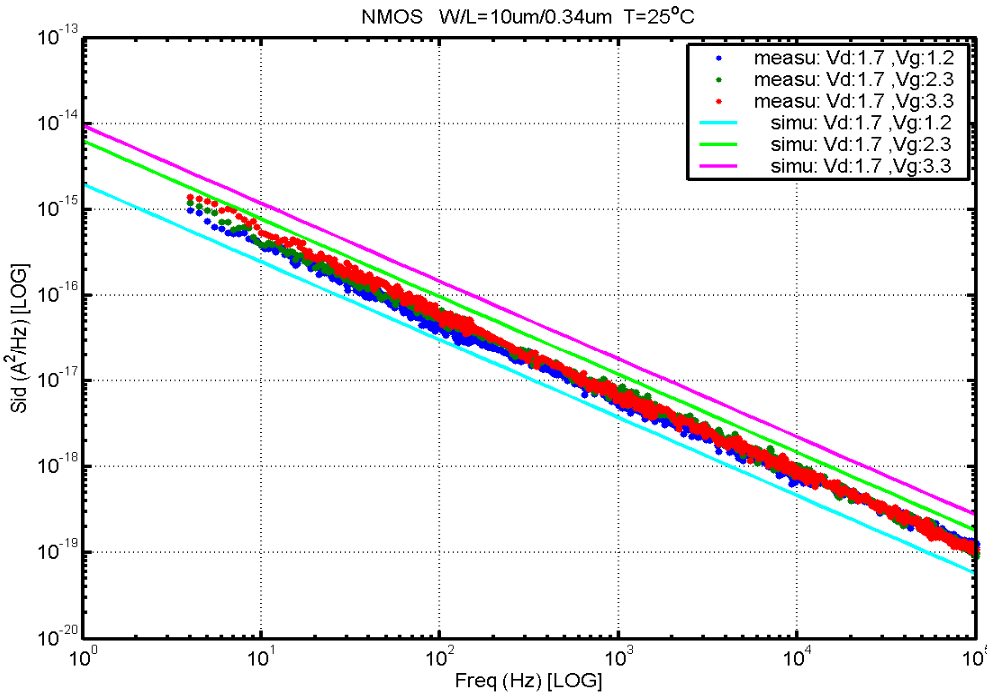

Figure 7-2 NMOS W/L=10µm/0.34µm @ Vd=1.7V

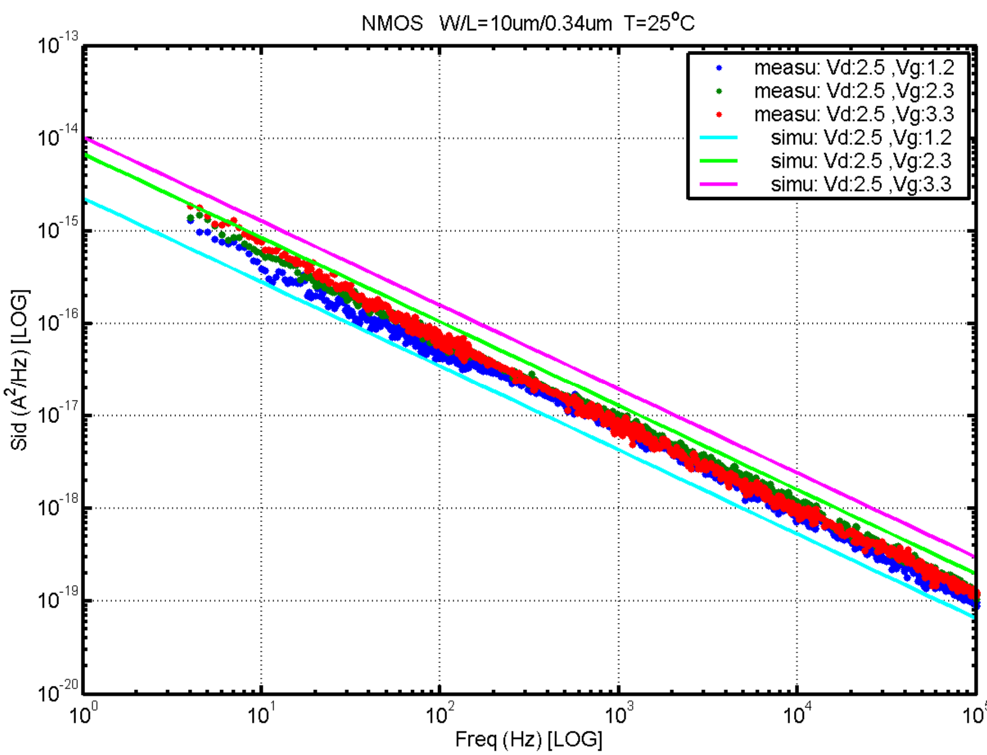

Figure 7-3 NMOS W/L=10µm/0.34µm @ Vd=2.5V

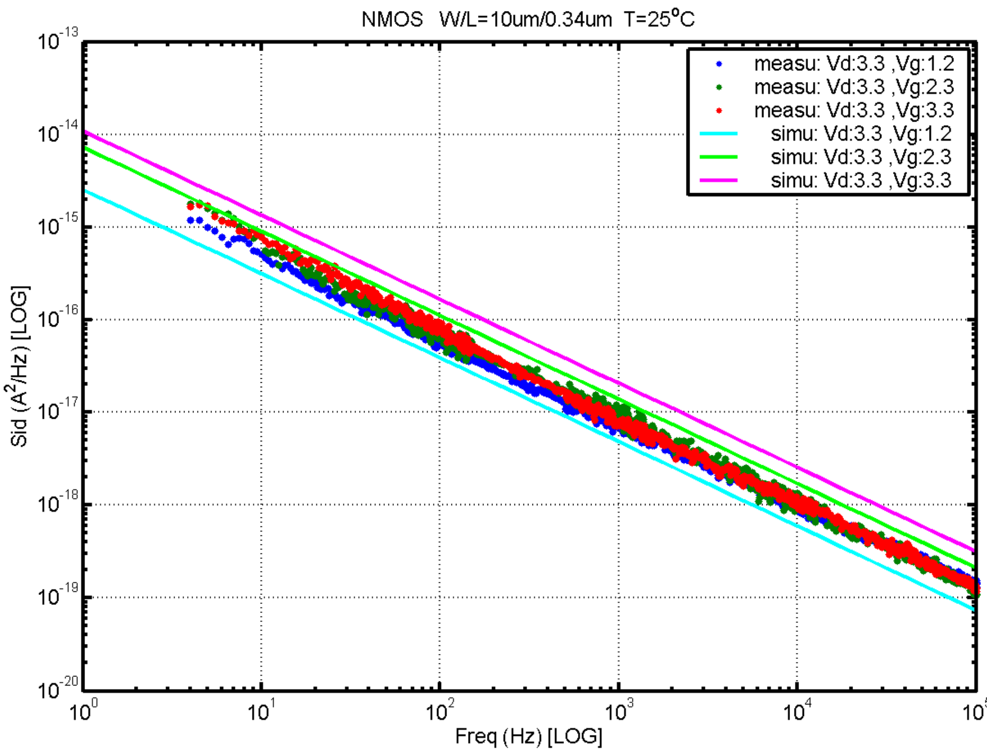

Figure 7-4 NMOS W/L=10μm/0.34μm @ Vd=3.3V

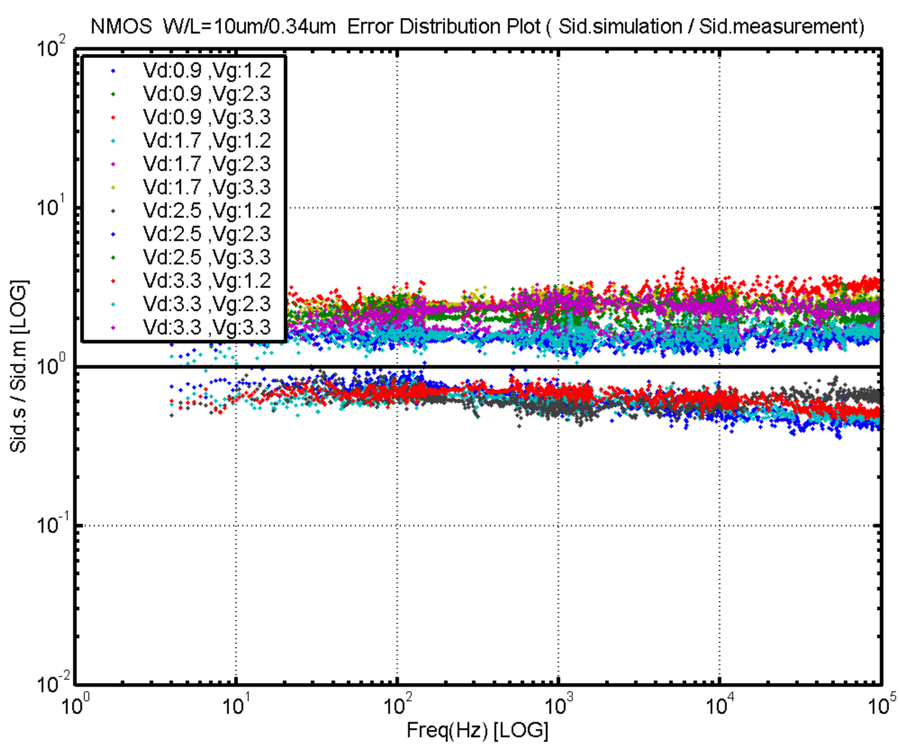

Figure 7-5 NMOS W/L=10μm/0.34μm Error Distribution Plot

NMOS, W=10µm L=0.5µm

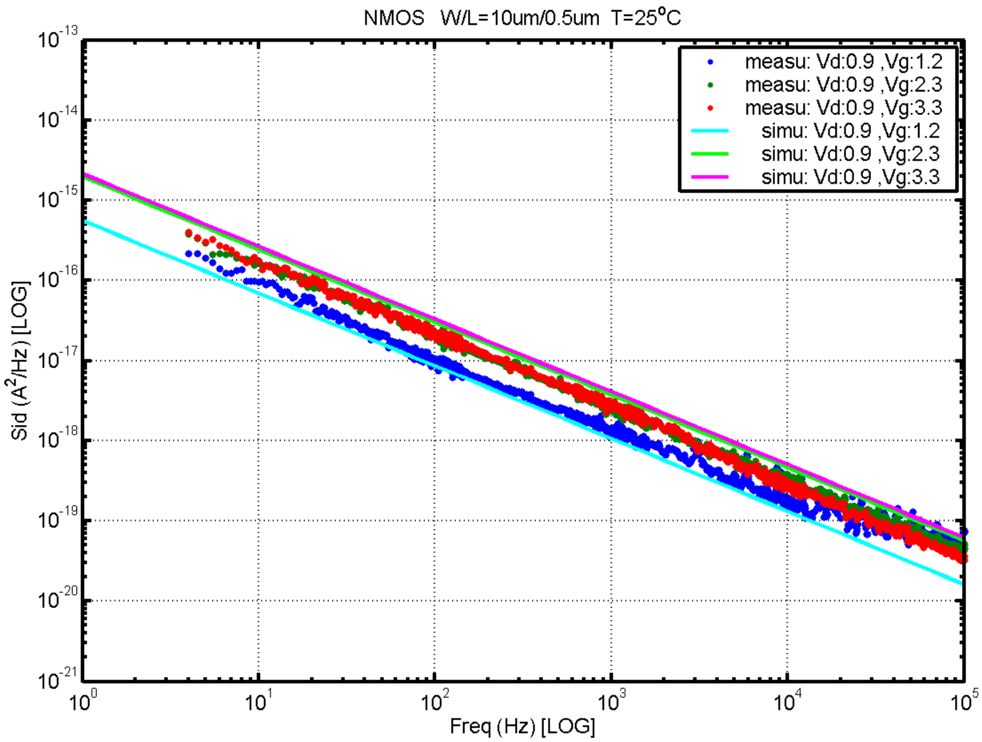

Figure 7-6 NMOS W/L=10µm/0.5µm @ Vd=0.9V

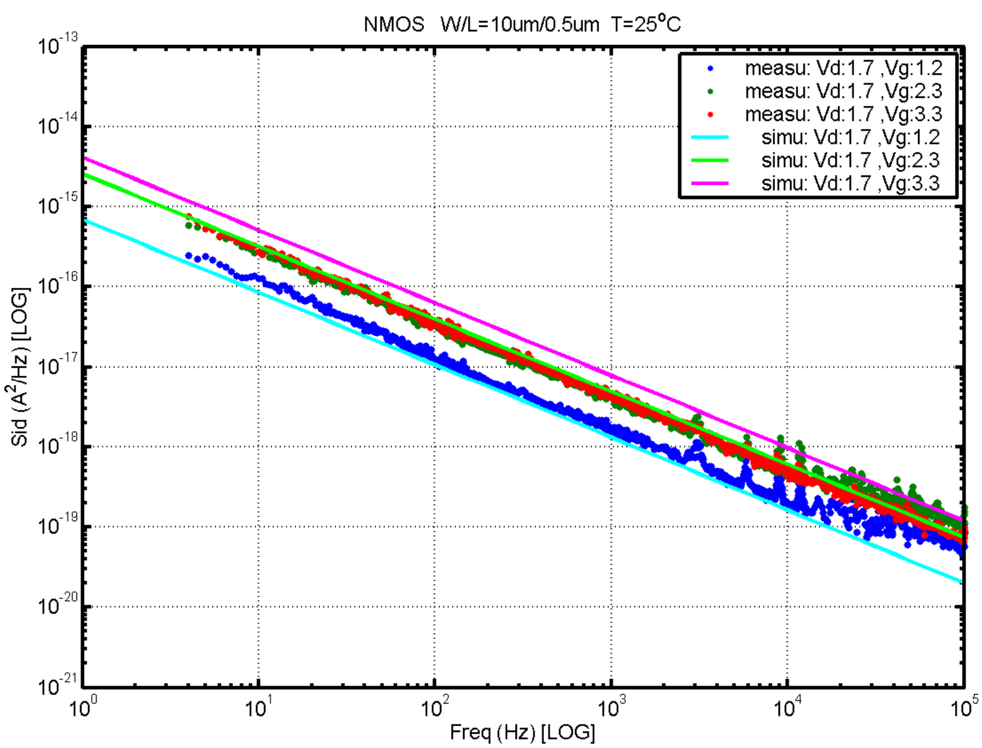

Figure 7-7 NMOS W/L=10µm/0.5µm @ Vd=1.7V

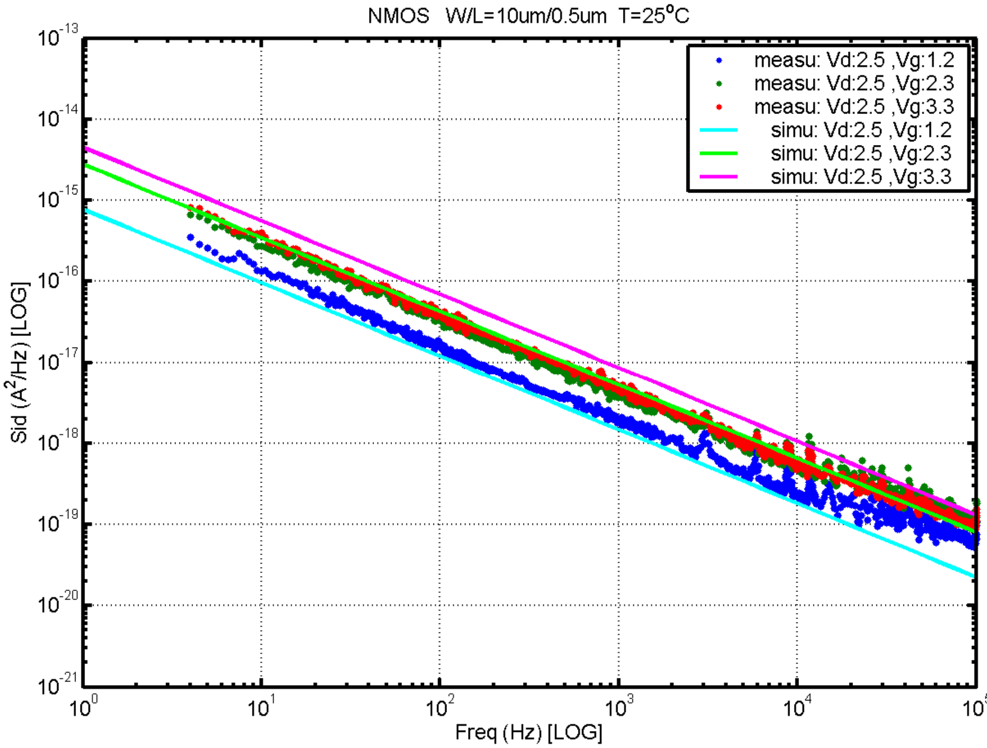

Figure 7-8 NMOS W/L=10μm/0.5μm @ Vd=2.5V

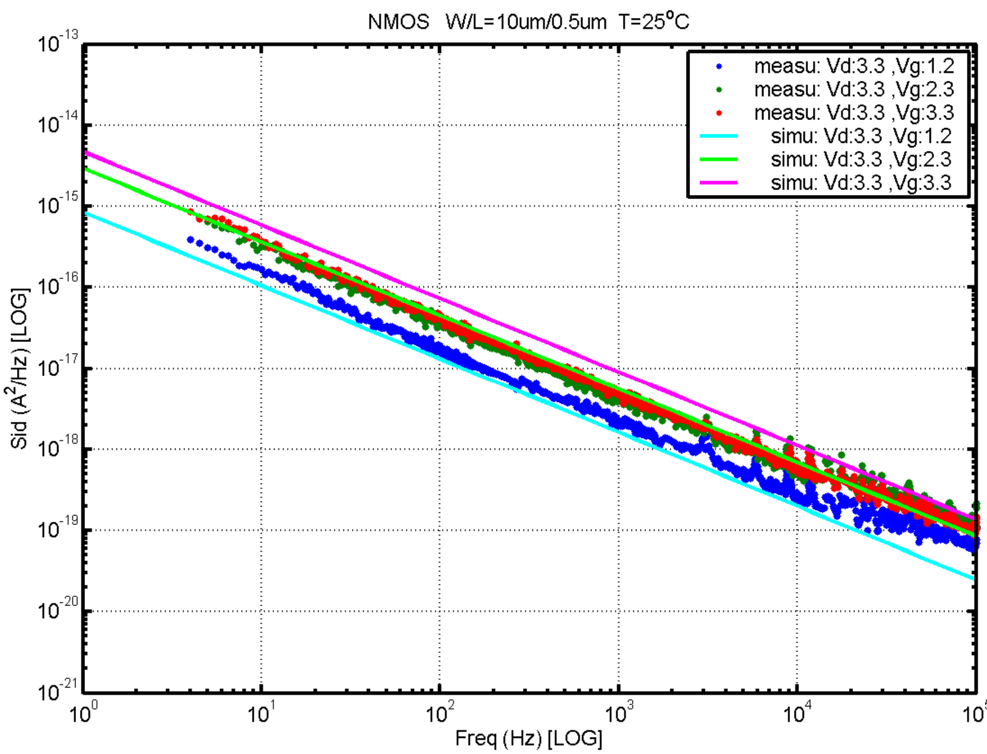

Figure 7-9 NMOS W/L=10μm/0.5μm @ Vd=3.3V

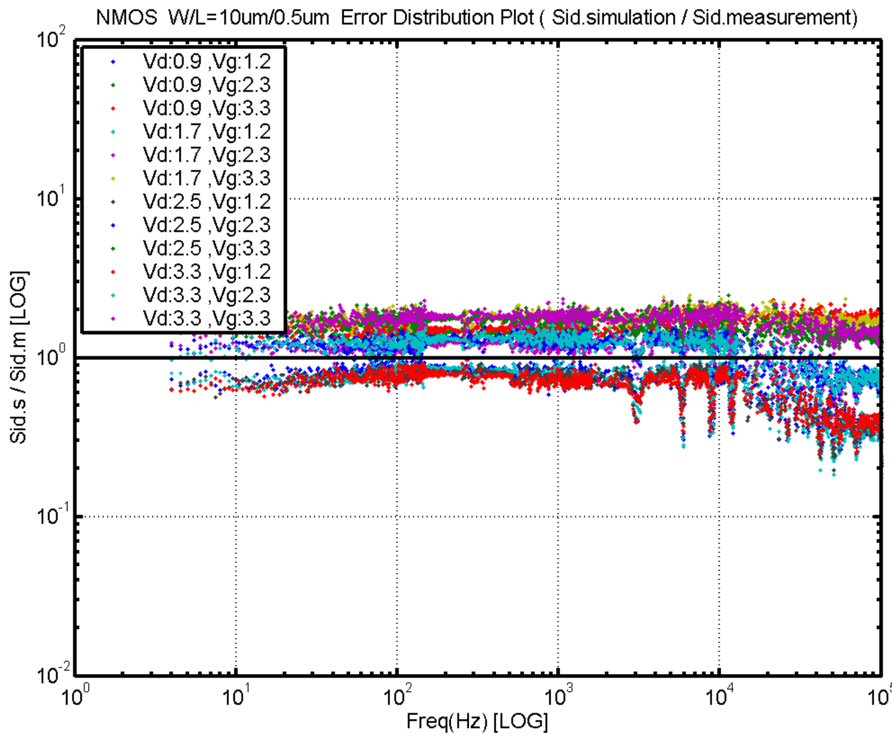

Figure 7-10 NMOS W/L=10μm/0.5μm Error Distribution Plot

NMOS, W=10μm L=1μm

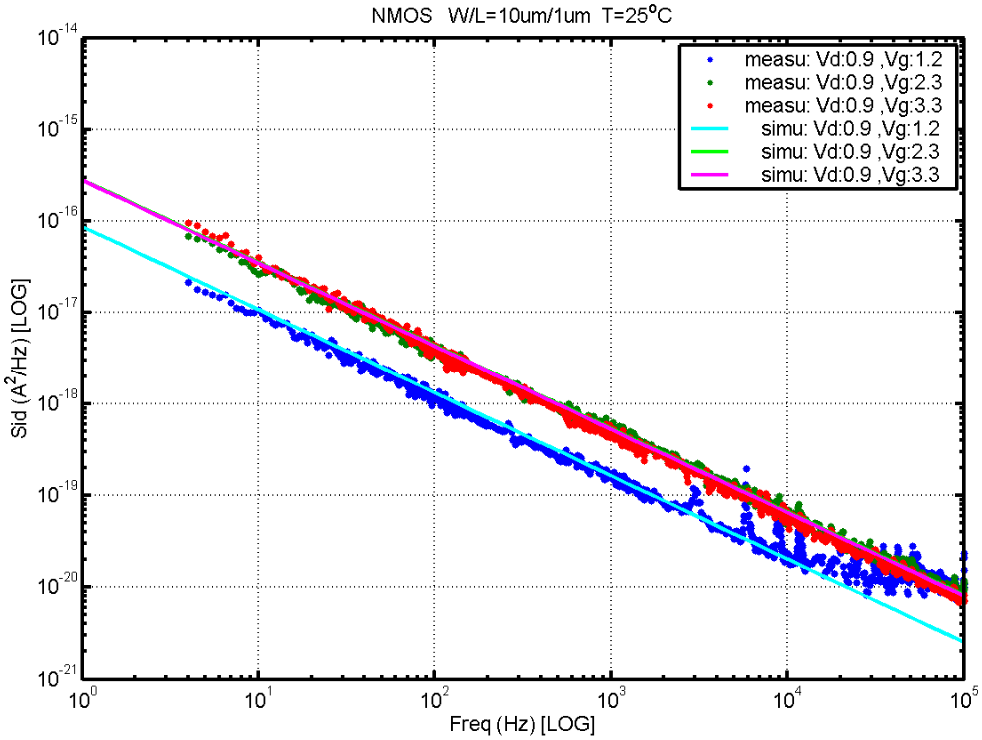

Figure 7-11 NMOS W/L=10μm/1μm @ Vd=0.9V

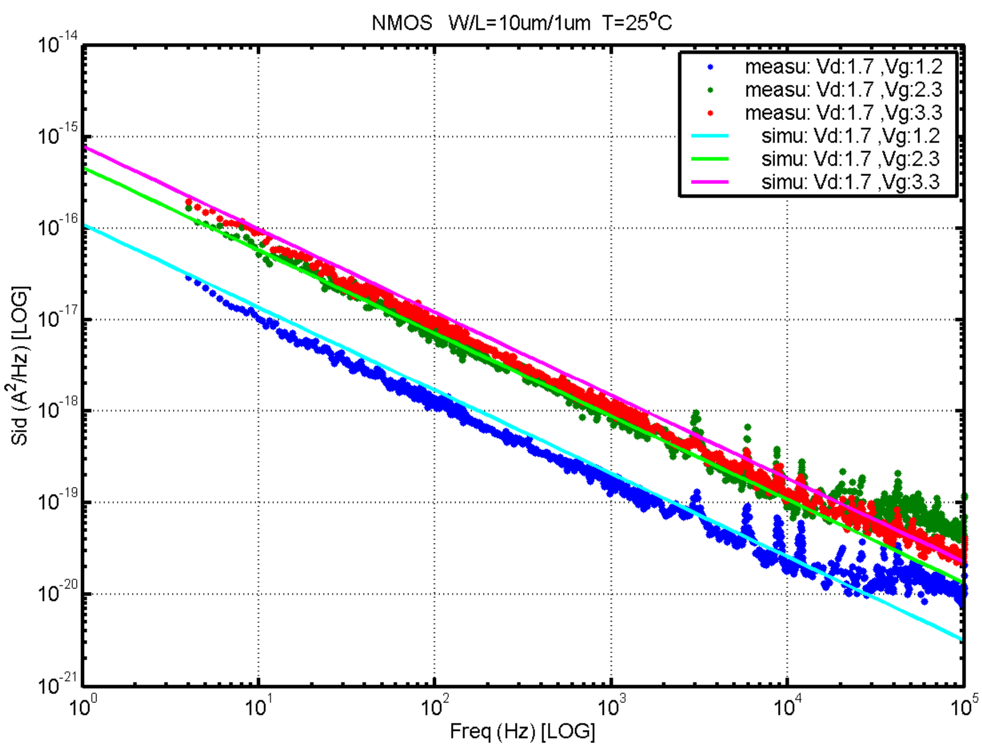

Figure 7-12 NMOS W/L=10μm/1μm @ Vd=1.7V

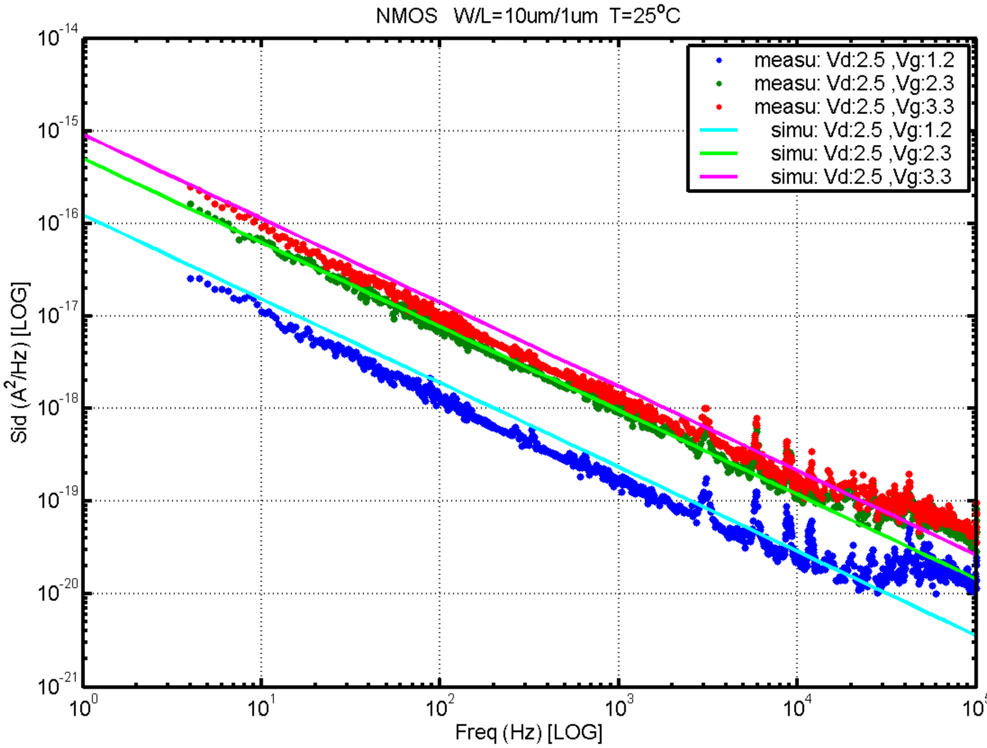

Figure 7-13 NMOS W/L=10μm/1μm @ Vd=2.5V

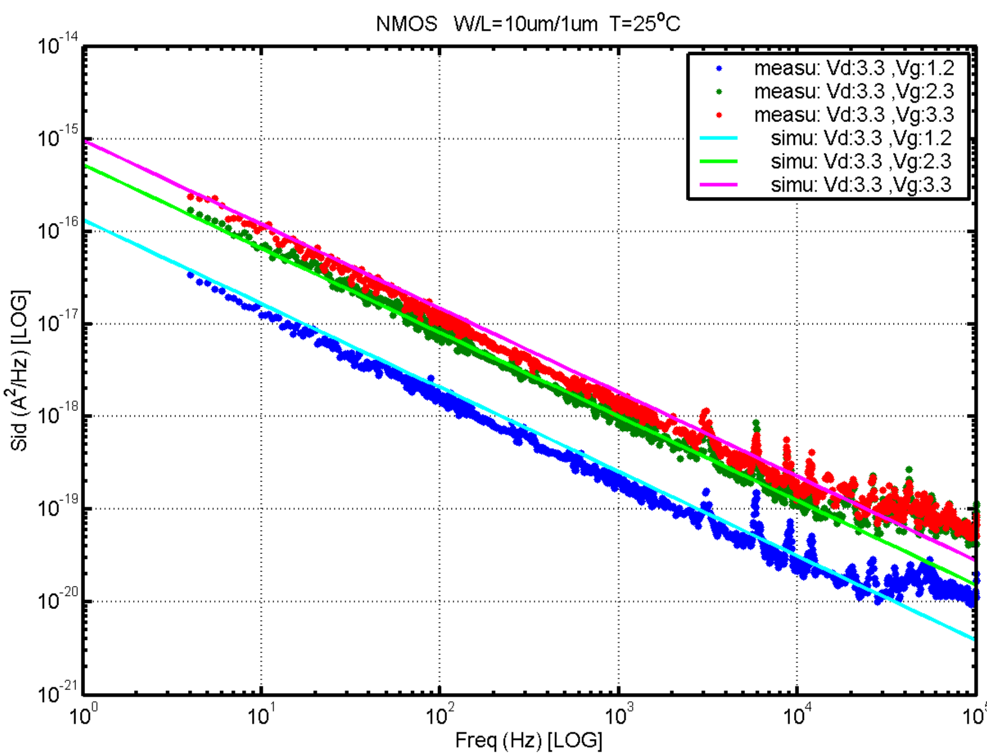

Figure 7-14 NMOS W/L=10μm/1μm @ Vd=3.3V

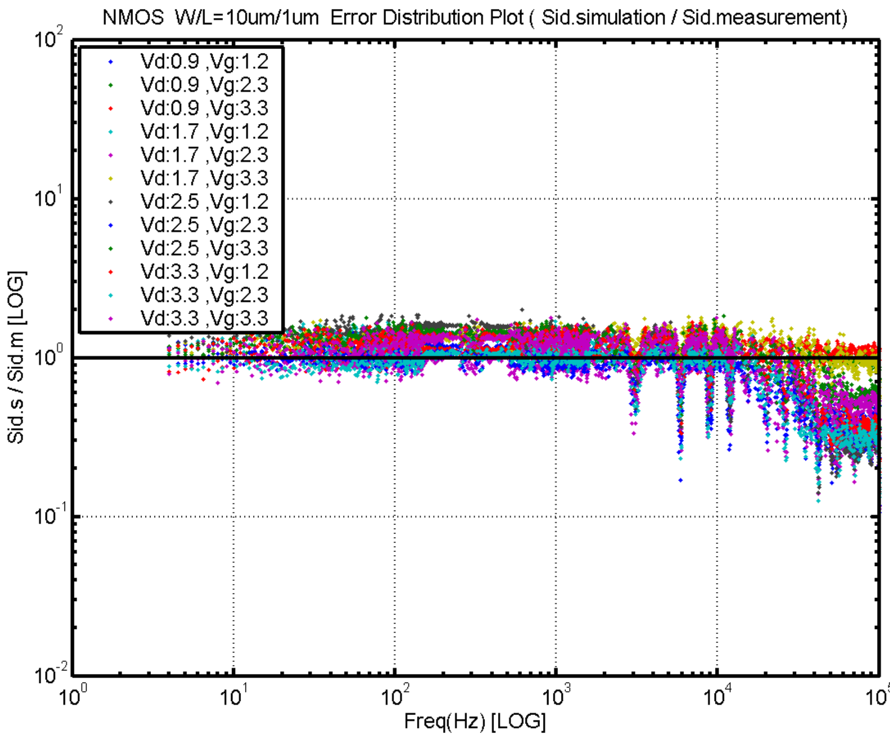

Figure 7-15 NMOS W/L=10µm/1µm Error Distribution Plot

NMOS, W=10μm L=10μm

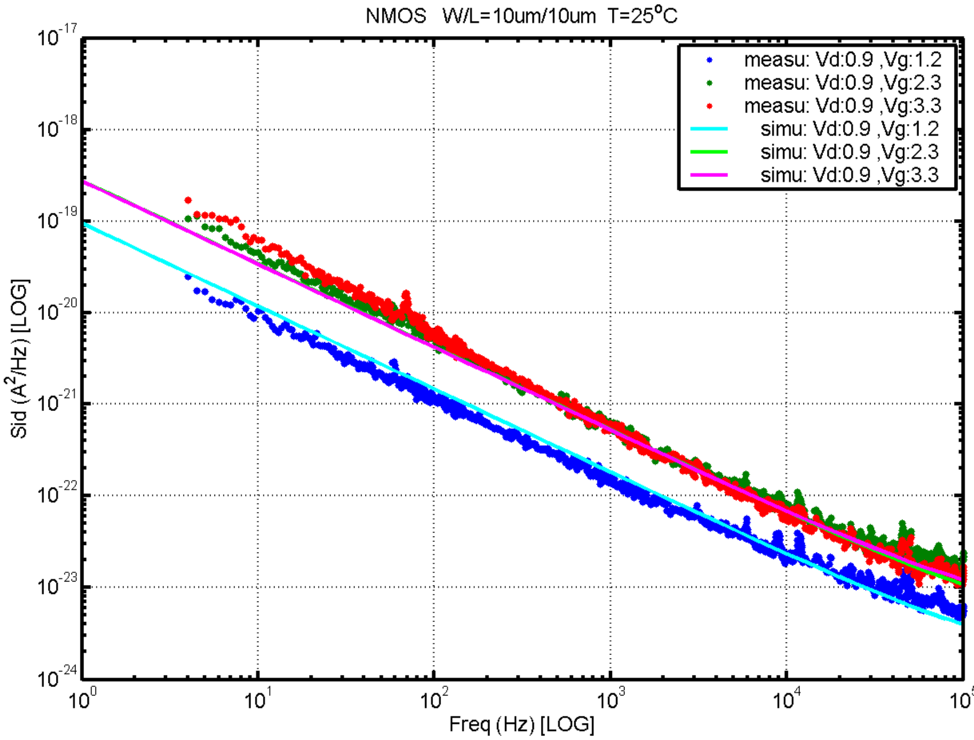

Figure 7-16 NMOS W/L=10μm/10μm @ Vd=0.9V

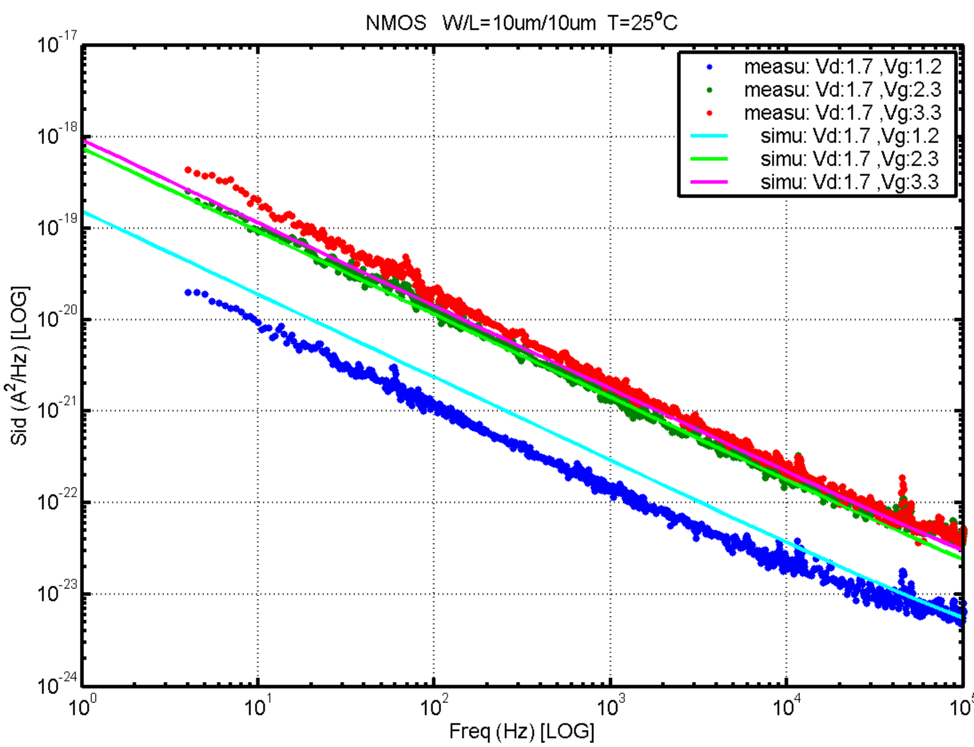

Figure 7-17 NMOS W/L=10μm/10μm @ Vd=1.7V

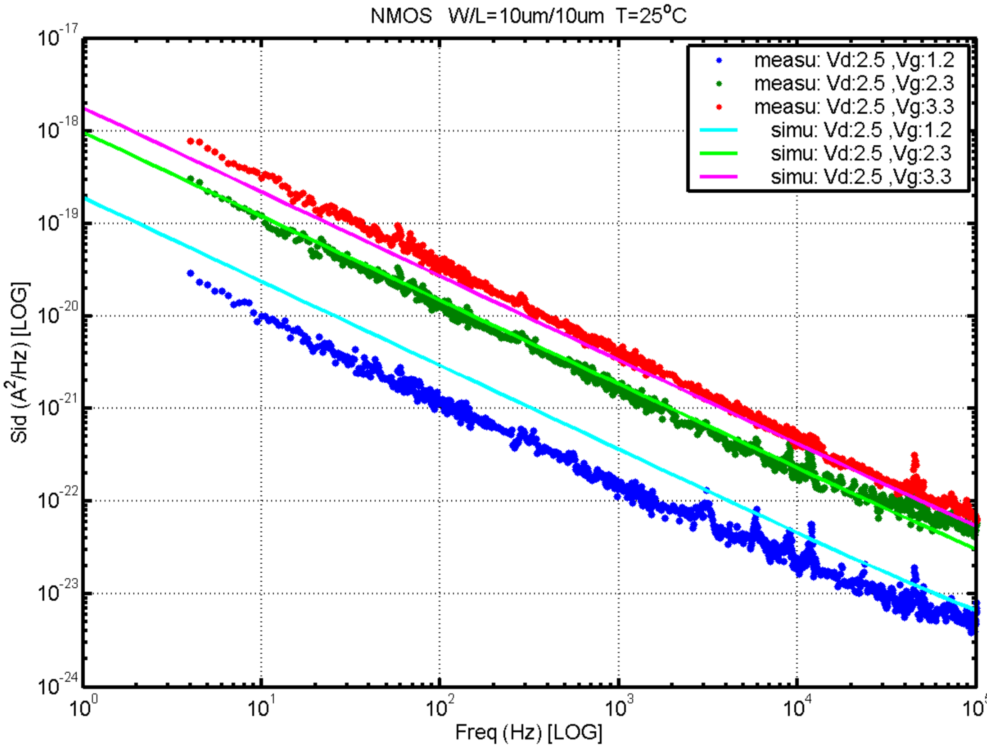

Figure 7-18 NMOS W/L=10µm/10µm @ Vd=2.5V

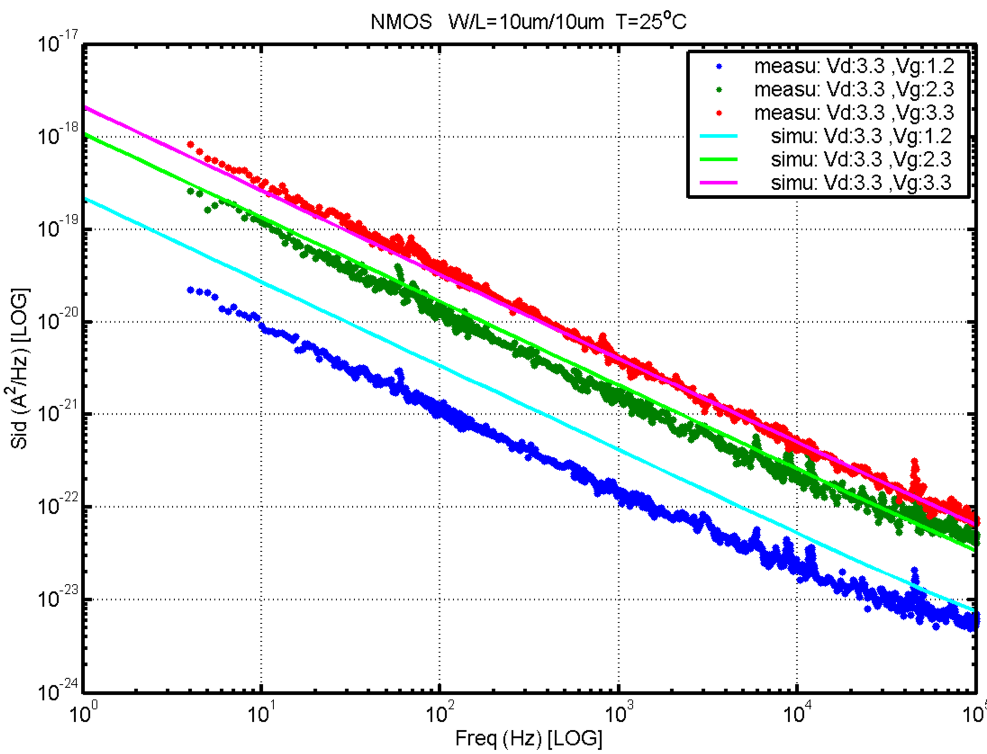

Figure 7-19 NMOS W/L=10µm/10µm @ Vd=3.3V

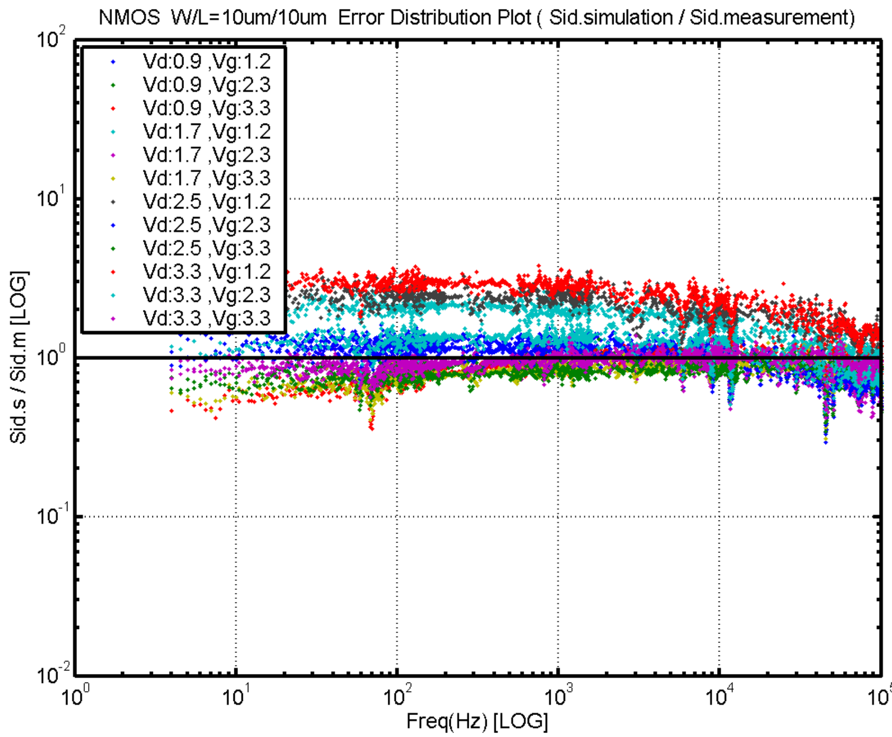

Figure 7-20 NMOS W/L=10μm/10μm Error Distribution Plot

PMOS, W=10μm L=0.34μm

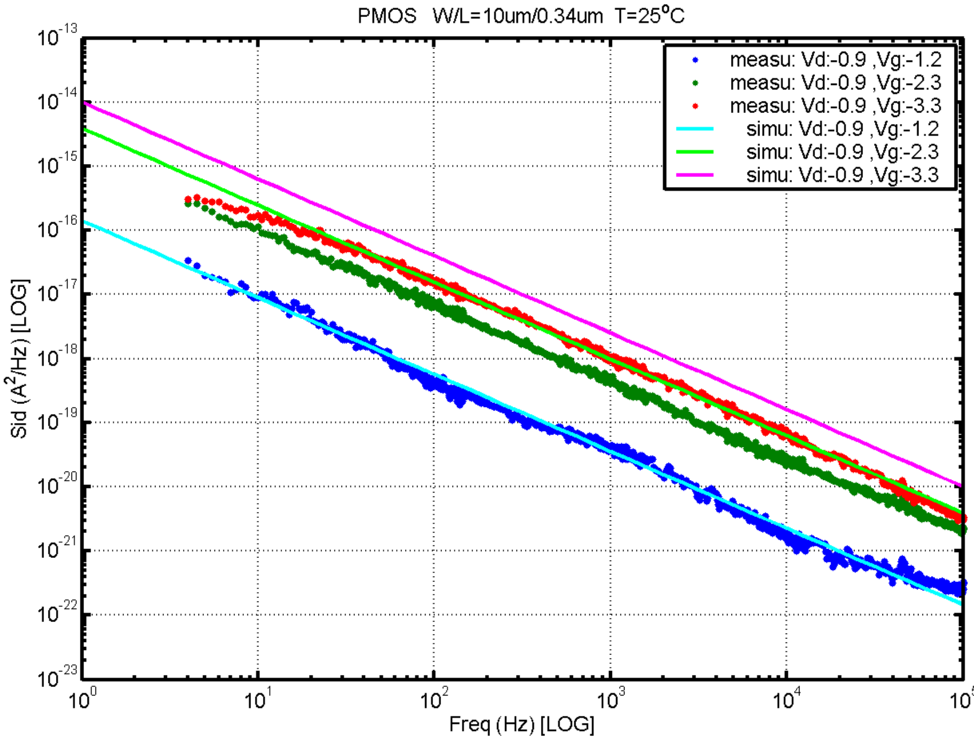

Figure 7-21 PMOS W/L=10μm/0.34μm @ Vd= -0.9V

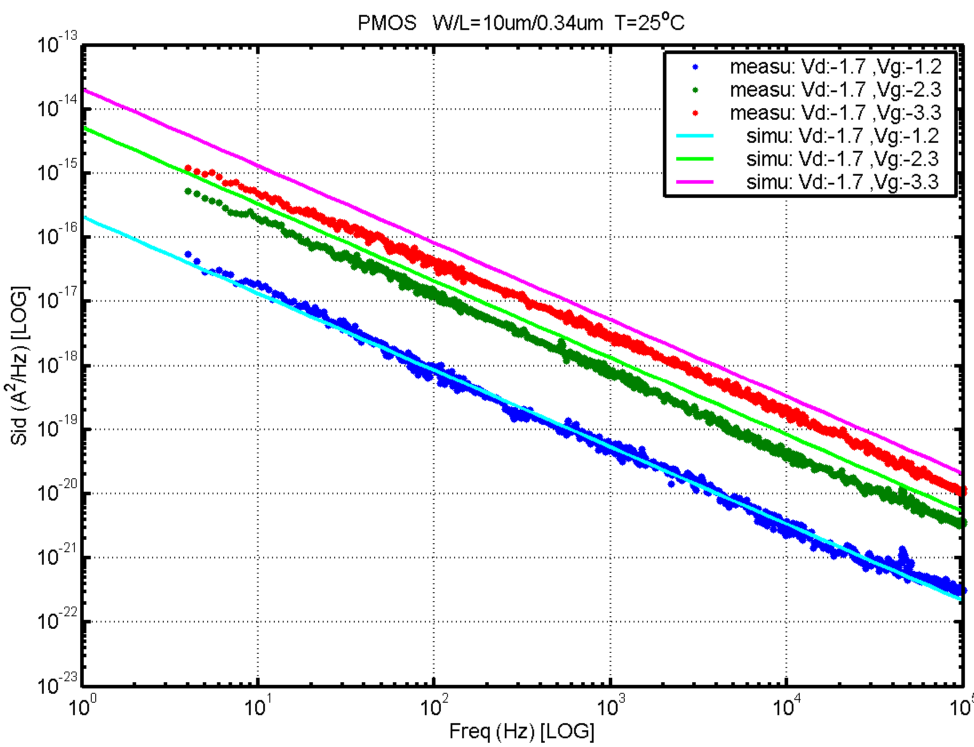

Figure 7-22 PMOS W/L=10μm/0.34μm @ Vd= -1.7V

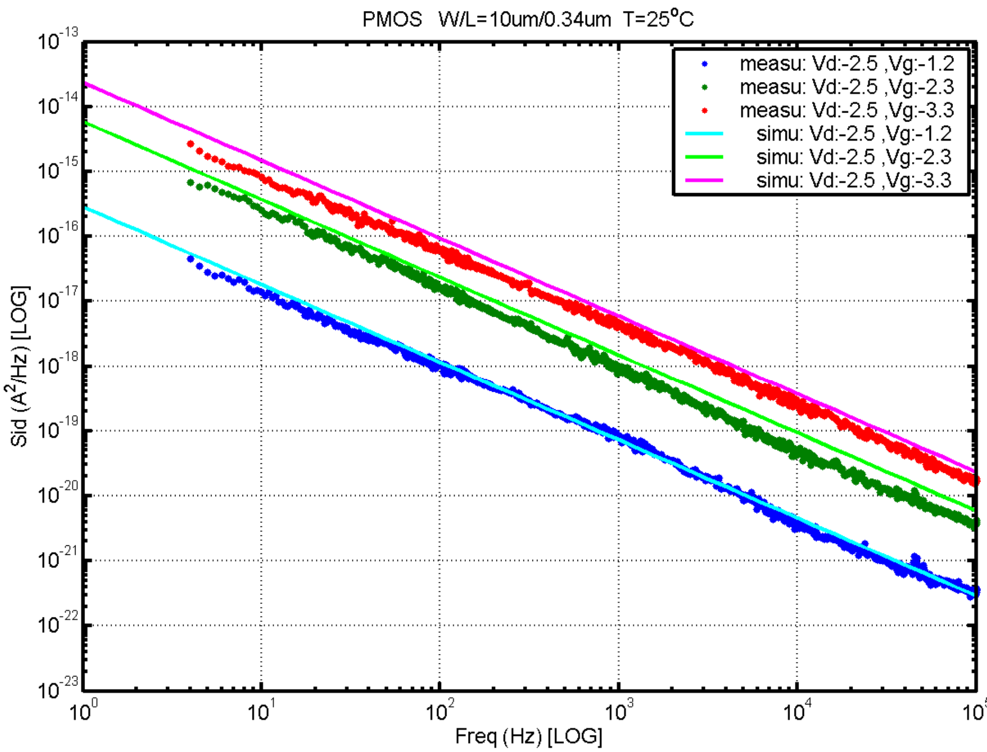

Figure 7-23 PMOS W/L=10µm/0.4µm @ Vd= -2.5V

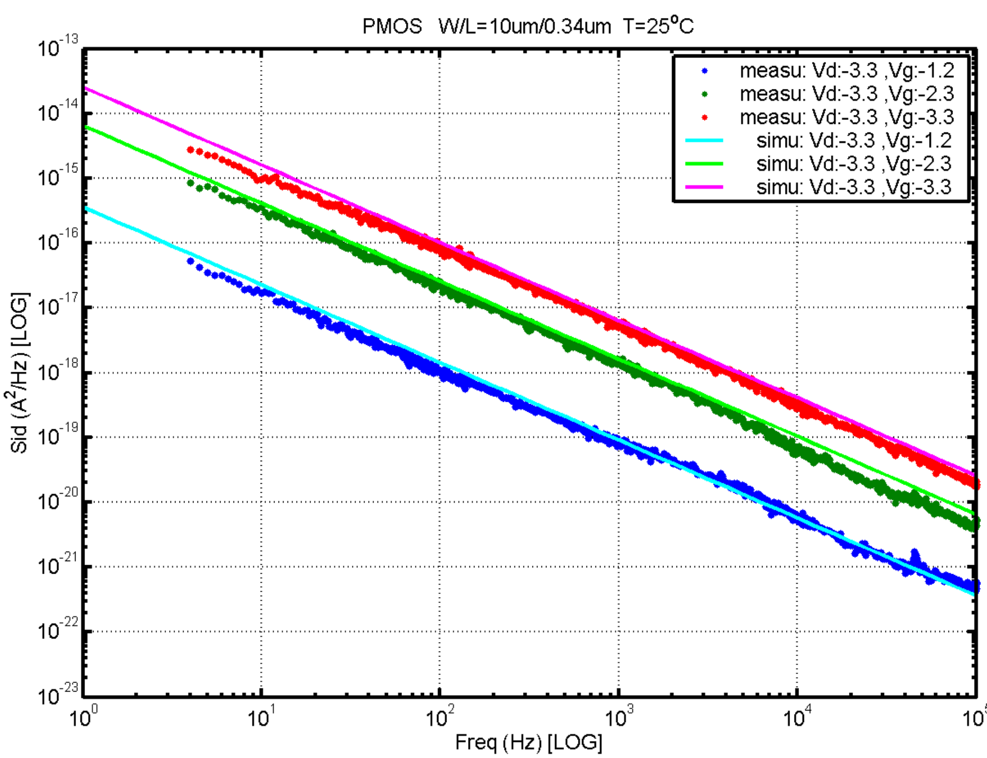

Figure 7-24 PMOS W/L=10µm/0.4µm @ Vd= -3.3V

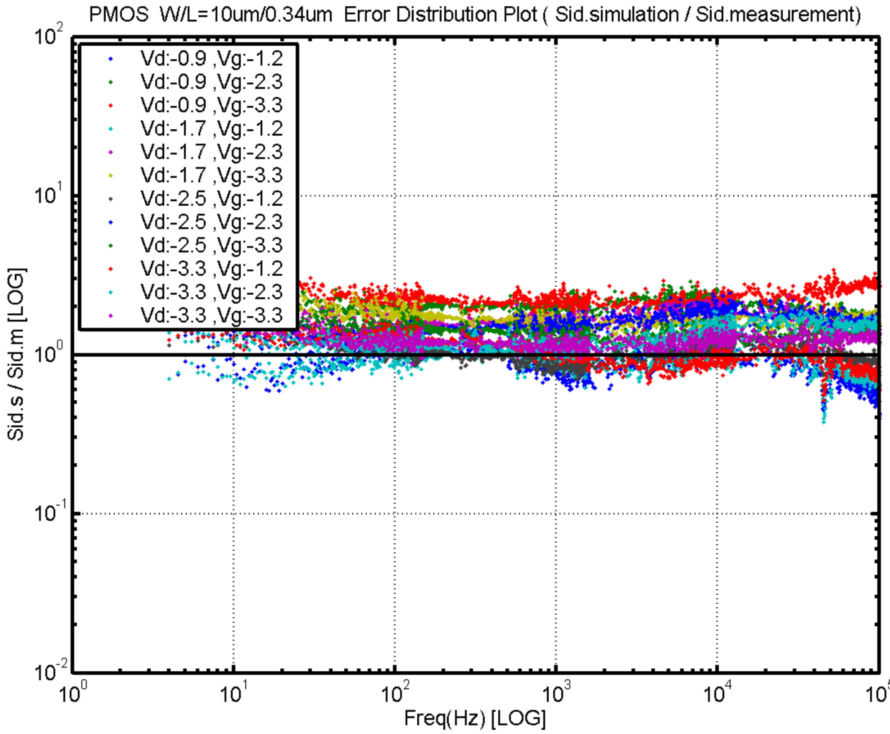

Figure 7-25 PMOS W/L=10μm/0.34μm Error Distribution Plot

PMOS, W=10µm L=0.5µm

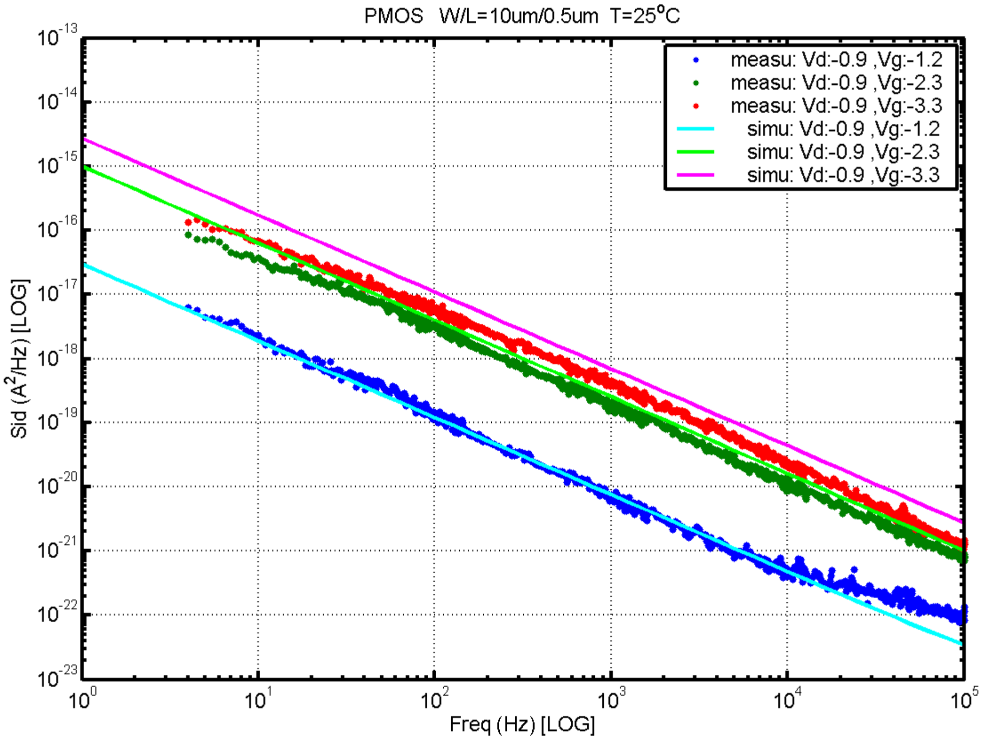

Figure 7-26 PMOS W/L=10µm/0.5µm @ Vd= -0.9V

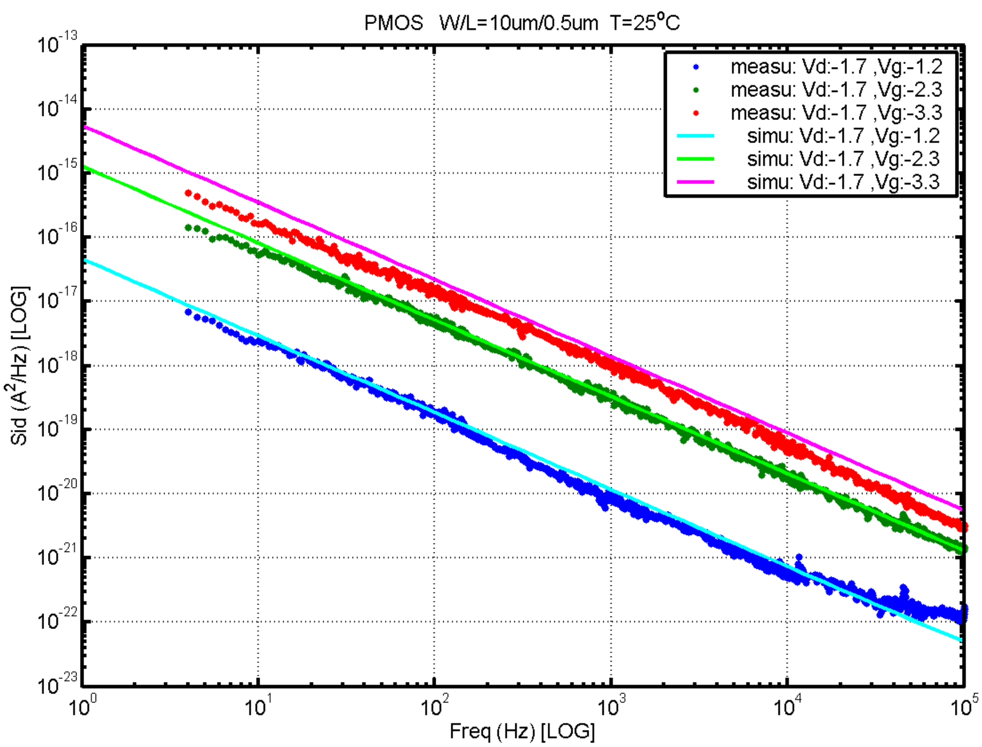

Figure 7-27 PMOS W/L=10µm/0.5µm @ Vd= -1.7V

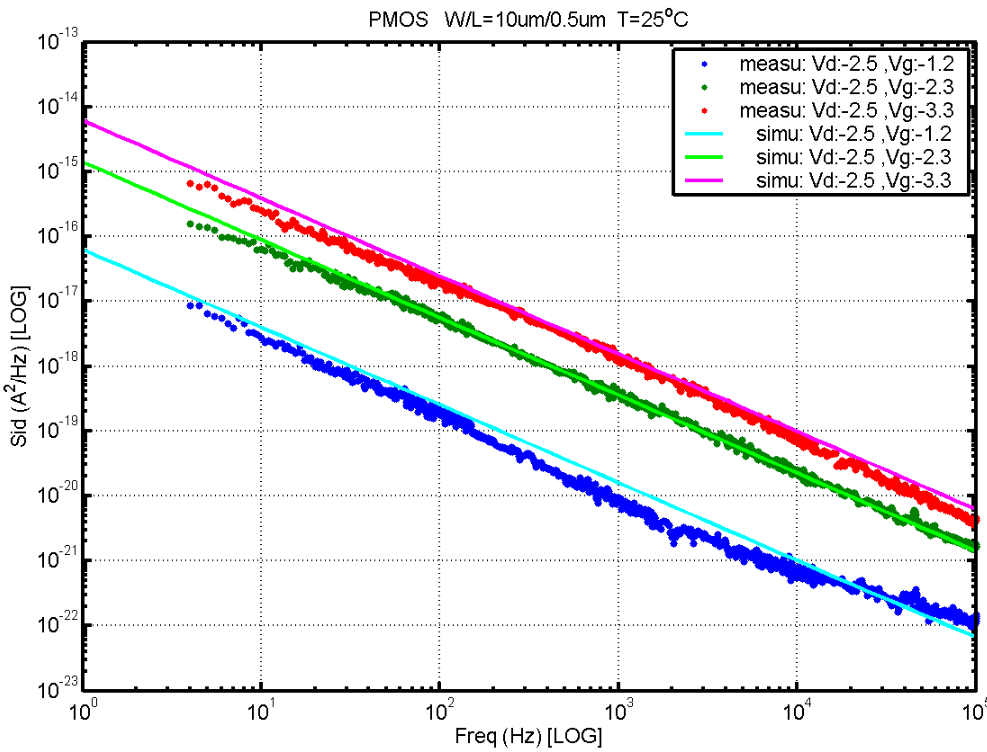

Figure 7-28 PMOS W/L=10µm/0.5µm @ Vd= -2.5V

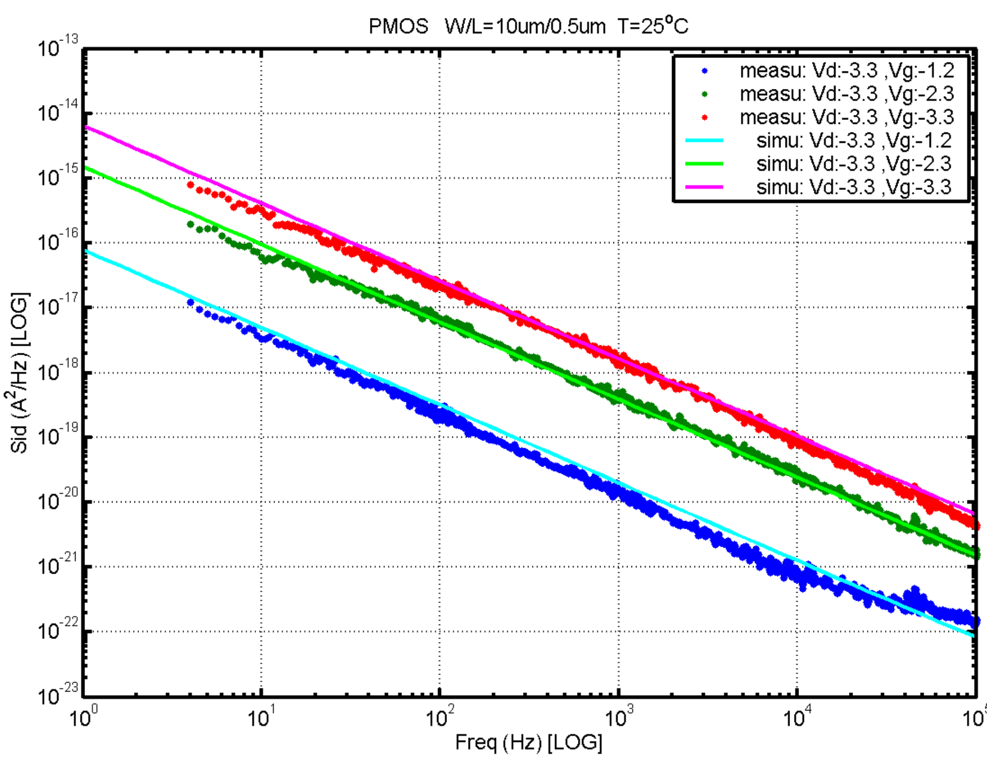

Figure 7-29 PMOS W/L=10µm/0.5µm @ Vd= -3.3V

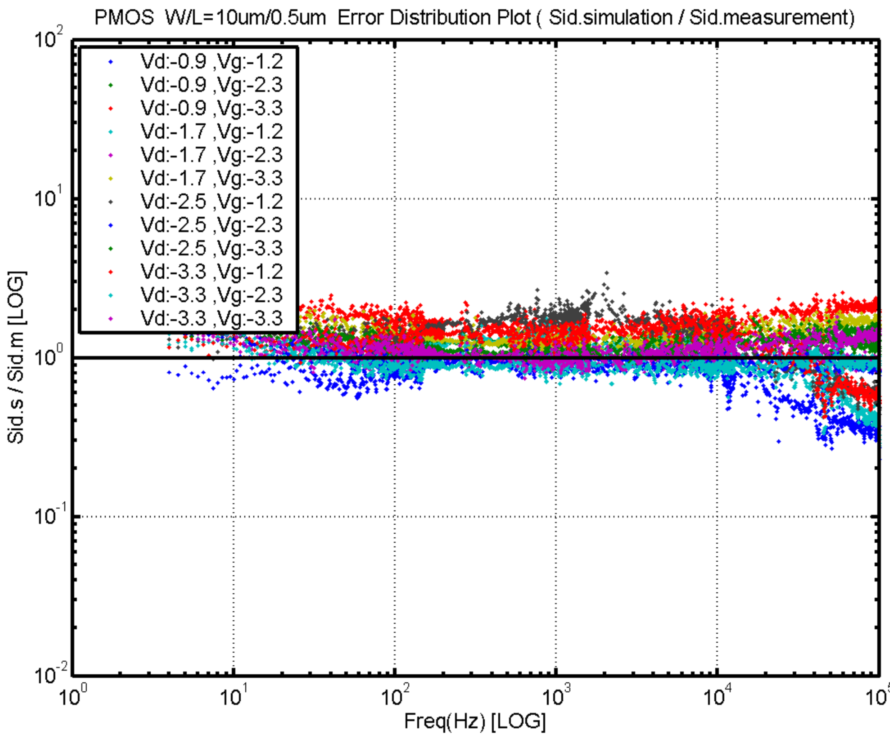

Figure 7-30 PMOS W/L=10µm/0.5µm Error Distribution Plot

PMOS, W=10μm L=1μm

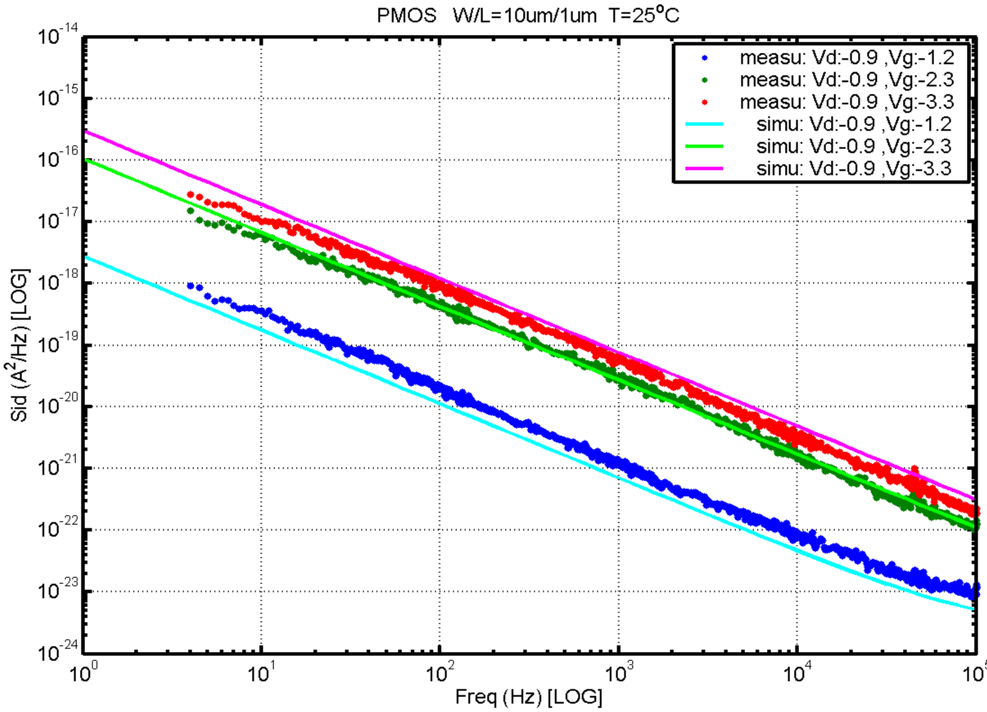

Figure 7-31 PMOS W/L=10μm/1μm @ Vd= -0.9V

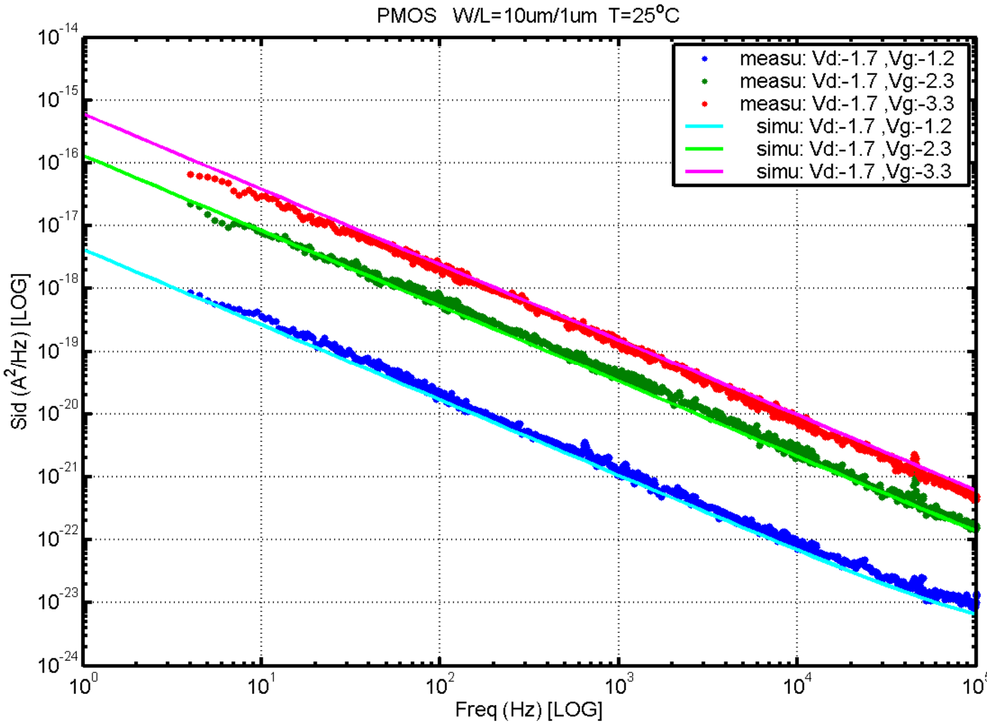

Figure 7-32 PMOS W/L=10μm/1μm @ Vd= -1.7V

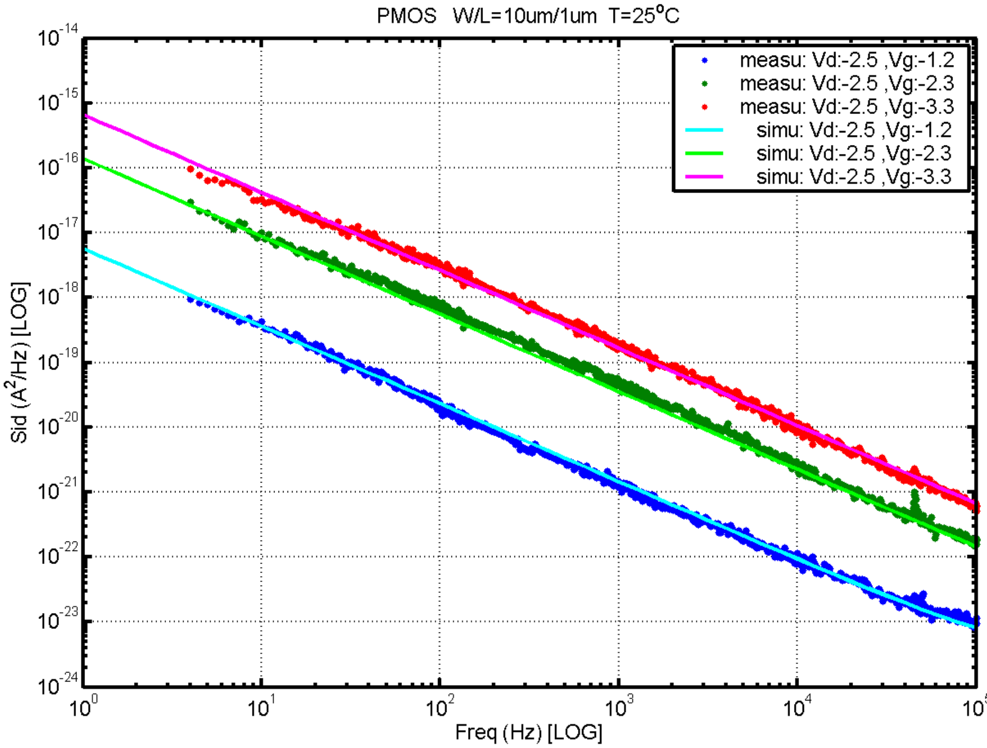

Figure 7-33 PMOS W/L=10µm/1µm @ Vd= -2.5V

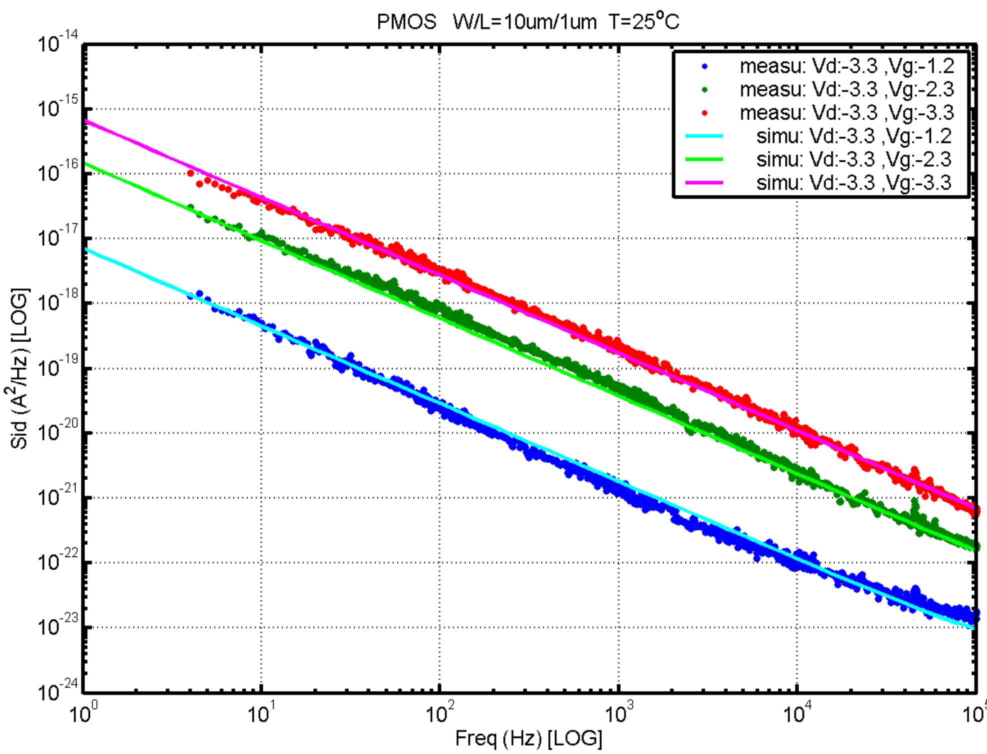

Figure 7-34 PMOS W/L=10µm/1µm @ Vd= -3.3V

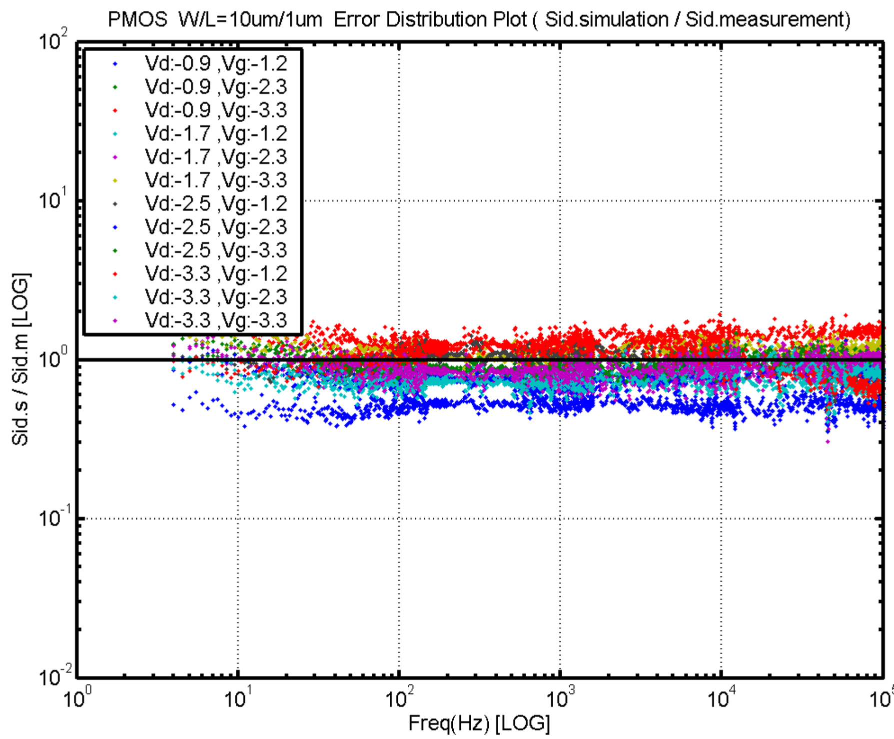

Figure 7-35 PMOS W/L=10µm/1µm Error Distribution Plot

PMOS, W=10µm L=10µm

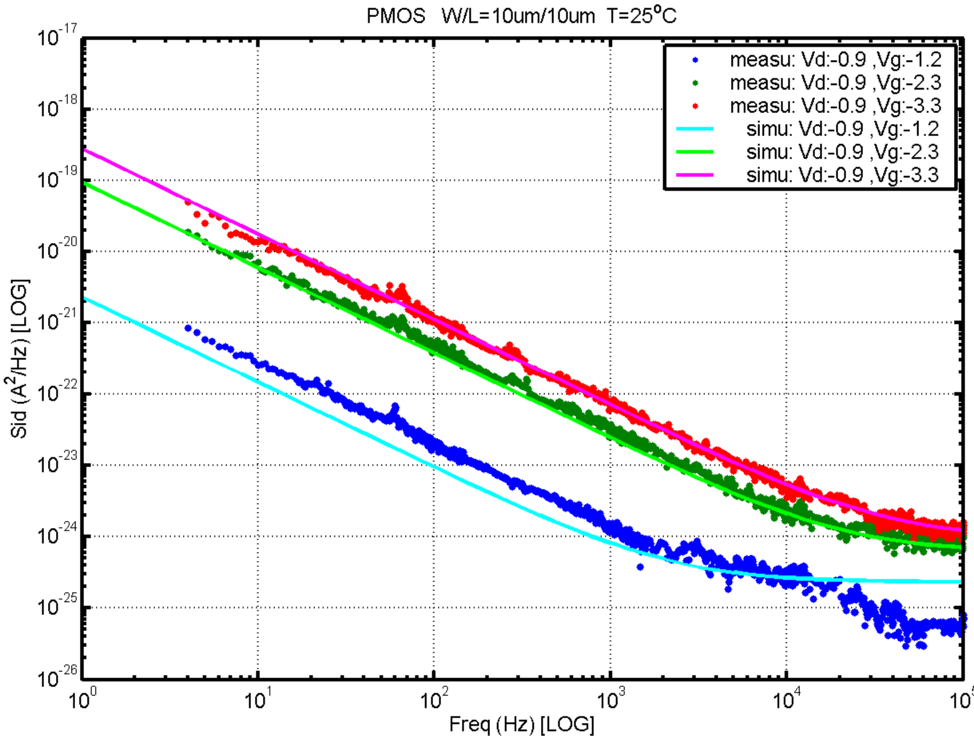

Figure 7-36 PMOS W/L=10µm/10µm @ Vd= -0.9V

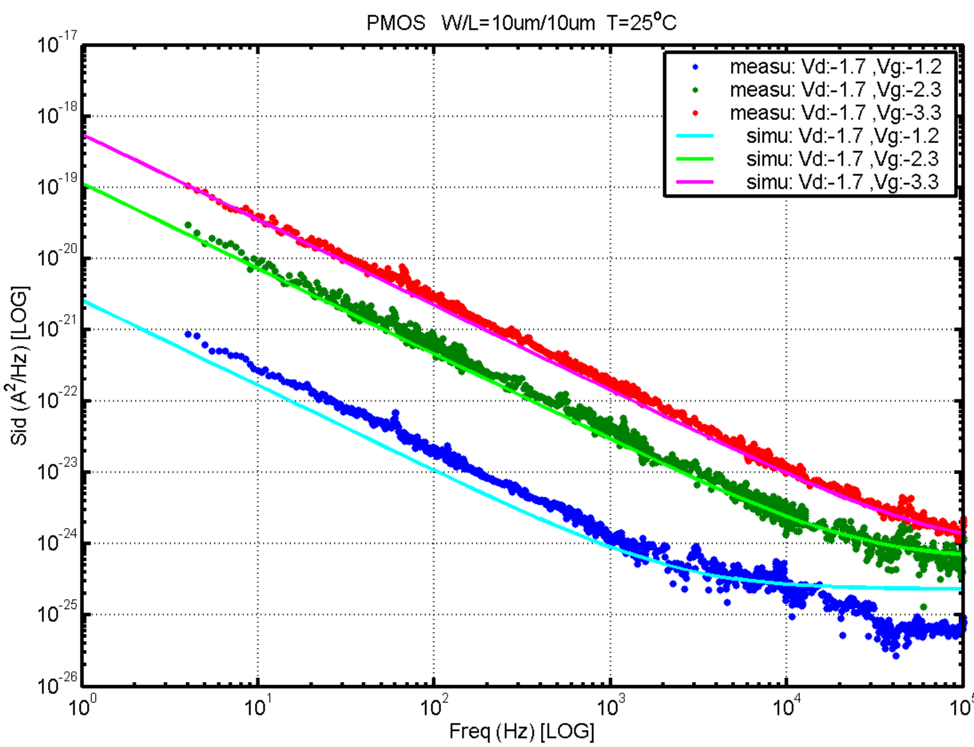

Figure 7-37 PMOS W/L=10µm/10µm @ Vd= -1.7V

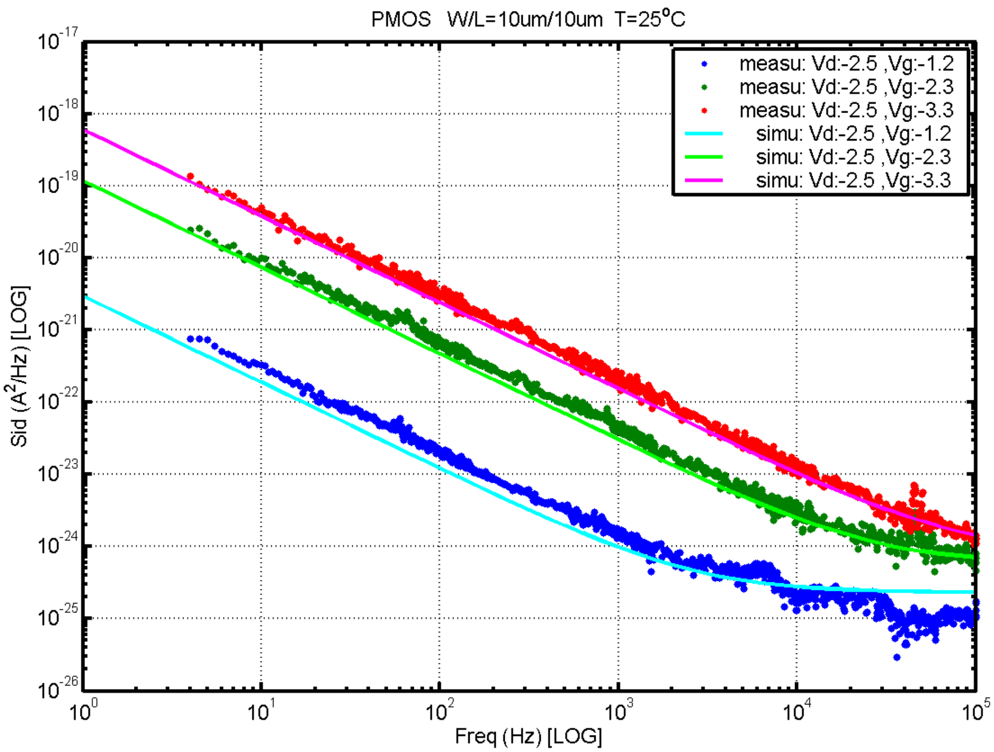

Figure 7-38 PMOS W/L=10µm/10µm @ Vd= -2.5V

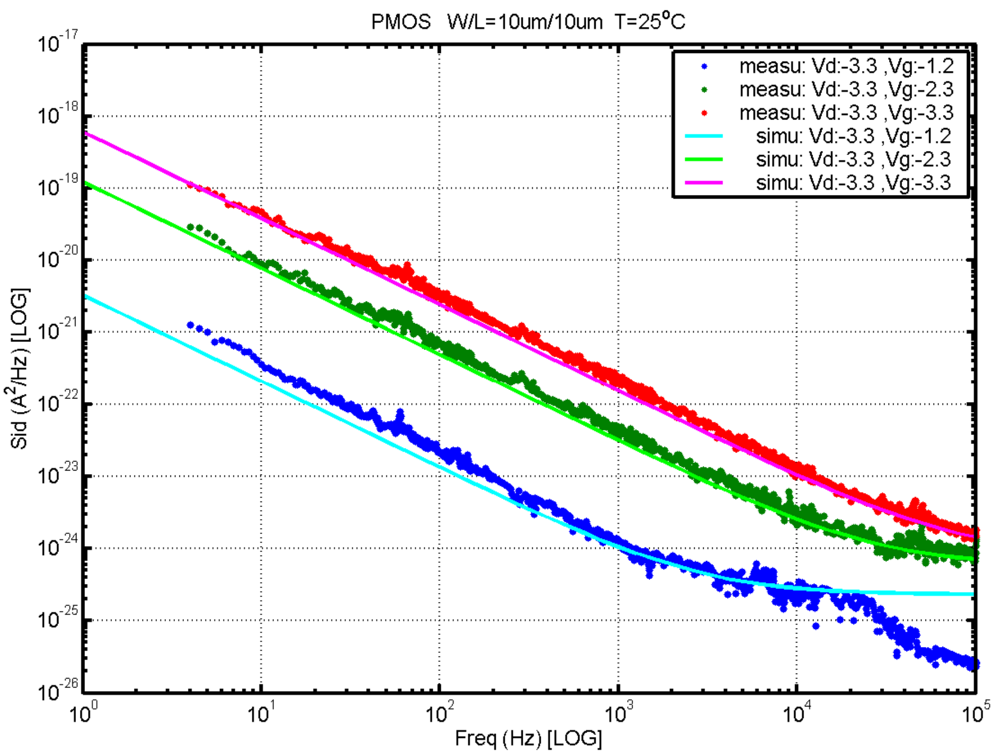

Figure 7-39 PMOS W/L=10µm/10µm @ Vd= -3.3V

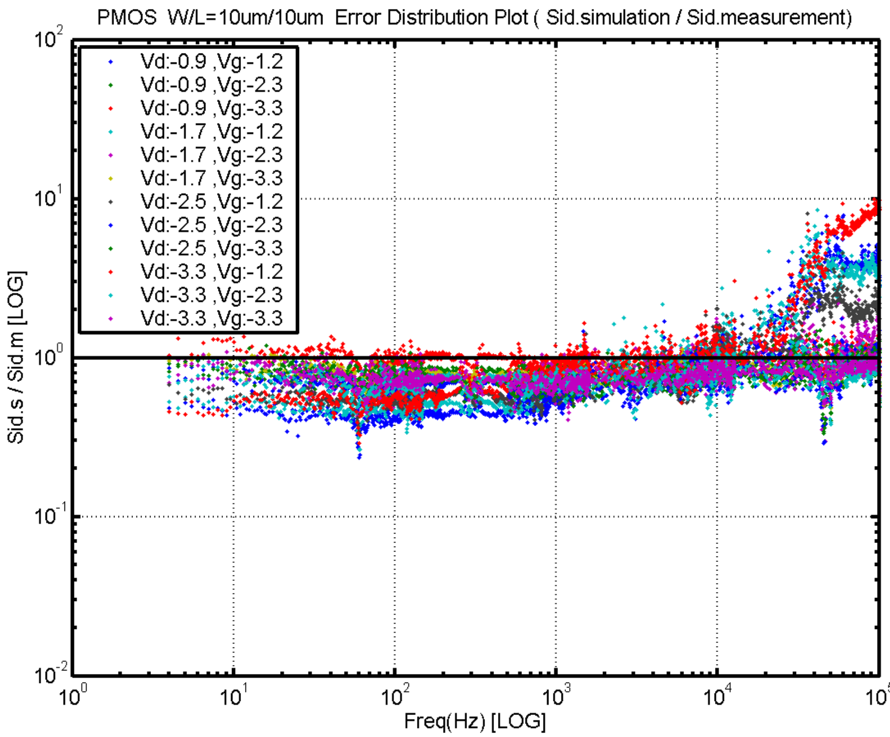

Figure 7-40 PMOS W/L=10µm/10µm Error Distribution Plot

7.2 Trend plots at frequency=100Hz

A. NMOS

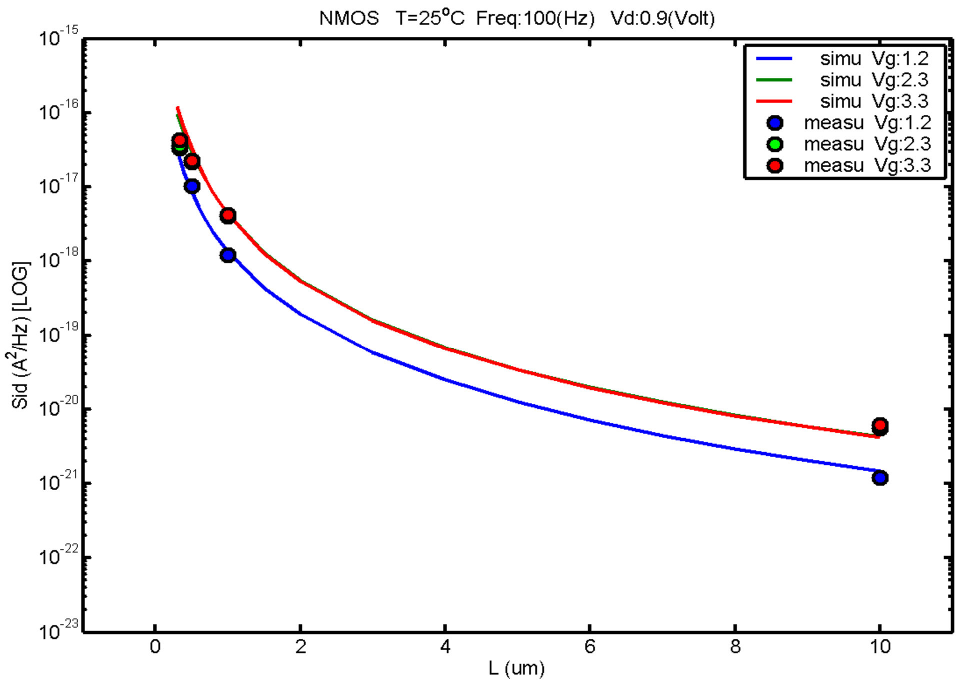

Figure 7-41 Noise vs Length for NMOS @ vd=0.9V

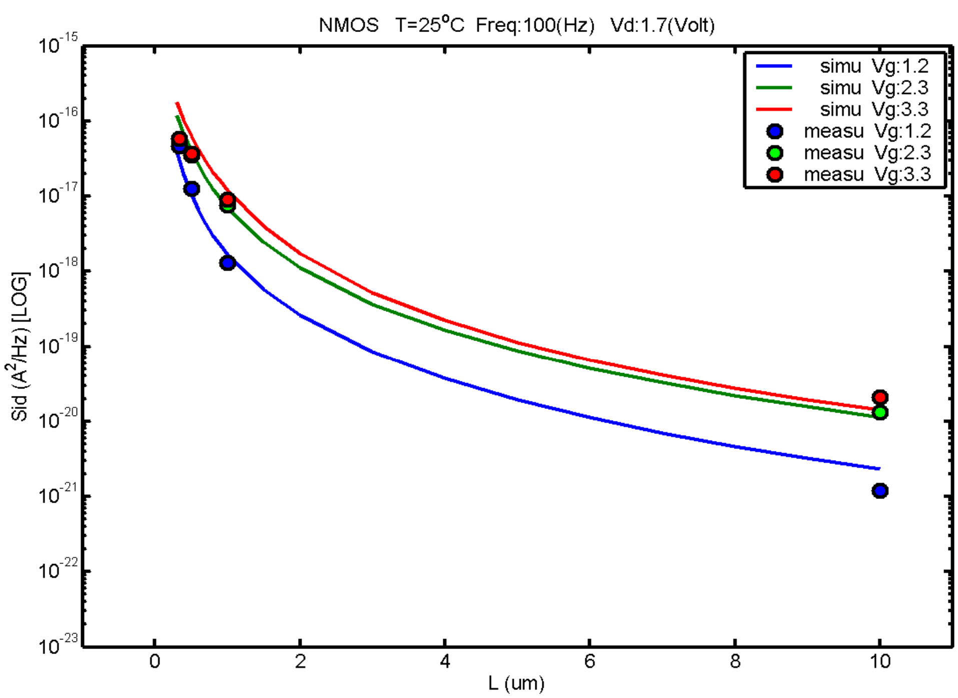

Figure 7-42 Noise vs Length for NMOS @ vd=1.7V

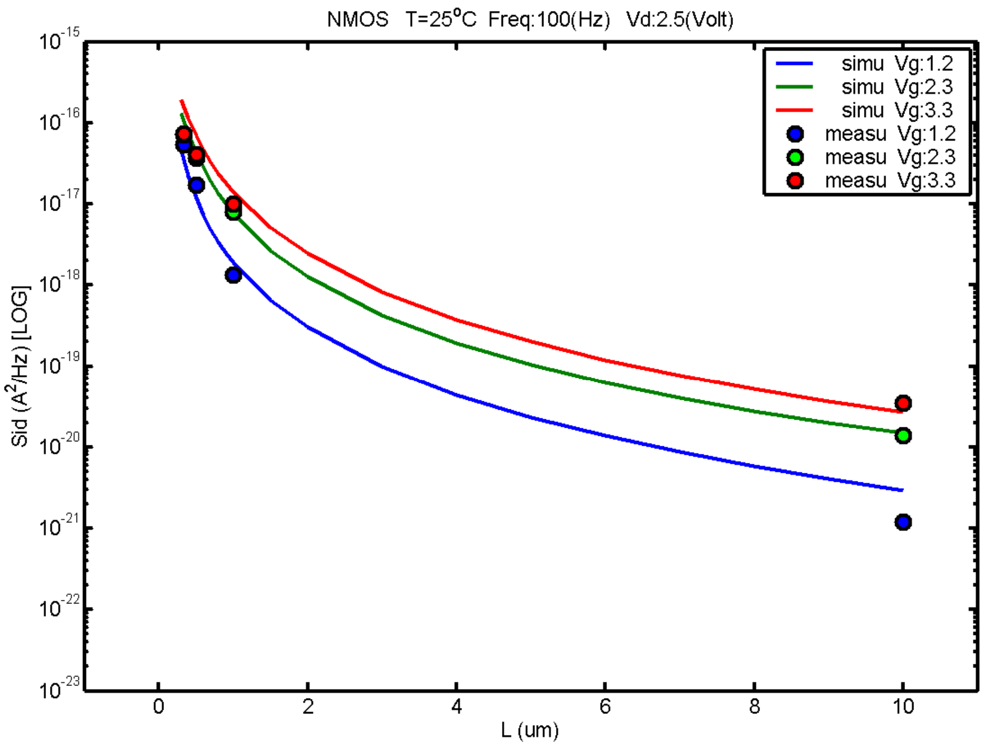

Figure 7-43 Noise vs Length for NMOS @ vd=2.5V

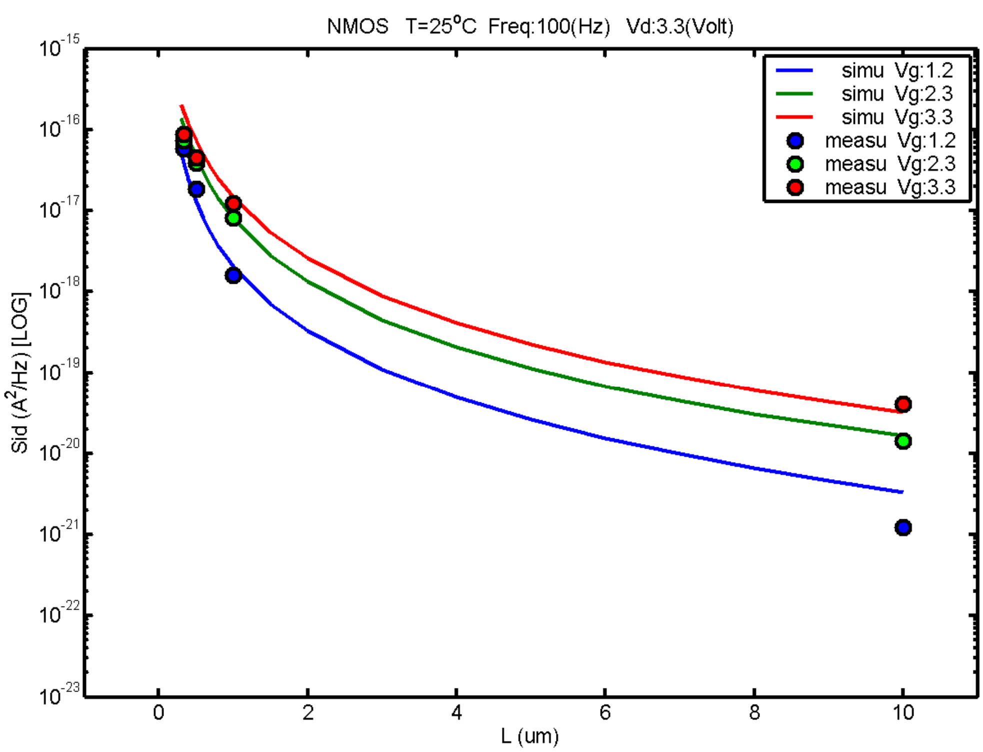

Figure 7-44 Noise vs Length for NMOS @ vd=3.3V

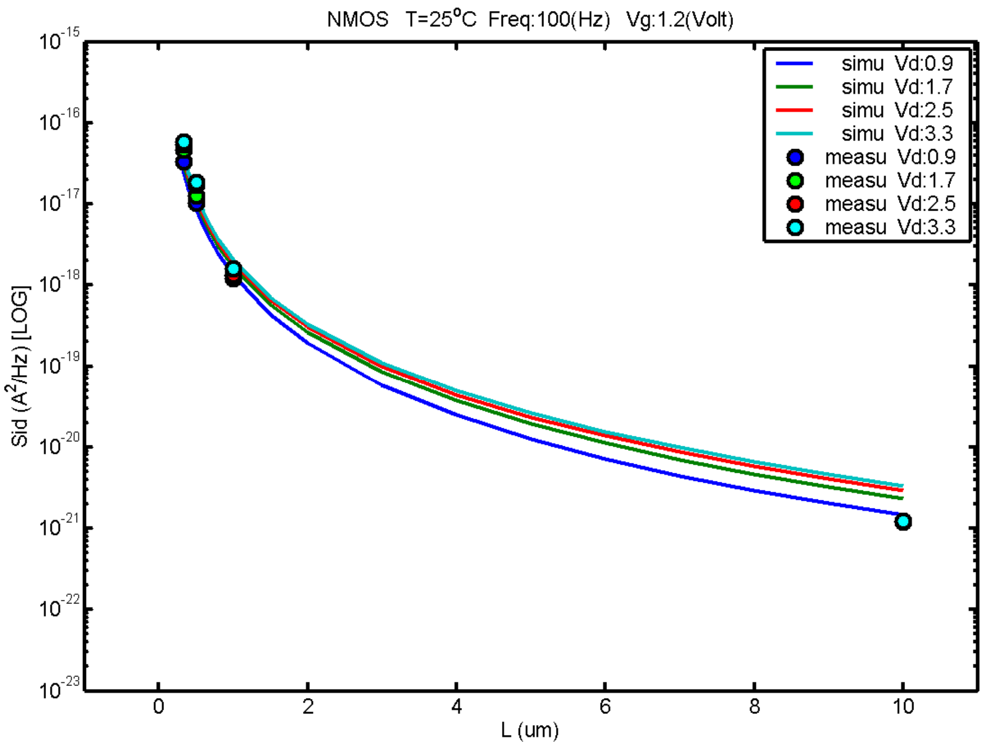

Figure 7-45 Noise vs Length for NMOS @ vg=1.2V

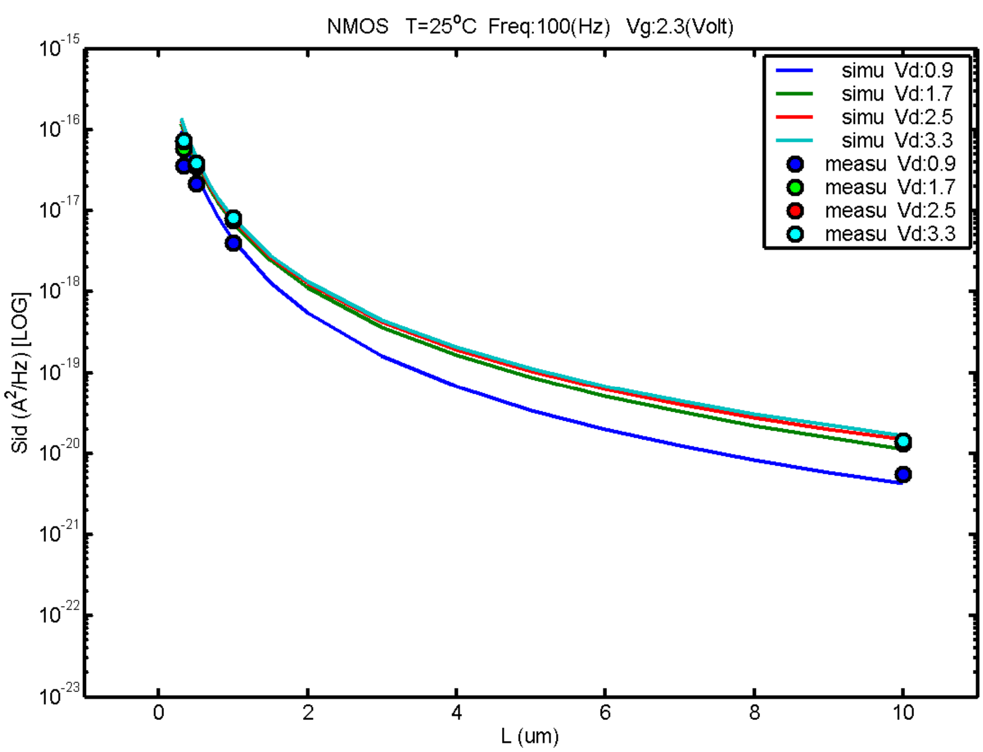

Figure 7-46 Noise vs Length for NMOS @ vg=2.3V

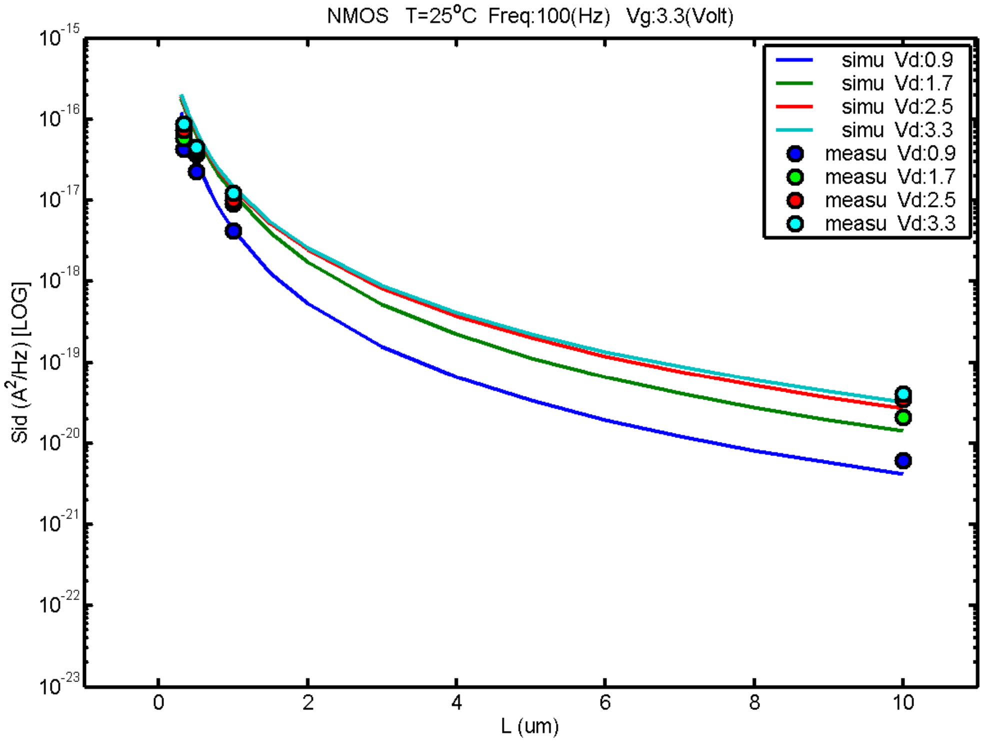

Figure 7-47 Noise vs Length for NMOS @ vg=3.3V

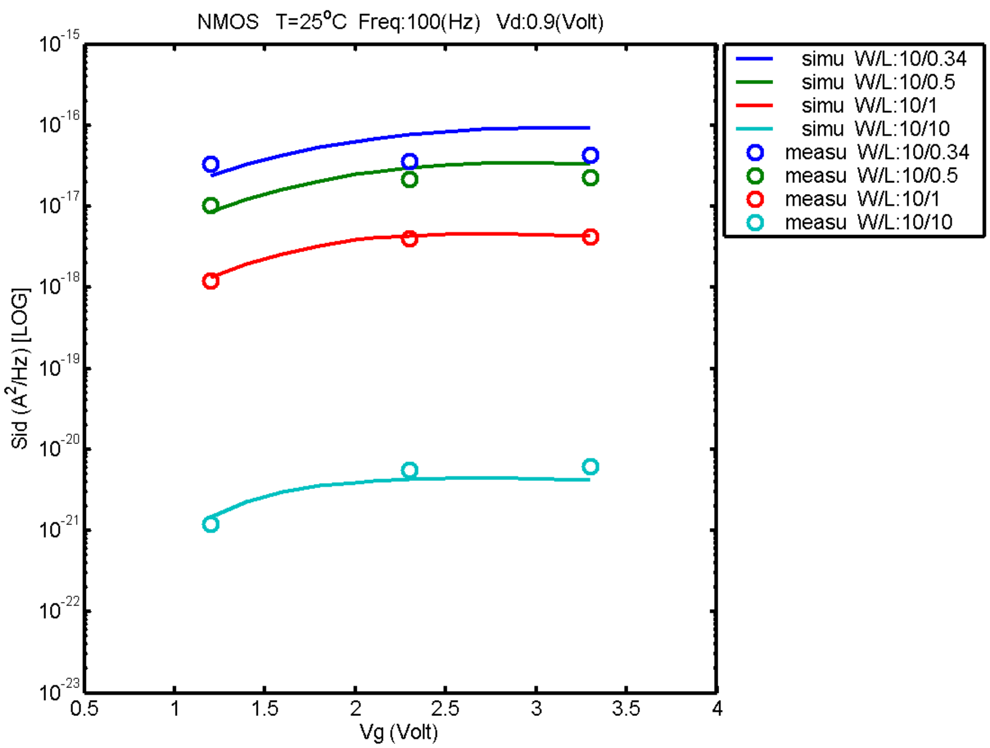

Figure 7-48 Noise vs Vg for NMOS @ vd=0.9V

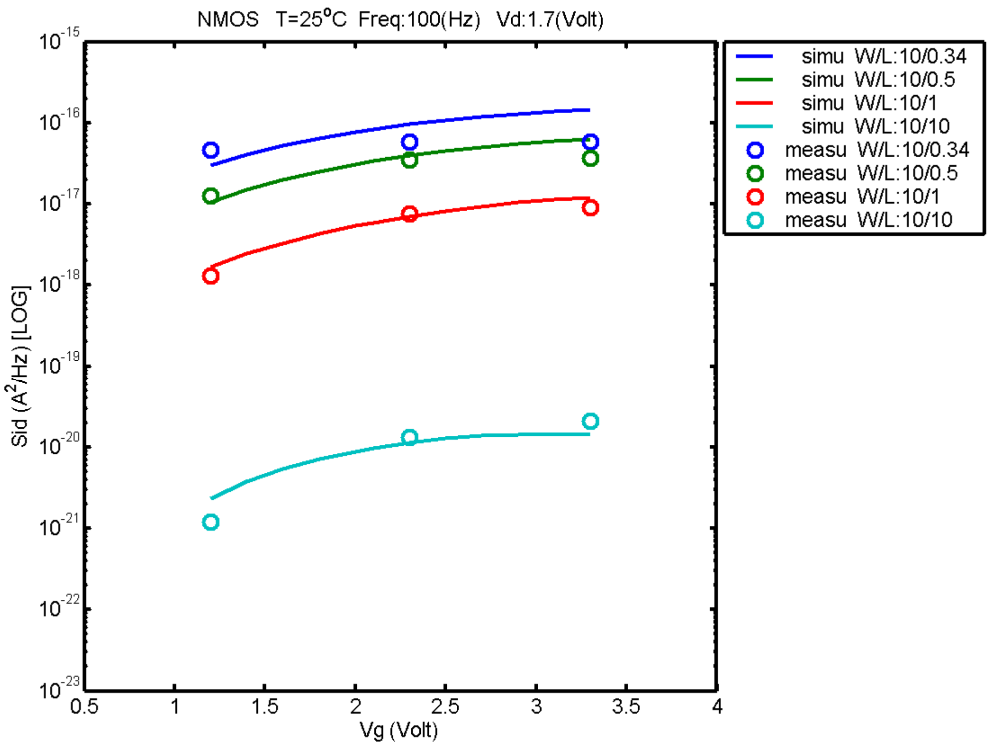

Figure 7-49 Noise vs Vg for NMOS @ vd=1.7V

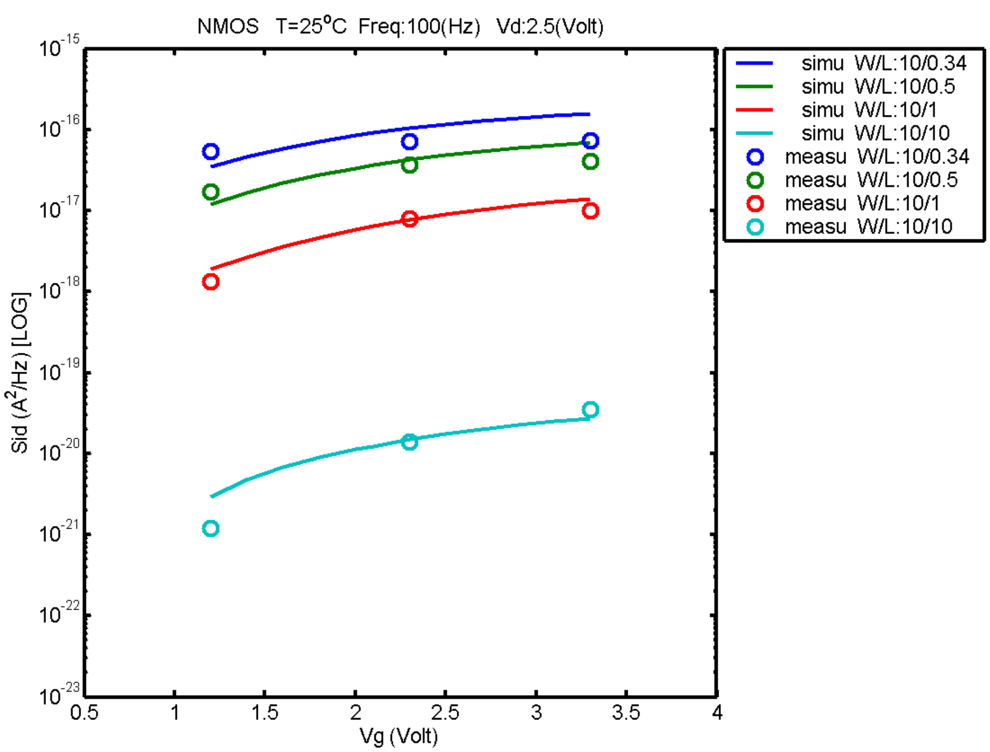

Figure 7-50 Noise vs Vg for NMOS @ vd=2.5V

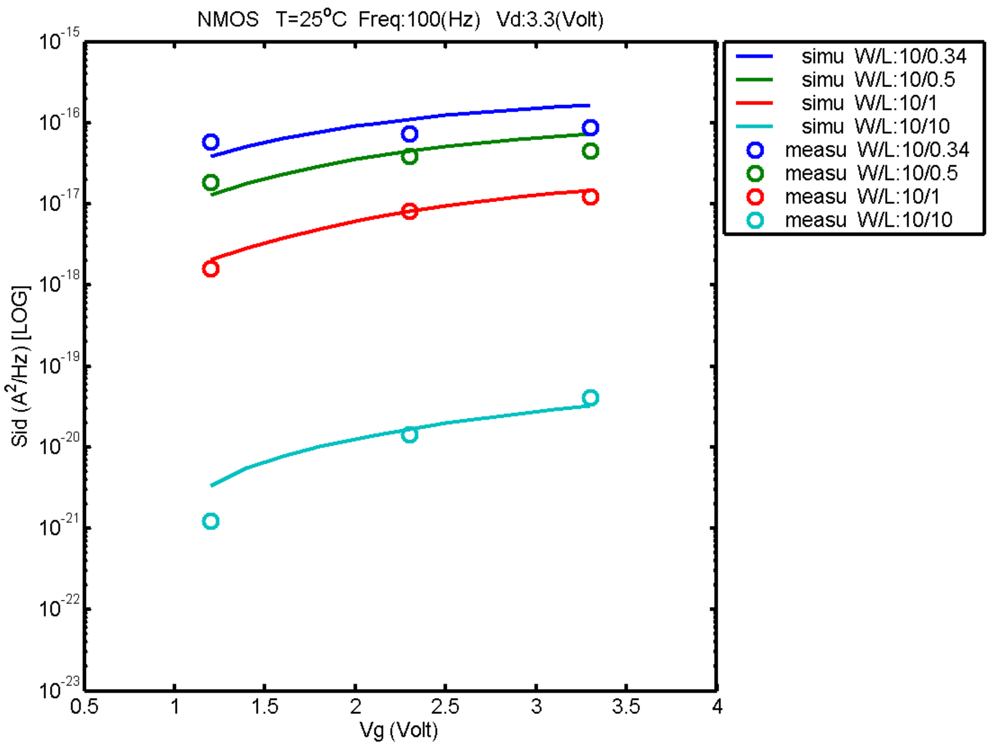

Figure 7-51 Noise vs Vg for NMOS @ vd=3.3V

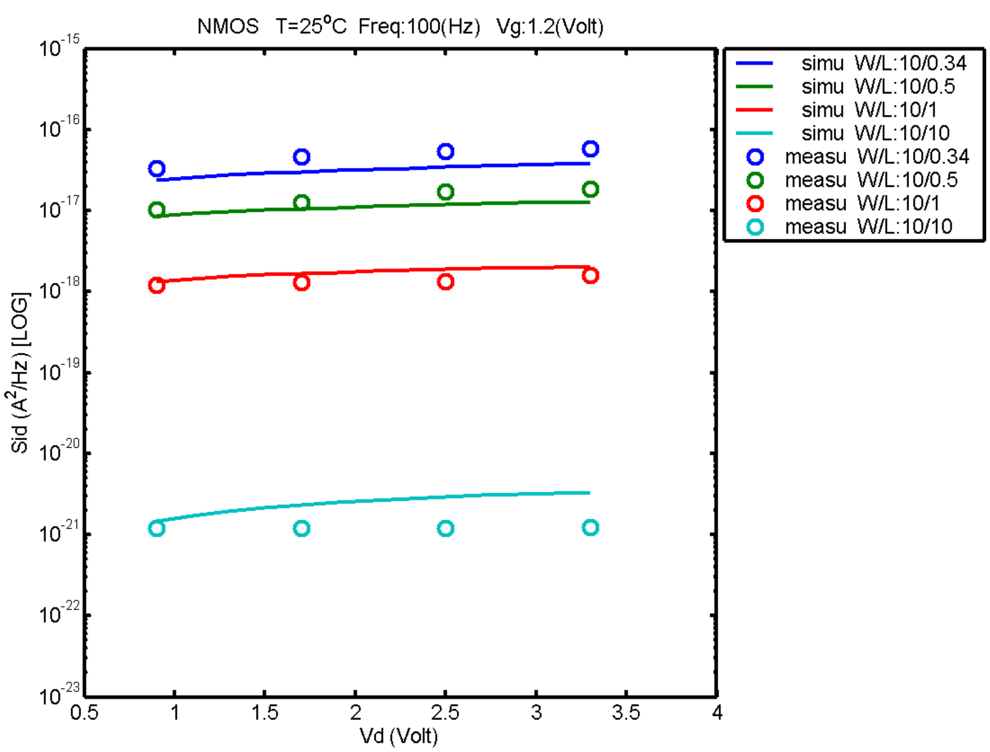

Figure 7-52 Noise vs Vd for NMOS @ vg=1.2V

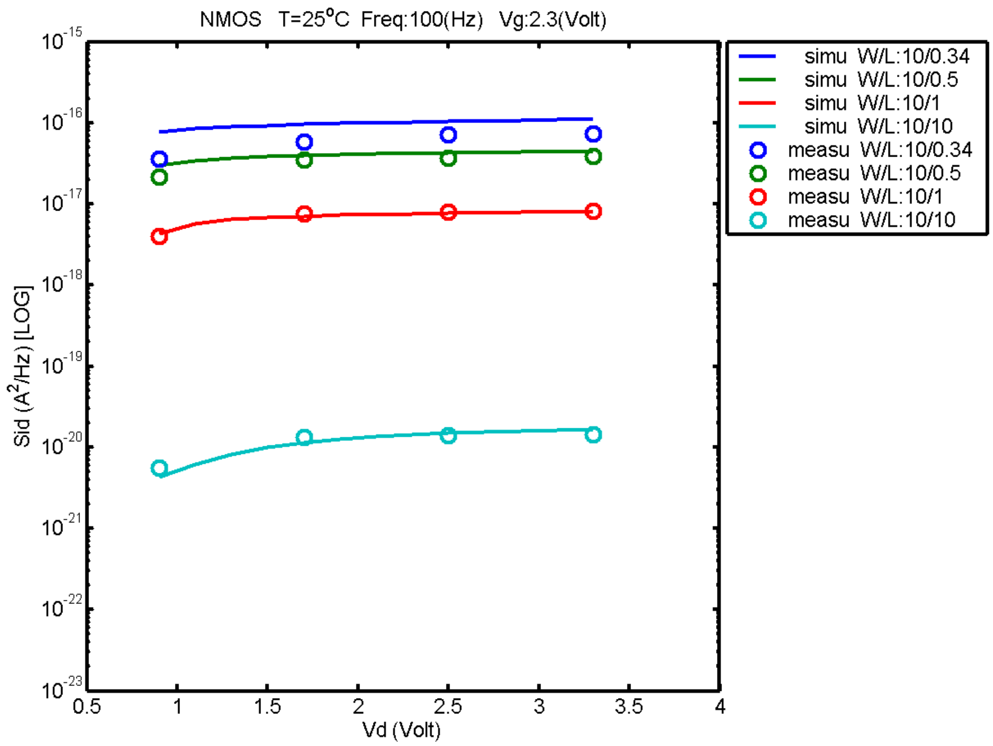

Figure 7-53 Noise vs Vd for NMOS @ vg=2.3

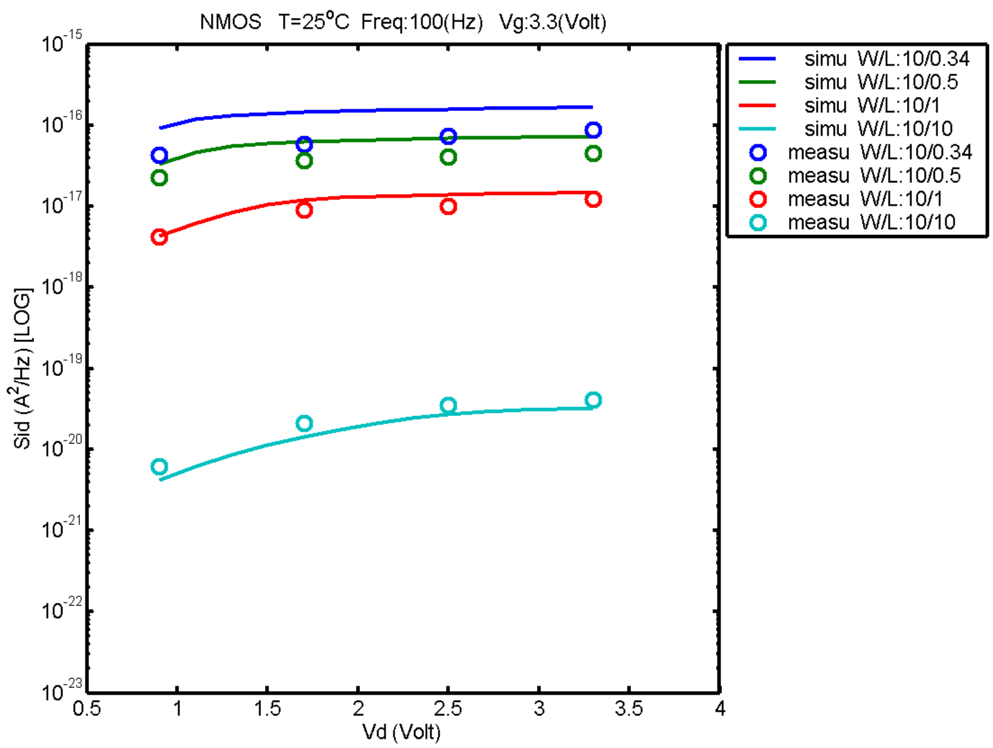

Figure 7-54 Noise vs Vd for NMOS @ vg=3.3V

B. PMOS

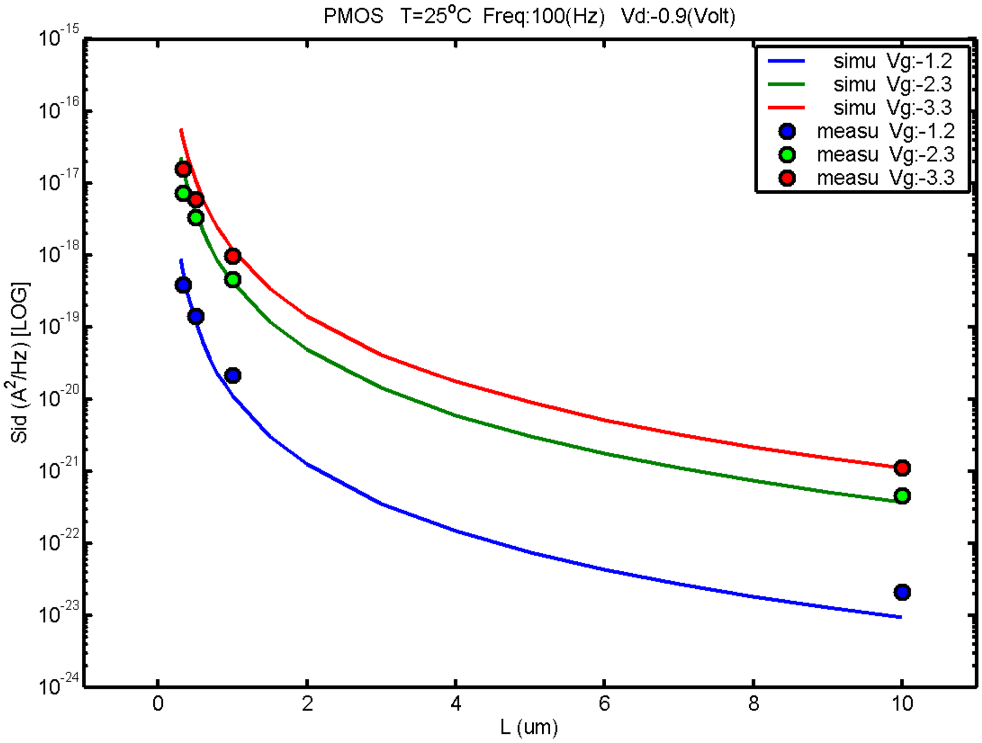

Figure 7-55 Noise vs Length for PMOS @ vd= -0.9V

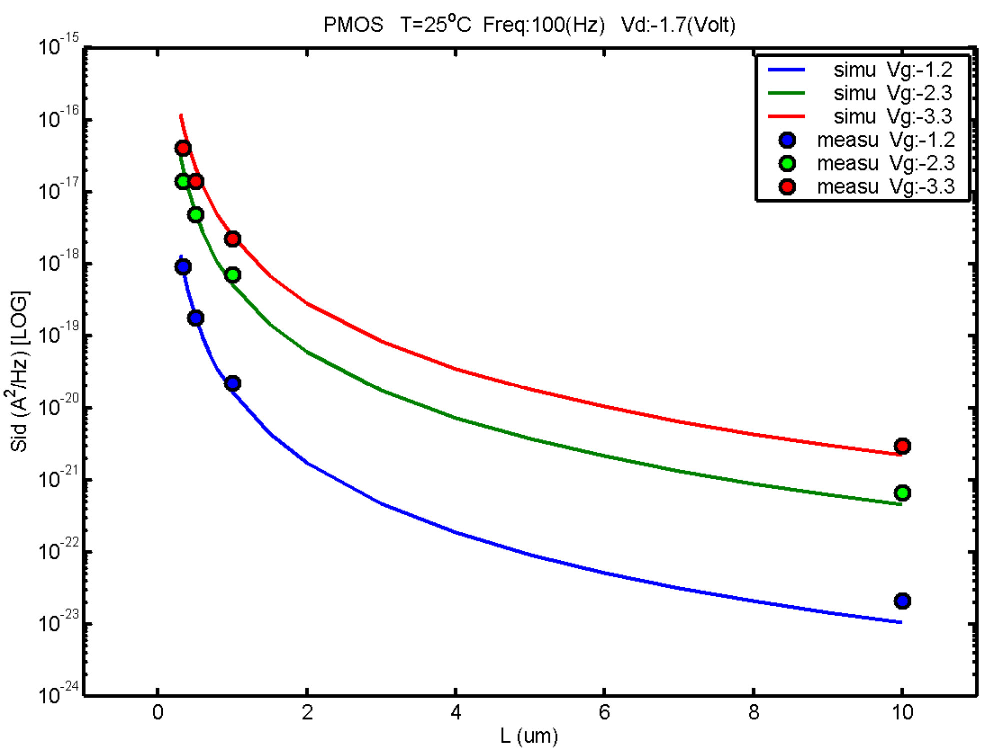

Figure 7-56 Noise vs Length for PMOS @ vd= -1.7V

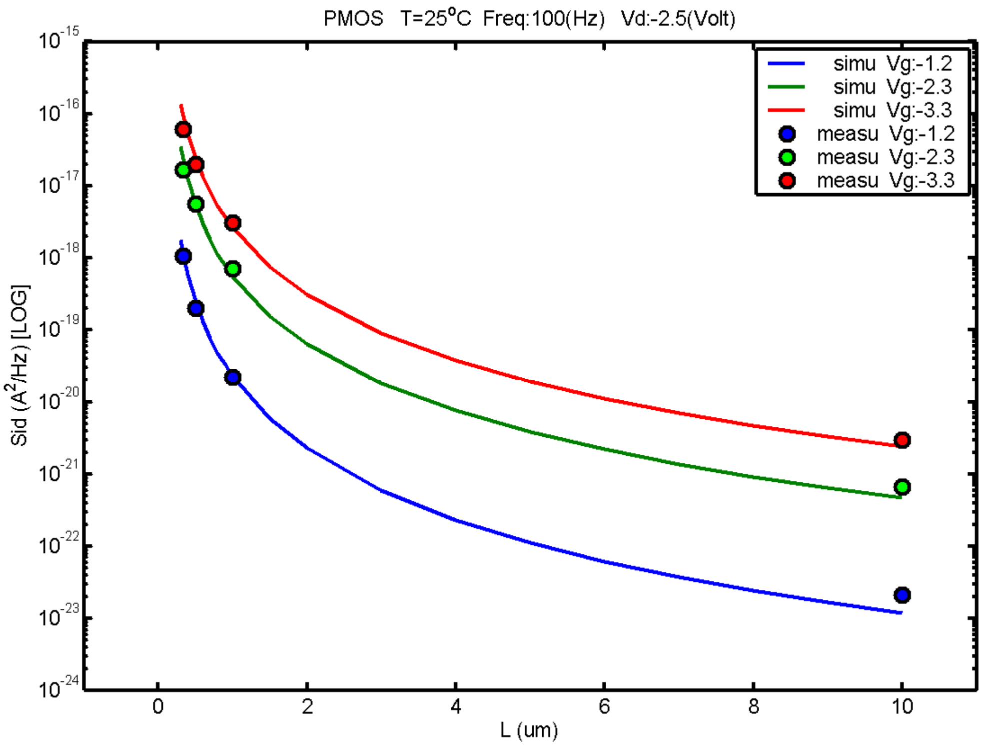

Figure 7-57 Noise vs Length for PMOS @ vd= -2.5V

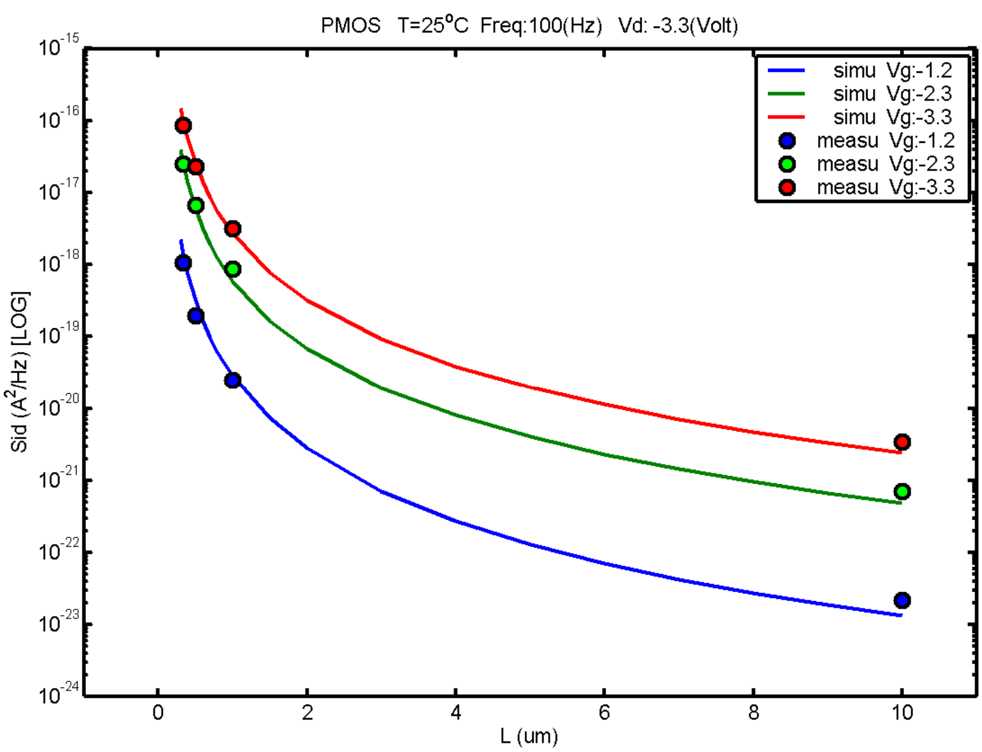

Figure 7-58 Noise vs Length for PMOS @ vd= -3.3V

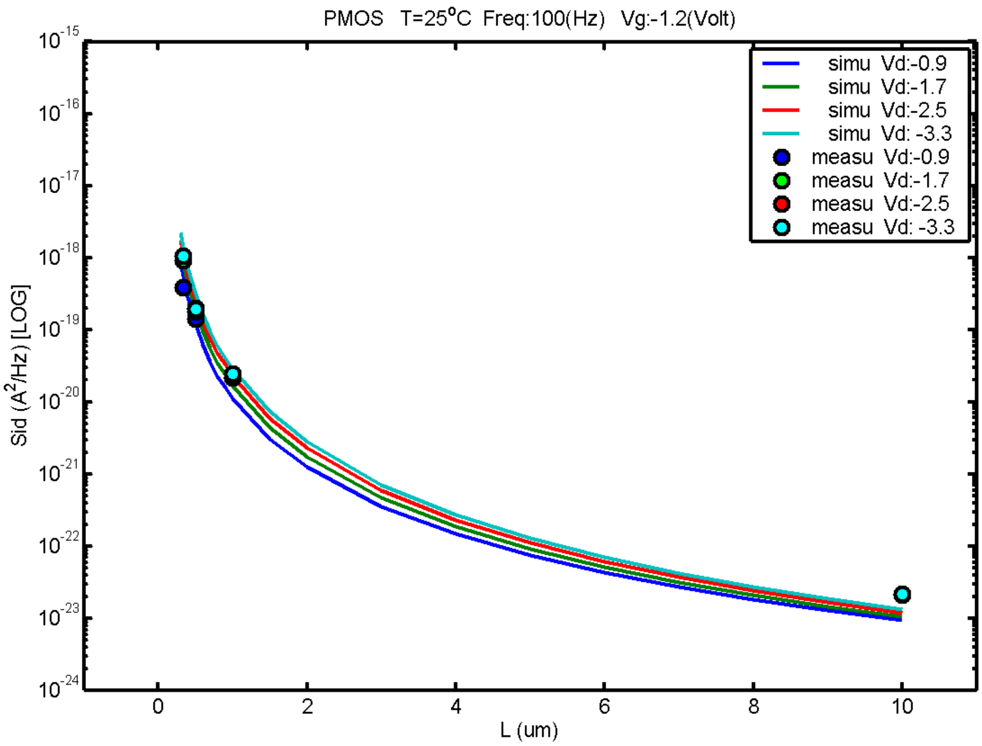

Figure 7-59 Noise vs Length for PMOS @ vg= -1.2V

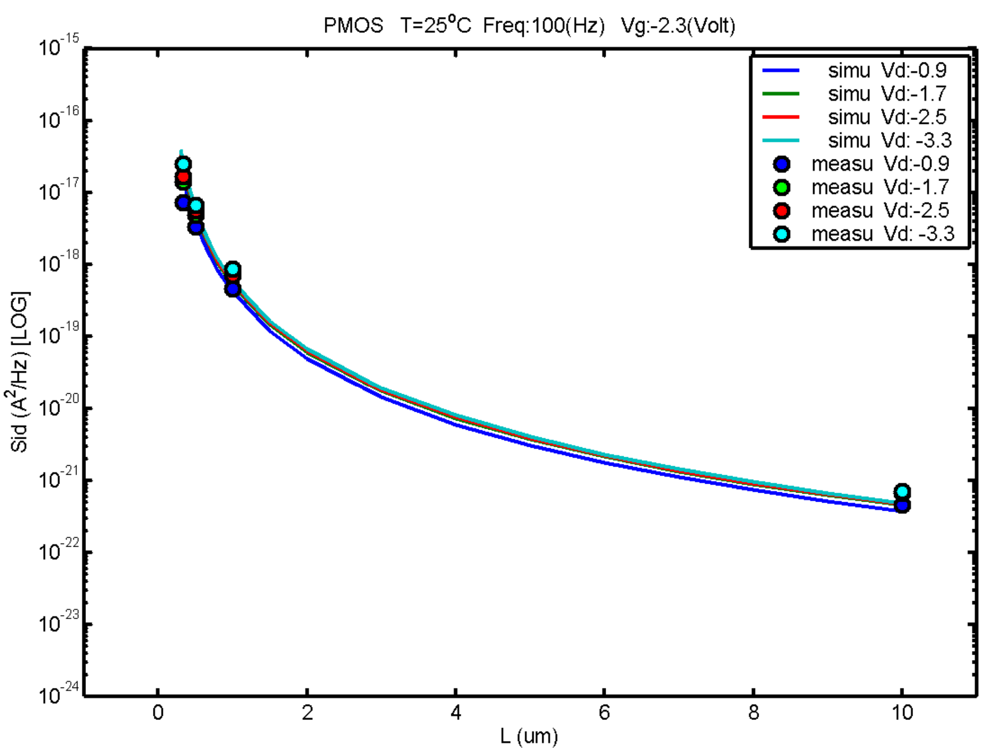

Figure 7-60 Noise vs Length for PMOS @ vg= -2.3V

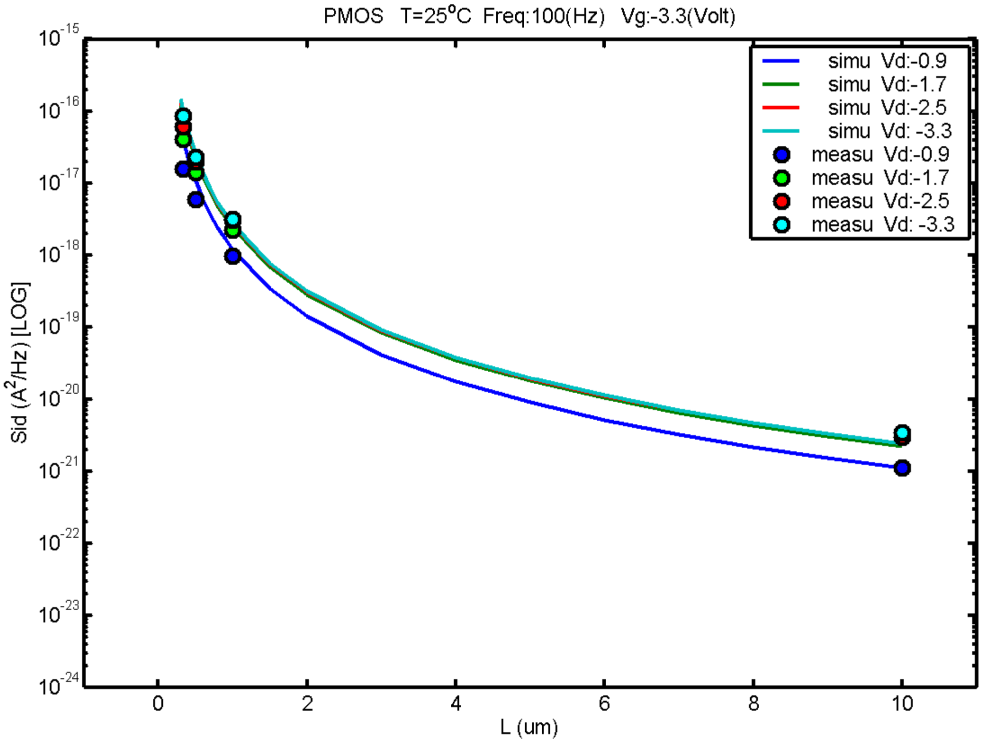

Figure 7-61 Noise vs Length for PMOS @ vg= -3.3V

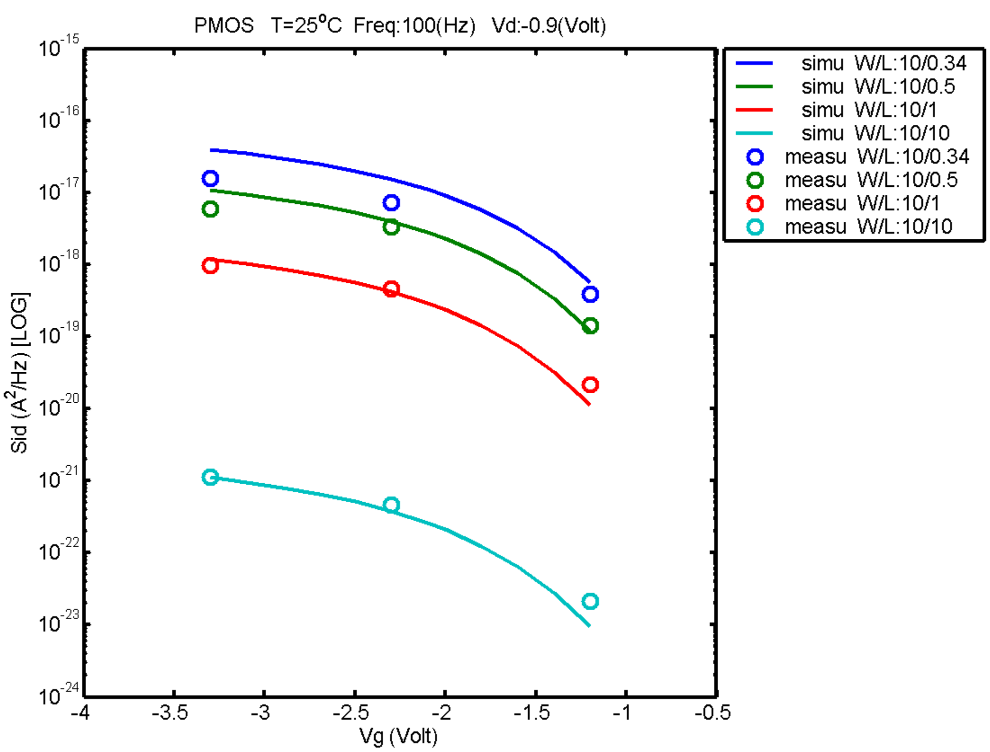

Figure 7-62 Noise vs Vg for PMOS @ vd= -0.9V

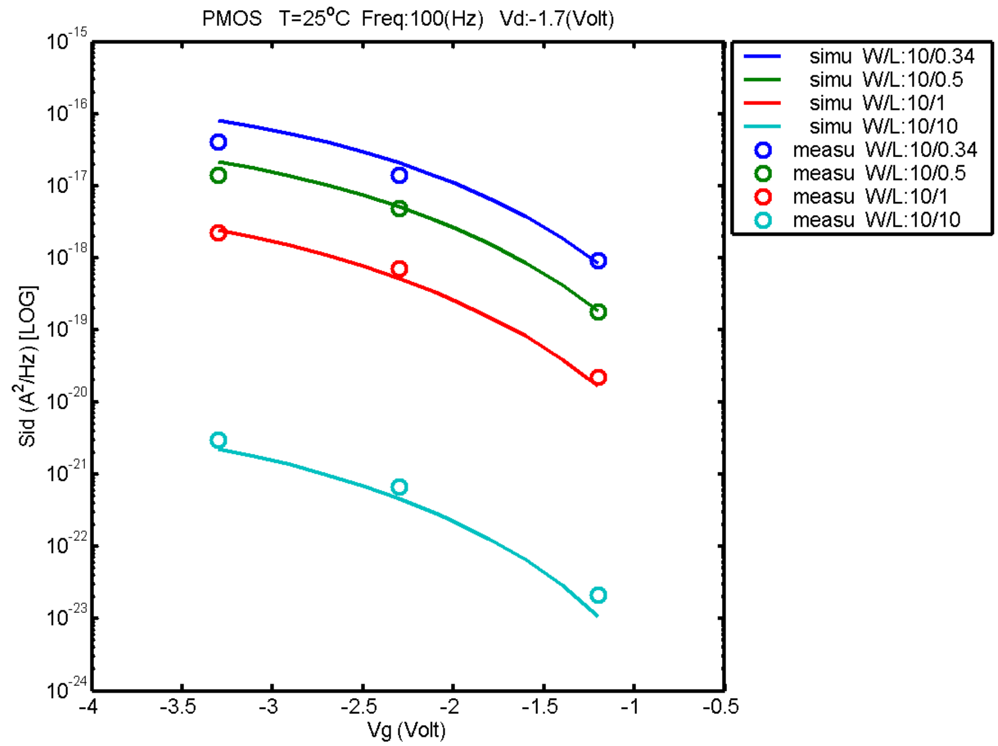

Figure 7-63 Noise vs Vg for PMOS @ vd= -1.7V

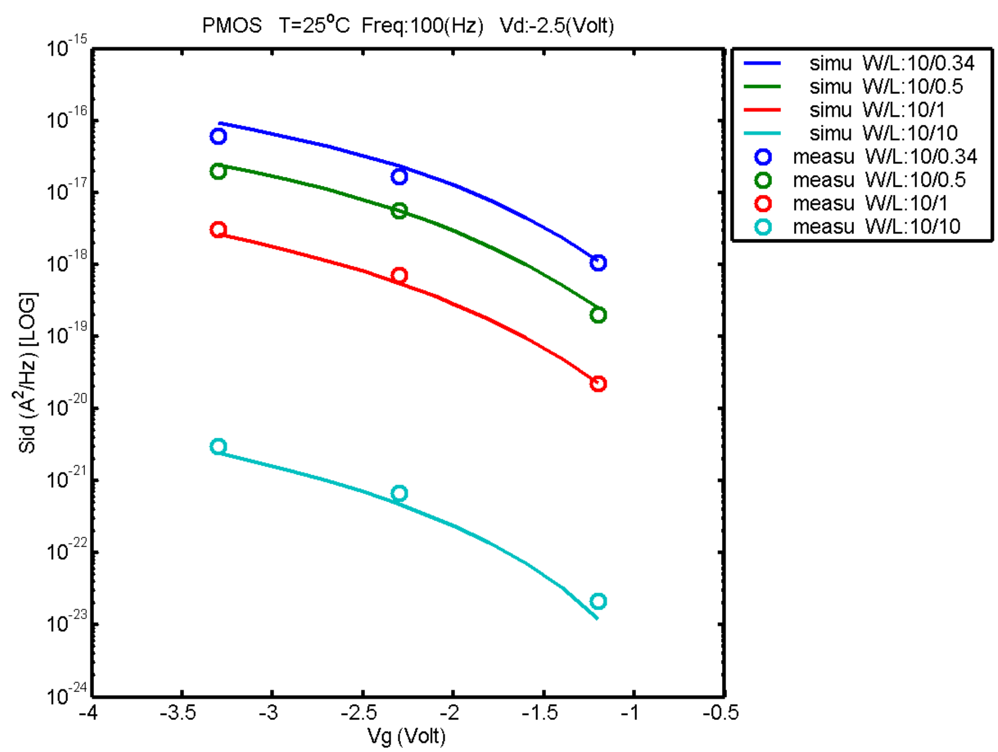

Figure 7-64 Noise vs Vg for PMOS @ vd= -2.5V

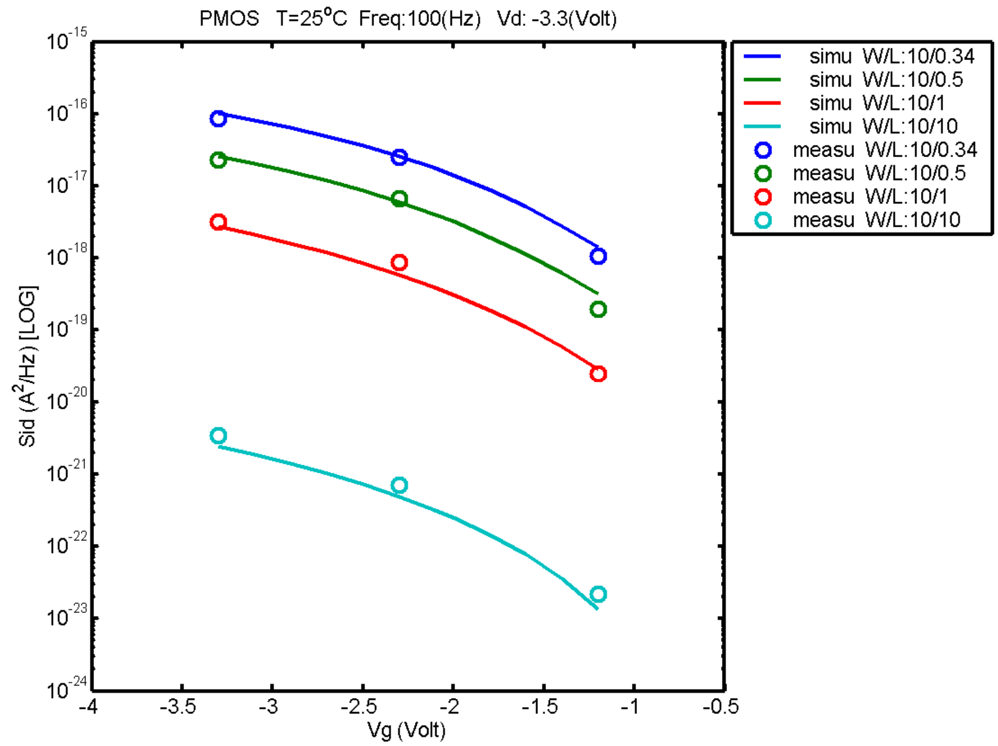

Figure 7-65 Noise vs Vg for PMOS @ vd= -3.3V

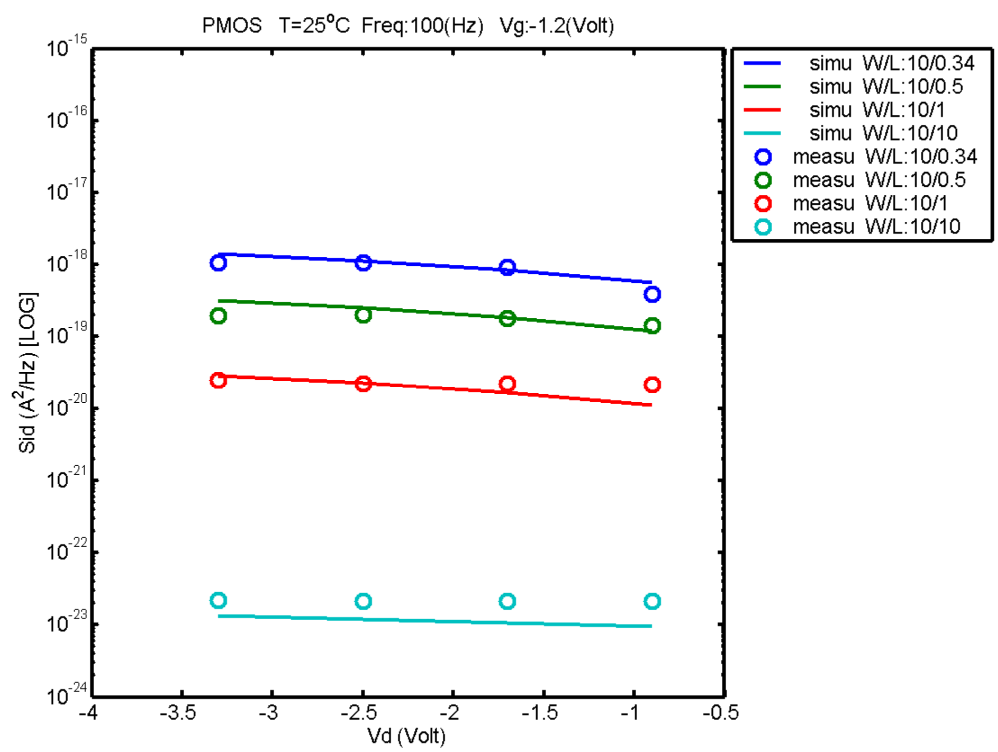

Figure 7-66 Noise vs Vd for PMOS @ vg= -1.2V

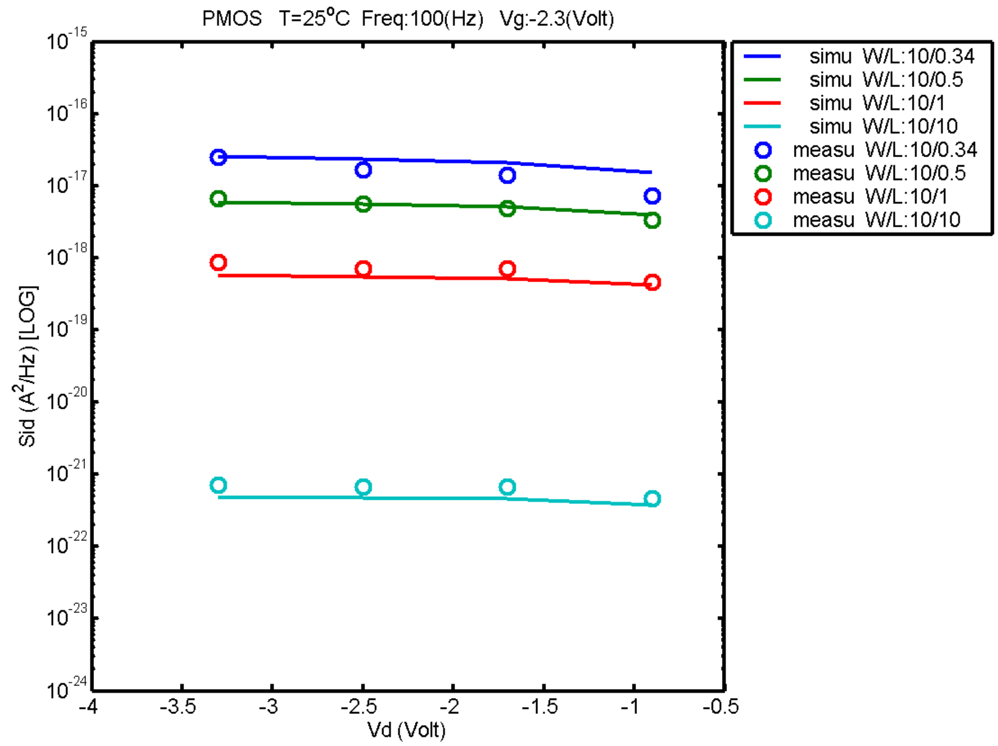

Figure 7-67 Noise vs Vd for PMOS @ vg= -2.3V

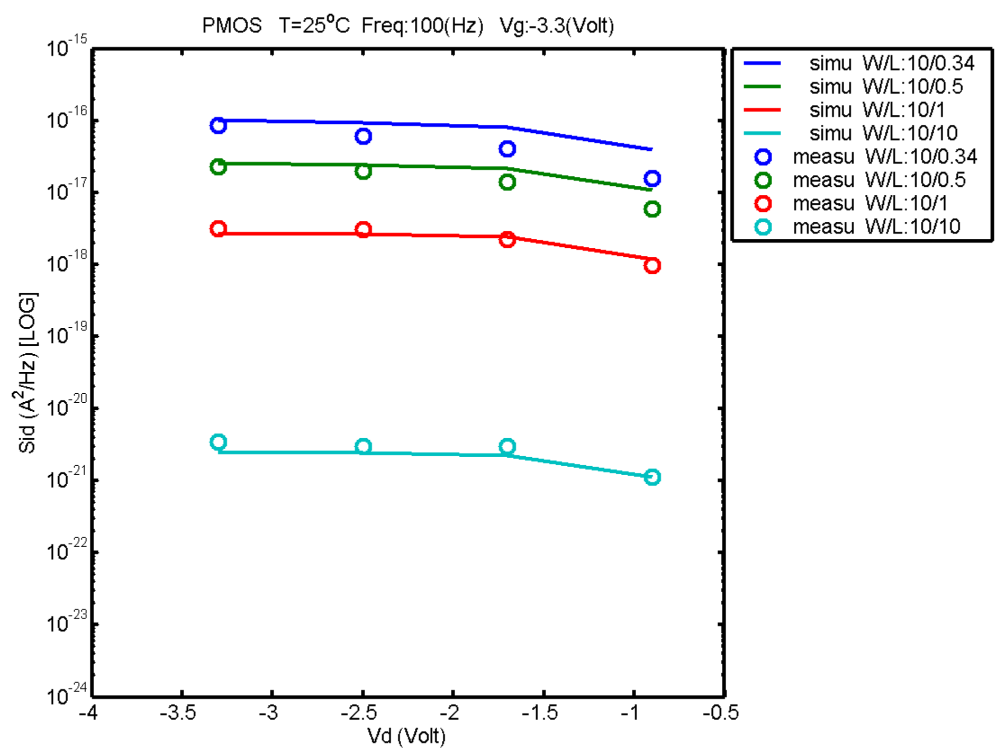

Figure 7-68 Noise vs Vd for PMOS @ vg= -3.3V

8 Appendix

8.1 Verify corner plot.

Table 8-1 Verify wafer information

Mask ID	A11001	A11001	A11001
Lot ID	D60K6	D5N0S	D63LJ
Wafer Number	#1 and #3	#1 and #2	#8 and #9

Threshold voltage in linear region V_{TLN}

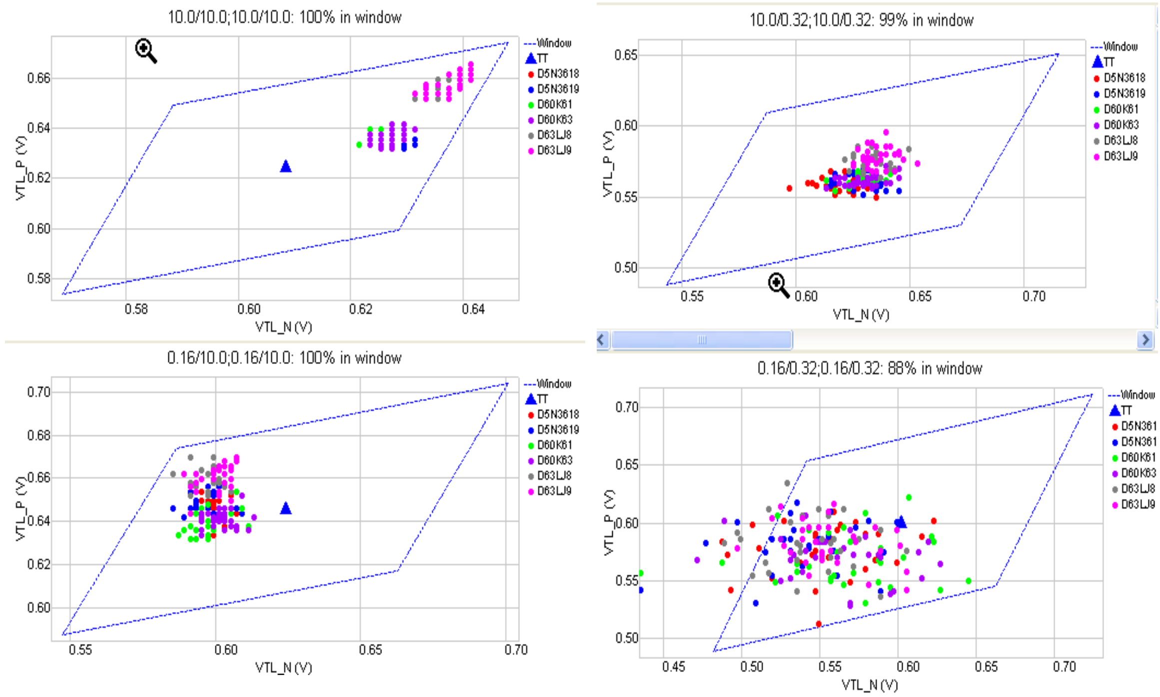

Figure 8-1 Corner graph of threshold voltage at low drain bias

Threshold voltage in Saturation region V_{Tsat}

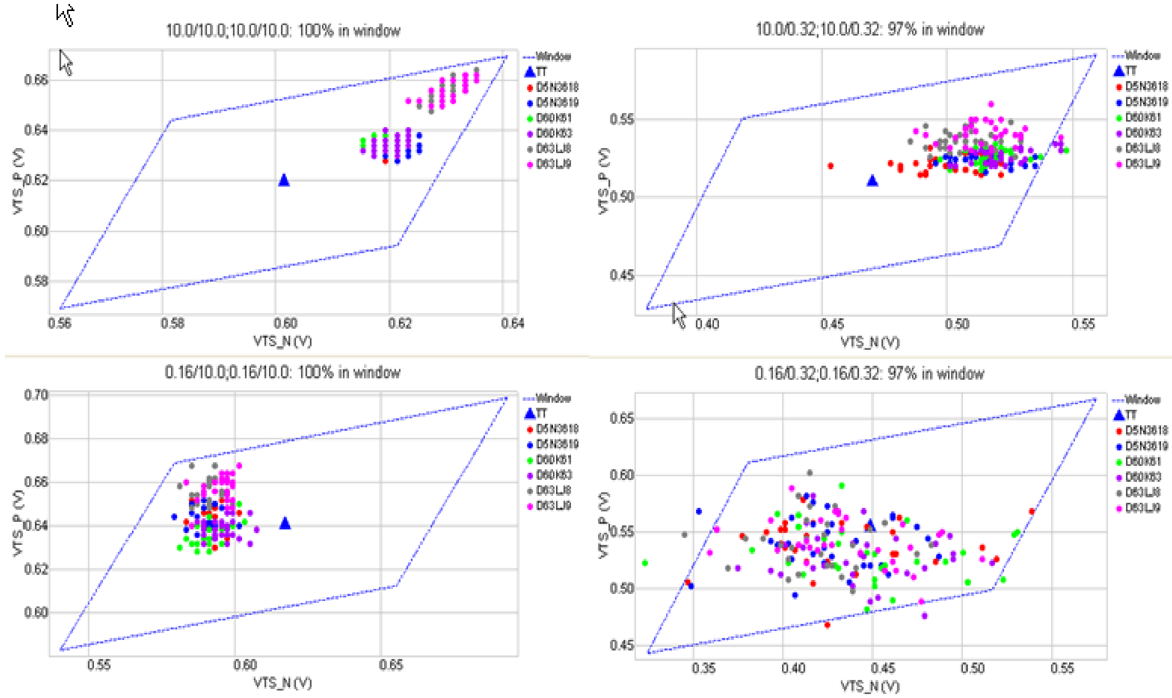

Figure 8-2 Corner graph of threshold voltage at high drain bias

Linear Current I_{LIN}

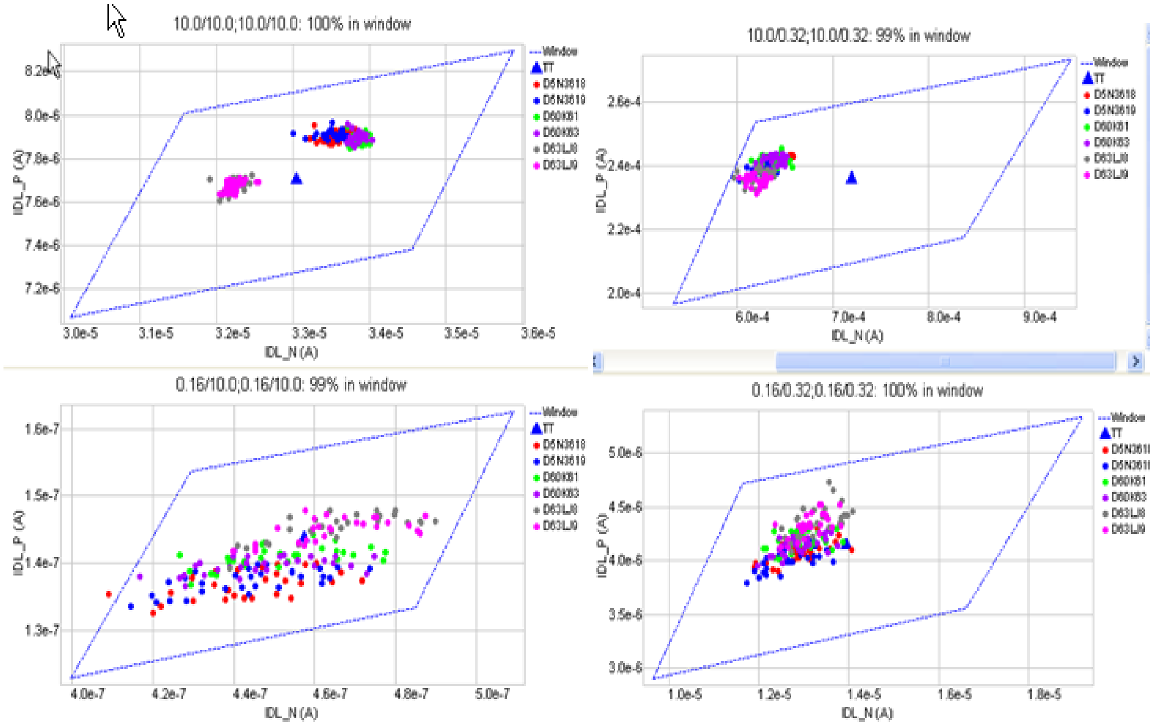

Figure 8-3 Corner graph of linear current

Saturation Current I_{ON}

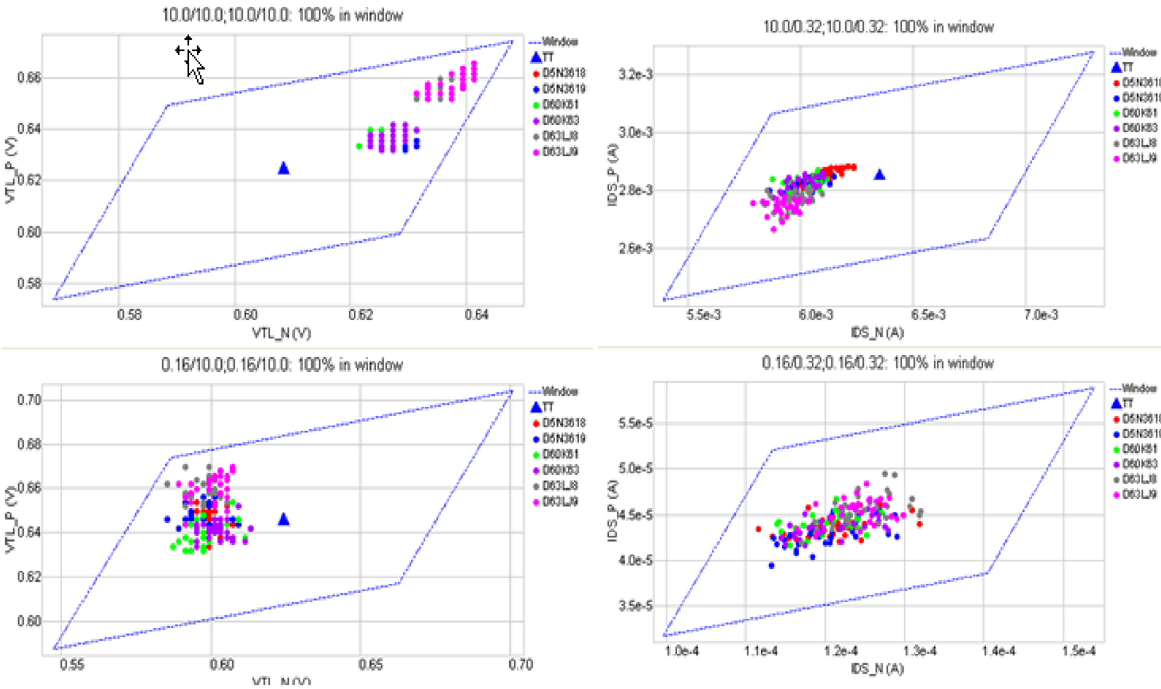

Figure 8-4 Corner graph of on-current

8.2 HSPICE Warning Message

Table 8-2 Hspice Warning Message

Item	Warning Message	Explanation
1	None	None

8.3 ELDO Warning Message

Table 8-3 Eldo Warning Message

Item	Warning Message	Explanation
1	None	None

8.4 SPECTRE Warning Message

Table 8-4 Spectre Warning Message

Item	Warning Message	Explanation
1	None	None